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Corso di Laurea Magistrale in Informatica

Tesi di Laurea Magistrale

Logic Programming for Inference on Quantum Computers

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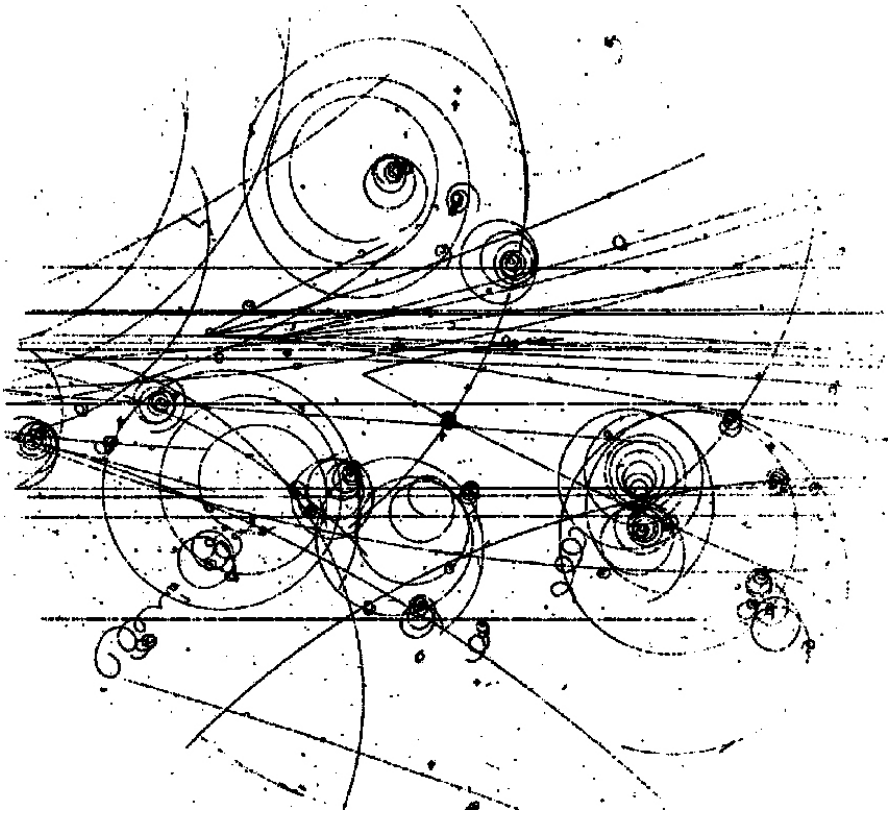

A tutti quelli a cui voglio bene.
To everyone I love.

"If you think you understand quantum mechanics,
you don't understand quantum mechanics."
— Richard P. Feynman

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ABSTRACT

In this thesis we revisit and extend the work of Scott Pakin presented in [1]. This work presents a rewriting pipeline to move from a high-level language, Prolog, to a formulation compatible with solving on quantum annealers, a particular type of adiabatic quantum computing, that is, quantum hardware specialized in solving optimization problems.

In this thesis we start from Pakin's work by making it compatible again with the frameworks used to interface with quantum annealers. Once the compatibility of the pipeline has been restored, we extend it both upstream and downstream.

Upstream: we rewrite an ontology, a particular type of knowledge base, in Prolog. We then present some solutions to make this translation compatible with Pakin's pipeline, which does not implement the full set of Prolog features. Finally, we study the size of the problem obtained as a function of the ontology we translate, and the query used to interrogate the knowledge base.

Downstream: we show that the output of the pipeline is not limited to resolution through annealing, but that a formulation as an optimization problem also allows the application of algorithms developed for the other type of quantum architecture (the quantum gate-based one). We show how to move from one formulation to another and how to perform some experiments to study this approach's behavior.

Now we briefly show how the work is organized. The first three chapters provide a theoretical introduction. In Chapter 1 we provide some basics of quantum mechanics; the goal of this chapter is to argue that physical transformations, and therefore quantum gates, must be reversible. In Chapter 2 we introduce the study of computer science, defining computational complexity classes and showing where quantum computers fit into this analysis. The chapter concludes with the presentation of the two paradigms of quantum computation: quantum gate-based computing and adiabatic quantum computing. Chapter 3 presents ontologies, showing the main characteristics of these knowledge bases and discussing the complexity of performing reasoning over them.

We then move to two chapters in which we present the tools used in the rest of the work. In Chapter 4 we show how to set up a de-

velopment environment with all the software and libraries needed to carry out our experiments. Chapter 5 presents Pakin's pipeline for translating from Prolog to an optimization problem. We examine the characteristics of each step of the pipeline and present our contribution to restore compatibility between the pipeline and the other software it interfaces with.

The last two chapters concern the experiments carried out in this work. Chapter 6 shows the translation of an ontology into Prolog and from Prolog to an optimization problem. Chapter 7 deals with the discretized version of quantum annealing implementable on quantum gate-based systems. We show how to start from the result of Pakin's pipeline and how to obtain a formulation suitable for this paradigm.

In the conclusions (Chapter 8) we review the results achieved and provide some directions for future work.

CONTENTS

ACRONYMS

AQC
: Adiabatic Quantum Computing.

KB
: Knowledge Base.

QA
: Quantum Annealing.

QAOA
: Quantum Approximation Optimization Algorithm.

Part I

THEORETICAL BASES

1 QUANTUM MECHANICS

In this chapter we will explore the basics of quantum mechanics in order to understand what we can or cannot do with a quantum computer. The reference architecture for this work is the quantum gate-based quantum computer. In the next pages we will try to justify why the algorithms that run on this hardware need to be reversible.

The chapter starts from the very beginning with the definition of a quantum system and presents the basis to understand the evolution of a quantum system, trying to justify why every evolution needs to be reversible and what exactly reversibility means. At the end of the chapter we will put all of our new knowledge together to derive the famous Schrödinger equation.

With all this work we will be able to imagine the function of a quantum gate and understand the limitations that are imposed when we develop an algorithm for a quantum computer.

This chapter is inspired by [2] specifically on chapters 1, 2, 3 and 4 of the original book. Because of this, no citations will appear throughout the text as it would only be redundant.

1.1 EXPERIMENTS

We start our introduction to quantum mechanics with an experiment. Experiments are not only an excuse to introduce the topic, but the essential key of physics, both classical and modern.

Theory and models need to adapt to the experiments, and when the experimental results are in contradiction with the actual model it means that the model needs to be changed to respect the behavior of the world.

1.1.1 Spin

We analyze the experiment of an electron in a magnetic field. An electron is an electrically charged particle; when some electrons are shot in an electric field all of them are influenced by Coulomb's law; if all electrons have the same initial velocity the beam of electrons remains intact.

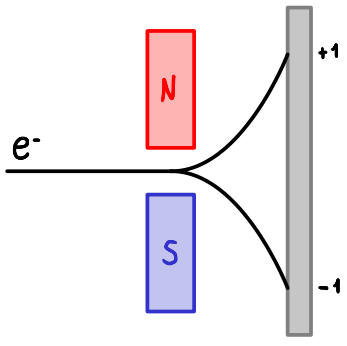


Figure 1.1: Experiment's schema

What happens to the same beam in a magnetic field? Again electrons are deflected by a force, but this time the beam splits. If the initial velocity was parallel to the x axis and the magnetic field is oriented along the z axis some electrons are deflected upward, some downward, but the intensity of the deflection is

the same for all the electrons. This means that no electrons are deflected less or more than the others and the beam splits exactly into two parts.

Starting from this experiment we can make a measuring instrument: this apparatus $\mathcal{A}$ can be oriented along an arbitrary axis, and in the previous configuration $\mathcal{A}$ displays $+1$ if the electron is deflected upward, -1 otherwise. We call this number the spin of the electron.

Repeatability of measure

If we measure the spin of an electron and $\mathcal{A}$ displays $+1$, we can confirm the experiment's results by measuring the spin again and we obtain spin $+1$ every time. This means that the measurements are repeatable (an essential property to construct models and make predictions). We can think, and it will be clear later why it is useful, that the first experiment prepares the spin $+1$ and the others confirm this result.

Spin is a quantum property and all the visual representations such as the rotation of the electron around its axis would lead to misrepresentation. Spin and rotation, however, have some similarities. Let's analyze what would happen if we consider a charged sphere in a magnetic field with the laws of classical physics. We consider a sphere rotating around its axis, and this axis is parallel to the z axis. The x or y component of the angular momentum is zero. Measuring the component along a generic axis, oriented like the versor $\hat{n}$[1], we would obtain a result proportional to the projection of $\hat{z}$ on $\hat{n}$. This projection can be found with the scalar product $\hat{z} \cdot \hat{n} = \cos\theta$[2], where θ is the angle between the axes.

Now we consider the quantum version of this phenomenon. Let's start by measuring the z component of the spin and assume that the result is $\sigma_z = +1$; if we rotate the apparatus $\mathcal{A}$ around, for example, the x axis, we can measure σ_x. This component would not be zero, and $\mathcal{A}$ keeps displaying only $+1$ or -1. The single result is not helpful, but we can repeat the experiment, namely:

1. orienting $\mathcal{A}$ along the z axis and preparing a spin $\sigma_z = +1$
2. rotating $\mathcal{A}$ around x
3. measuring the x component of the spin

statistically we would observe the same number of $\sigma_x = +1$ and $\sigma_x = -1$.

If we start the experiments with a spin prepared as $\sigma_z = +1$ and then orient $\mathcal{A}$ along a generic axis $\hat{n}$ each measure would be binary and unpredictable, but the mean of the measures tends to $\hat{z} \cdot \hat{n} = \cos\theta$ where θ is the angle between $\hat{z}$ and $\hat{n}$. In the most general case we can start with the apparatus oriented like m and prepare the spin $\sigma_m = +1$, then we rotate $\mathcal{A}$ around $\hat{n}$ without interfering with the spin and measure again; we would obtain the statistical result $\langle \sigma_n \rangle = \hat{m} \cdot \hat{n}$[3].

The result of a single measure is non deterministic, but we can make predictions over the mean values of the measures: the expected values behave as the single results of the classic experiment.

Invasive experiment

Considering now a sequence of three measures: starting with $\mathcal{A}$ oriented along z we prepare the spin $\sigma_z = +1$, then we rotate $\mathcal{A}$ to measure σ_x

1 A versor is a vector of magnitude 1 (unit vector), it is normally used to specify a direction.
2 We can use directly the angle because we are considering versors.
3 The Dirac bracket $\langle\ \rangle$ denotes the statistical mean of a quantity. We call that expectation value.

obtaining, let's say, $+1$ (the reasoning is the same if we obtain -1); lastly returning with $\mathcal{A}$ parallel to z we cannot make any prediction on the single result, the initial configuration (with $\sigma_z = +1$) is lost forever, the only result we can predict is that $\langle \sigma_z \rangle = 0$.

1.1.2 Qubit

We have introduced the spin referring to electrons in a magnetic field. However, we can study the spin without examining the associated electron; we have isolated a simple physical system, the simplest we can study.

Spin belongs to a class of simple physical systems called *qubit*; in all of these systems the result of a measure is binary. We will see that, even if the result of a measure is equal to the classical *bit*, the qubit system is described in a very different way compared to its classical alter ego.

1.1.3 Boolean Logic

In this paragraph we try to understand why we need two different ways to describe a classical and a quantum state space. To do so we analyze the results of some logical propositions, both basic and composed via logical connectives.

Starting with the classical case we consider a bag of colored and numbered balls. We can construct the state space by enumerating all states, namely taking each ball from the bag and annotating the pair number–color. The basic propositions we analyze are:

- The extracted ball is red.
- The number on the extracted ball is even.

If we consider a particular state we can say if a proposition is true or false; we can also define two subsets of balls, the first with all the red balls (for this subset the first proposition is true), the second one with the balls that show an even number (subset that makes the second proposition true). Considering now disjunction and conjunction:

- The extracted ball is red OR even.
- The extracted ball is red AND even.

Again it is simple to associate a truth value to these propositions if we consider a single state; also we can construct two subsets that satisfy the propositions from the subsets we defined before: the new subsets are respectively the union and intersection of the old ones.

In the quantum world the situation is very different. Let's start from propositions that can be verified with a simple experiment:

- The z component of the spin is $+1$.
- The x component of the spin is $+1$.

If we want to check the first proposition we can orient the apparatus $\mathcal{A}$ along z and make a measurement; the same procedure can be followed for the second proposition. The disjunction and conjunction of these propositions are:

- The z component of the spin is $+1$ OR the x component is $+1$.
- The z component of the spin is $+1$ AND the x component is $+1$.

The disjunction is not commutative

Starting with the disjunction. Considering a state prepared, without our knowledge, with $\sigma_z = +1$. If our first measure is along the z axis, $\mathcal{A}$ will always display $+1$ and we can immediately conclude that the proposition is true. If we start measuring the x component, we have a 50% chance that $\mathcal{A}$ displays $+1$ or -1; also this measurement destroys the initial state and the measure of σ_z becomes non predictable. In this scenario we have a 25% chance of deducing that the proposition is false; Figure 1.2 shows all the possible measurement results in this case. The logical value of a proposition depends on the order in which we perform the measurements.

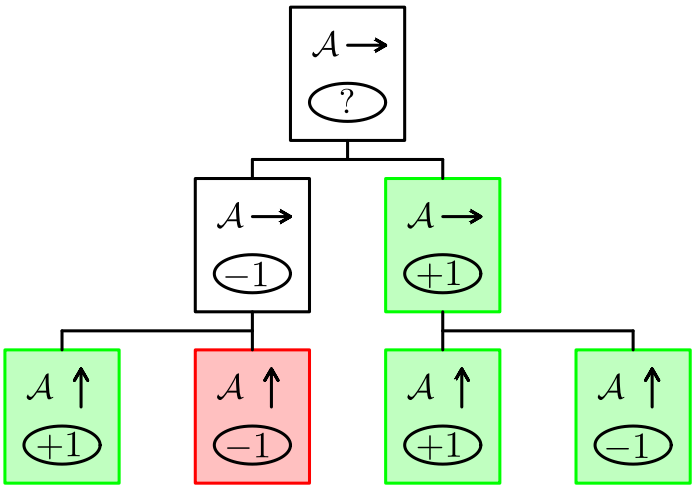


Figure 1.2: The apparatus $\mathcal{A}$ is represented as a box, the arrow represents the direction along which the apparatus is oriented, the display (ellipse) shows the result of measurement. We have highlighted in green the cases in witch we can immediately conclude that the disjunction of the propositions is true

The conjunction loses its meaning

The conjunction is even worse: no matter the order of the measurements, the second one destroys the result of the first. The disjunction is true if at least one of the sub-propositions is true, and if we find a spin component that is $+1$ we can always confirm this result with another measurement. In the conjunction the two sub-propositions must be true *at the same time*, but with the second measurement we lose all the knowledge of the first one. We can never conclude that the conjunction is true.

1.2 QUANTUM STATES

In the previous section we have understood that a state space of a quantum system cannot be represented in the same way as a classical state space. Now we present a formal mathematical model to describe the state space for spin.

Axiom 1. *The state space for a quantum system is a complex vector space.*

This is a physical axiom, which means that it is true because there are a lot of experiments that confirm this model and none that shows a contradiction.

1.2.1 Vector Spaces

A vector space is a mathematical and abstract construction that can have multiple dimensions (even infinite) and has, as components, integers, real or complex numbers, or other elements. An example that shows well how abstract a vector space can be is the complex-valued continuous function of variable x; the set of these functions generates a vector space.

In quantum mechanics the state space is described by a vector space having as element $|A\rangle$ called *ket*. The properties of this space are: *Hilbert space*

- the sum of two kets is a ket;
- addition is commutative;
- addition is associative;
- existence of identity element for addition;
- existence of inverse elements for addition;
- existence of identity element for scalar multiplication;
- linearity property.

1.2.2 Bra and Ket

An example of ket that we will find often is the column vector of two dimensions:

$$|A\rangle = \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix}$$

where α_1 and α_2 are complex numbers. With this simple example of ket it is easy to verify the validity of all previously described properties.

If, for complex numbers, exists the complex conjugate, for every ket there exists a *bra*. The set of bra generates a dual conjugate space with respect to the state space of ket. We denote a bra as $\langle A|$. If $|A\rangle$ is the ket of the previous example the corresponding bra is a row vector having as elements the complex conjugate of $|A\rangle$:

$$\langle A| = (\alpha_1^*, \alpha_2^*)\,.$$

Name and symbol associated with elements of Hilbert spaces become clear *Inner product*

when we define the product *bra-ket*, this is the corresponding scalar product of an ordinary vector and is called inner product. Considering bra and ket of two dimensions we can evaluate the inner product by adding the products of corresponding components:

$$\langle A \,|\, B \rangle = (\alpha_1^*, \alpha_2^*) \cdot \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} = \alpha_1^* \beta_1 + \alpha_2^* \beta_2 .$$

Having the inner product we can define:

VERSOR normalized vector $|A\rangle$ in which $\langle A \,|\, A \rangle = 1$;

ORTHOGONAL VECTOR vectors that have a null inner product: $\langle A \,|\, B \rangle = 0$.

We are familiar with these concepts in two and three dimensions, the first one is a vector of length one, the second is the right angle between two vectors. This representation is misleading in our case, we cannot imagine a ket like an arrow and the state space is completely abstract even if there are properties and operations in common between this space and the 3D space that we are familiar with.

We have lost the geometric interpretation, and it seems that we have defined two completely abstract and useless concepts, we will see next that these are key concepts in the description of quantum systems and have a precise and important physical meaning.

Orthonormal basis

By having a vector space is possible to build a set of orthogonal versors that generates all vectors in the given space. This set is called orthonormal basis and the cardinality of the set is equal to the dimension of the space.

Formally having a basis $\mathscr{B} = \{ |i_1\rangle, |i_2\rangle, \ldots, |i_N\rangle \}$ of a space with N dimensions, we can write a generic vector in that space as

$$|A\rangle = \sum_{n=1}^{N} \alpha_n |i_n\rangle = \sum_{n=1}^{N} |i_n\rangle \langle i_n \,|\, A\rangle \tag{1.1}$$

this is the linear combination of the basis versors; where kets $|i_n\rangle$ are the versors in the basis and α_n are the vector components. We can obtain those components with the inner product between the vector $|A\rangle$ and the basis versors:

$$\alpha_i = \langle i \,|\, A \rangle . \tag{1.2}$$

1.2.3 Hidden variables

In a classical system we can measure all the variables associated to a physical system and then make a deterministic prediction of the evolution of that system. From the experiments described in the first section we have learned that a quantum system is not completely predictable even if we can make all the measurements that we want[4]. We can ask ourselves if our measurements aren't enough, if there are other variables that can make the prediction completely deterministic. About that topic we don't have any experimental proof, the opinion of physicists is divided in two main visions:

4 We remember that a measure along one axis destroys our knowledge about the result along another axis.

OPINION ONE : there are hidden variables and, if we manage to measure them, the prediction of results become deterministic. These variables can be

- very difficult to measure
- unknowable to us because also we are constituted by quantum material.

OPINION TWO : hidden variables don't exist, we already know all the information about a given system and quantum mechanics is intrinsically non deterministic.

No hidden variables

Probably no experiment could determine which vision is correct, but this doubt doesn't worsen our comprehension of the physical world. We can simply choose one vision and build our model coherently. We choose the simpler one, without hidden variables, all that we have to model are the quantities that we can measure and the measurements allow us to know all the information about a given system.

Even if we have lost complete determinism, knowing the state of a system gives us some information about the system and the successive measurements. In the next section we will see what we can deduce about spin.

1.2.4 Spin states

Let's start enumerating all possible spin states along the coordinate axes. If we rotate the apparatus $\mathcal{A}$ around z, we can obtain $\sigma_z = \pm 1$; we call these states *up* and *down* and label them with kets $|u\rangle$ and $|d\rangle$. Orienting $\mathcal{A}$ along x, we obtain *left* $|l\rangle$ and *right* $|r\rangle$. Lastly, along the y axis, we measure the states *in* $|i\rangle$ and *out* $|o\rangle$.

Spin space states have two dimensions

The hypothesis that there aren't hidden variables allows us to represent the space state in a simple way: each spin state can be represented as a ket in a two-dimensional complex vector space.

To express a vector we need a basis; we choose $\mathscr{B} = \{\, |u\rangle , |d\rangle \,\}$[5] and try to obtain all states as a linear combination (*superposition*) of the basis vectors. A generic state $|A\rangle$ can be expressed as:

$$|A\rangle = \alpha_u |u\rangle + \alpha_d |d\rangle$$

where α_u and α_d are the components of $|A\rangle$ along $|u\rangle$ and $|d\rangle$, and can be obtained by projection: $\alpha_u = \langle u | A\rangle$ and $\alpha_d = \langle d | A\rangle$ (as in Equation 1.2).

Probability amplitudes

$|A\rangle$ components are complex numbers and their physical meaning is: having a spin prepared in the state $|A\rangle = \alpha_u |u\rangle + \alpha_d |d\rangle$[6]; $\alpha_u^* \alpha_u$ is the probability of measuring $\sigma_z = +1$, while $\alpha_d^* \alpha_d$ is the probability that a measurement of σ_z will yield -1. Formally we can denote the probability of measuring $+1$ and -1 as P_u and P_d respectively and write:

$$\begin{aligned} P_u &= \langle A | u\rangle \langle u | A\rangle \\ P_d &= \langle A | d\rangle \langle d | A\rangle . \end{aligned} \tag{1.3}$$

5 We will show that these vectors are in fact orthogonal and why they need to be.

6 From now on we use "prepared" or "measured" as synonyms: every measurement is invasive and can change the spin state, so no matter what was the previous state, after a measurement the state is the one we have measured.

Components α_u and α_d are called probability amplitudes, and their physical meaning is given by the square of the magnitude. This is the actual probability, and we want the sum of all probabilities to be one. This is equivalent to requiring that $|A\rangle$ is normalized: $\langle A \,|\, A\rangle = 1$.

Now we will show why $|u\rangle$ and $|d\rangle$ have to be orthogonal:

$$\langle u \,|\, d\rangle = 0$$
$$\langle d \,|\, u\rangle = 0.$$

We try to give an idea with a *reductio ad absurdum*: if $|u\rangle$ and $|d\rangle$ were not orthogonal, the projection of one on the other would not be null. This means that if we orient $\mathcal{A}$ along z and measure $\sigma_z = +1 = |u\rangle$, we would have $\alpha_d = \langle d \,|\, u\rangle \neq 0$, which is a contradiction to experimental results. If $\alpha_d \neq 0$, then $\alpha_d^* \alpha_d > 0$; we started with a state prepared as $\sigma_z = +1$ and ended with a nonzero probability of measuring $\sigma_z = -1$: this is absurd.

Orthogonal states are mutually exclusive

We can extend the reasoning to a general and key concept of quantum mechanics: two orthogonal states are distinct and mutually exclusive. If the system is in the first state, the probability of finding it in the second is zero.

Now we are ready to express spin states as linear combinations of the basis vectors $\mathscr{B} = \{\, |u\rangle, |d\rangle \,\}$. The representation of the basis vectors themselves is naturally easy:

$$|u\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \tag{1.4}$$

$$|d\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{1.5}$$

To construct vector *right*, let's consider a spin prepared in the state $|r\rangle$. If we measure σ_z, we have a 50% chance of obtaining $+1$ (and 50% for -1); this means that for $|r\rangle$ we have $\alpha_u^* \alpha_u = \alpha_d^* \alpha_d = 1/2$. A vector that satisfies this constraint is:

$$|r\rangle = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}. \tag{1.6}$$

The reasoning is the same for state *left*; we also add the constraint that a state *left* cannot be *right* and vice versa: $\langle r \,|\, l\rangle = \langle l \,|\, r\rangle = 0$. We can express *left* as:

$$|l\rangle = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix}. \tag{1.7}$$

Lastly, the constraints to find explicit forms for *in* and *out* are:

- states must be orthogonal: $\langle i \,|\, o\rangle = \langle o \,|\, i\rangle = 0$;
- if we have a spin prepared as *in* or *out*:
  - equiprobability of measuring $\sigma_z = +1$ and $\sigma_z = -1$;
  - equiprobability of measuring $\sigma_x = +1$ and $\sigma_x = -1$.

Two vectors that satisfy these constraints are:

$$|i\rangle = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} \end{pmatrix} \tag{1.8}$$

$$|o\rangle = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{i}{\sqrt{2}} \end{pmatrix}. \tag{1.9}$$

This last derivation shows why it is important that the state space is complex: if we only accepted real components for our vectors, the system of equations we have implicitly defined would not have any solution[7].

1.3 OBSERVABLES

We have learned that in classical mechanics we can trust our intuition, and we can do one or more measurements to know exactly the state of a system: a measurement does not perturb the state, which is the same before, during, and after the measurement.

In quantum mechanics the situation is more complex; our intuition is misleading, and we need mathematical tools to describe what we can measure: the observables. These tools are mathematical operators called *machines* (**M**) and have as both input and output state vectors.

Axiom 2. *Machines associated with observables are described by linear operators.*

We will show that machines are Hermitian operators, so let's start defining these operators and describing their properties[8].

1.3.1 Hermitian operator

Formally, machines modify a state vector in this way:

$$\mathbf{M}|A\rangle = |B\rangle$$

The linearity of machines implies that:

$$\mathbf{M}|A\rangle = |B\rangle \Rightarrow \mathbf{M}z|A\rangle = z|B\rangle$$

and:

$$\mathbf{M}(|A\rangle + |B\rangle) = \mathbf{M}|A\rangle + \mathbf{M}|B\rangle.$$

If we choose a basis to represent machines and state vectors, we can write explicitly the linear operator as an $N \times N$ matrix, where N is the dimension of the vector space of the state vectors. A generic machine that transforms spins can be expressed as:

$$\mathbf{M} = \begin{pmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{pmatrix}.$$

7 To avoid confusion, we point out that $|i\rangle$ is the ket of state *in*. i, instead, is the imaginary unit.
8 The reason why we need this kind of operator will be clear in Section 1.3.2.

When we fix a basis, we are forced to express all state vectors and operators in that basis, but now we have a set of rules to define the application of the operator to a state vector, i.e. the matrix multiplication:

$$\mathbf{M}\,|A\rangle = \begin{pmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{pmatrix} \times \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} = \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} = |B\rangle$$

Eigenvalues and eigenvectors

When we consider a linear operator, we can search for eigenvalues and eigenvectors (if they exist). Eigenvectors are vectors that don't change their direction when multiplied by the operator; their magnitude is scaled by a constant factor called the eigenvalue. Formally:

$$\mathbf{M}\,|\lambda\rangle = \lambda\,|\lambda\rangle$$

where $|\lambda\rangle$ is the eigenvector and λ the eigenvalue.

Considering the transformation between ket $|A\rangle$ and $|B\rangle$: $\mathbf{M}\,|A\rangle = |B\rangle$, taking into account the dual space of bras and searching for a machine that transforms the bra $\langle A|$ into $\langle B|$, we cannot simply use the matrix having as elements the complex conjugate of $\mathbf{M}$; the correct operator is the *Hermitian conjugate* of $\mathbf{M}$, which is the transpose of the matrix having as elements the complex conjugates of $\mathbf{M}$. We denote the Hermitian conjugate with the dagger †:

$$\mathbf{M}^\dagger = [\mathbf{M}^*]^T = [\mathbf{M}^T]^*.$$

We can now write:

$$\mathbf{M}\,|A\rangle = |B\rangle \Rightarrow \langle A|\,\mathbf{M}^\dagger = \langle B|\,.$$

An operator that is equal to its Hermitian conjugate is called a *Hermitian operator*. Formally, $\mathbf{M}$ is Hermitian if and only if

$$\mathbf{M} = \mathbf{M}^\dagger.$$

Hermitian operators have some important properties:

- all eigenvalues are real;
- eigenvectors form a *complete set*: all vectors obtained with the application of the operator can be expressed as a linear combination of eigenvectors;
- if λ_1 and λ_2 are different eigenvalues, the associated eigenvectors are orthogonal;
- if two eigenvalues are equal (*degeneracy*), it is always possible to find two associated eigenvectors that are orthogonal.

Fundamental theorem

The last three properties can be summed up in the following way:

Theorem 1. *The eigenvectors of a Hermitian operator form an orthonormal basis.*

1.3.2 Principles of quantum mechanics

Let's introduce the first four principles of quantum mechanics, the ones about observables[9].

Principles 1. *Observables in quantum mechanics are described by linear operators* **L**.

L must also be a Hermitian operator: we can consider this proposition an axiom itself or deduce it from the other principles.

Principles 2. *The results of a measurement can only be the eigenvalues associated with the observable operator.*

Calling λ_i a generic eigenvalue and $|\lambda_i\rangle$ the associated eigenvector, if the system is in the *eigenstate* $|\lambda_i\rangle$, the measurement always returns λ_i. Since all λ_i must be physical quantities they must be real, a peculiar property of Hermitian operators.

Principles 3. *Unambiguously distinguishable states are represented by orthogonal vectors.*

Distinguishable states can be separated without ambiguity by a measurement. For example, if we want to distinguish between $|u\rangle$ and $|d\rangle$, we measure σ_z: *up* and *down* are distinct. We cannot, instead, say if a certain system is in state *up* or *right*, because even if the system is in the state $|u\rangle$ we can still measure σ_x and find (with 50% chance) that the system is in state $|r\rangle$.

Overlap

The inner product is a measure of how much two states are indistinguishable; for that reason it is also called overlap. Two states are physically distinct if the overlap is zero.

$$\langle u \,|\, d\rangle = 0$$
$$\langle u \,|\, r\rangle \neq 0$$

Principles 4. *If the system is in state* $|A\rangle$ *and we measure the observable* **L**, *the probability of obtaining* λ_i *is:*

$$P(\lambda_i) = \langle A \,|\, \lambda_i\rangle \langle \lambda_i \,|\, A\rangle .$$

where λ_i is a generic eigenvalue of **L** and $\langle\lambda_i|$, $|\lambda_i\rangle$ are the bra and ket associated with that eigenvalue (eigenvector of λ_i).

1.3.3 Spin Operator

The principles tell us what properties a machine must have to represent an observable. Let's construct the spin operator σ.

Until now, we have measured spins with the apparatus $\mathcal{A}$, orienting $\mathcal{A}$ along the component of our interest. σ is a mathematical tool that allows us to make predictions about the result of a measurement with $\mathcal{A}$ (fourth principle); as we can rotate $\mathcal{A}$, we must also rotate σ (mathematically). For this spatial property, σ is called a *3-vector operator*.

9 The fifth, and last one, concerns the temporal evolution. It will be discussed later on (Section 1.4).

OPERATOR σ_z: Let's start with the simplest operator[10]. The second principle says that all eigenvectors of σ_z are $|u\rangle$ and $|d\rangle$, with associated eigenvalues $+1$ and -1. We can write this assertion as equations:

$$\sigma_z |u\rangle = |u\rangle$$
$$\sigma_z |d\rangle = -|d\rangle .$$

In matrix form:

$$\begin{pmatrix} (\sigma_z)_{11} & (\sigma_z)_{12} \\ (\sigma_z)_{21} & (\sigma_z)_{22} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$
$$\begin{pmatrix} (\sigma_z)_{11} & (\sigma_z)_{12} \\ (\sigma_z)_{21} & (\sigma_z)_{22} \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = - \begin{pmatrix} 0 \\ 1 \end{pmatrix} .$$

The solution of this system is[11]:

$$\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} .$$

OPERATOR σ_x: With the same reasoning, we can construct the operator along the x axis. We have already deduced the representations of *right* and *left* in Equations 1.6 and 1.7. The equations that allow us to construct σ_x are:

$$\begin{pmatrix} (\sigma_x)_{11} & (\sigma_x)_{12} \\ (\sigma_x)_{21} & (\sigma_x)_{22} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$
$$\begin{pmatrix} (\sigma_x)_{11} & (\sigma_x)_{12} \\ (\sigma_x)_{21} & (\sigma_x)_{22} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{-1}{\sqrt{2}} \end{pmatrix} = - \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{-1}{\sqrt{2}} \end{pmatrix} .$$

The solution of this system is:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} .$$

OPERATOR σ_y: The last direction is along the y axis. Considering the expressions for *in* and *out* given in Equations 1.8 and 1.9, and following the second principle, we can write:

$$\sigma_y |i\rangle = |i\rangle$$
$$\sigma_y |o\rangle = -|o\rangle .$$

We can rewrite this in matrix form, and the solution we would obtain is:

$$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} .$$

Pauli matrices

We have obtained a matrix representation of the three spin operators σ_z, σ_x, and σ_y:

$$\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \qquad \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \qquad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} . \tag{1.10}$$

These famous and important matrices are named after their inventor, Wolfgang Ernst Pauli.

10 This is because we have chosen $\mathscr{B} = \{ |u\rangle , |d\rangle \}$ as the basis.
11 It is easy to verify that this operator is also linear.

1.3.4 Theory and experiments

Thanks to the operators σ_z, σ_x, and σ_y, if we know the state vector, we can statistically predict the result of a measurement of the spin along one of the three coordinate axes. What can we say about a measurement taken by orienting the apparatus $\mathcal{A}$ along a generic direction?

Considering $\mathcal{A}$ oriented along the unit vector $\hat{n}$, if σ behaves as a 3-vector, in order to obtain σ_n we only need the inner product:

$$\sigma_n = \vec{\sigma} \cdot \hat{n}$$

Expanding the components:

$$\sigma_n = \sigma_x n_x + \sigma_y n_y + \sigma_z n_z.$$

If we choose the basis $\mathscr{B} = \{ |u\rangle, |d\rangle \}$, we can use the Pauli matrices to express in matrix form the expression for σ_n:

$$\sigma_n = n_x \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + n_y \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} + n_z \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} n_z & n_x - i n_y \\ n_x + i n_y & -n_z \end{pmatrix}.$$

Given a direction (expressed by the unit vector $\hat{n}$), we can construct the matrix we have now made explicit, and then, after finding eigenvalues and eigenvectors, we can know all possible results of a measurement and obtain the probability associated with each result. For example, considering a direction in the x–z plane, the operator σ_n would be:

$$\sigma_n = \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}$$

where θ is the angle between $\hat{n}$ and z. For this matrix, the eigenvalues and eigenvectors are:

$$\lambda_1 = 1 \qquad |\lambda_1\rangle = \begin{pmatrix} \cos\frac{\theta}{2} \\ \sin\frac{\theta}{2} \end{pmatrix}$$

and

$$\lambda_2 = -1 \qquad |\lambda_2\rangle = \begin{pmatrix} -\sin\frac{\theta}{2} \\ \cos\frac{\theta}{2} \end{pmatrix}.$$

It should be pointed out that the theory is in agreement with experimental results[12]. Eigenvalues are $+1$ and -1, exactly the only results that the apparatus $\mathcal{A}$ can retrieve. The probability of obtaining a certain result can be evaluated as:

$$P(+1) = |\langle u | \lambda_1 \rangle|^2 = \cos^2\frac{\theta}{2}$$
$$P(-1) = |\langle u | \lambda_2 \rangle|^2 = \sin^2\frac{\theta}{2}$$

Expectation value

Lastly, let's calculate the average value for the measurement σ_n. From the first experiment we have seen in Section 1.1.1, we already know that the result of repeated measurements with $\mathcal{A}$ is $\cos\theta$. Let's verify if our model is coherent with the world.

12 If not, we must abandon this model and build another one.

Expected values are obtained as:

$$\langle L \rangle = \sum_i \lambda_i P(\lambda_i)$$

Specifically:

$$\langle \sigma_n \rangle = (+1)\cos^2\frac{\theta}{2} + (-1)\sin^2\frac{\theta}{2} = \cos\theta.$$

This is in complete agreement with the experimental results.

Before going on, we present, without proof, a useful theorem about expectation values:

Theorem 2. *To know the expectation value of an observable, we can simply place the operator associated with the observable between the bra and ket of the state vector:*

$$\langle \mathbf{L} \rangle = \langle A \,|\, \mathbf{L} \,|\, A \rangle \tag{1.11}$$

where $\mathbf{L}$ is an observable, $|A\rangle$ is a state vector, and $\langle A|$ is the corresponding bra.

1.3.5 Operator and Measure

Operators allow us to know the probability of measuring a certain spin given the direction of the measurement and the state vector. This probability is expressed by the state vector that we obtain when we apply the operator σ to the initial state.

It is important not to confuse the measurement act with the application of a machine that represents the observables. The spin state after the measurement is not the same as the one we obtain after the application of the operator. The operator is only an abstract mathematical construct that allows us to make statistical predictions about results, but doesn't have physical implications.

Let's consider an example to clarify the previous assertion. Having a spin prepared in the *up* state, its state vector is $|u\rangle = \binom{1}{0}$. If we apply the operator σ_z, we would obtain again $\binom{1}{0}$, and if we measure the spin with $\mathcal{A}$ oriented along z, it will always display $+1$, and we conclude that the state after the measurement is $|u\rangle$.

Consider now a spin prepared *right*, i.e. $|r\rangle = 1/\sqrt{2}\,|u\rangle + 1/\sqrt{2}\,|d\rangle$. Applying again the operator σ_z, the new state vector is $1/\sqrt{2}\,|u\rangle - 1/\sqrt{2}\,|d\rangle$. This vector tells us the probability of measuring $\sigma_z = +1$ (50%), but it is not the spin state after the measurement. Using the apparatus $\mathcal{A}$, we could measure:

- $+1$: the final state will be *up*;
- -1: the final state will be *down*.

No matter the result of the measurement, the final state will be different from the one we obtained by applying the operator.

1.4 TEMPORAL EVOLUTION

Let's explore the laws that describe the temporal evolution of a quantum system. In particular, we will see how the state vector can evolve over time.

1.4.1 Unitarity

In classical mechanics we are used to having a motion law that links different states of our system deterministically; this means being able to know precisely the following state given the previous one. A good law, however, doesn't allow us only to know the future, but also the past states that brought the system to the current state[13].

Reversibility

In other words, we want physical transformations to be reversible. This requirement is so important that we call this property the *minus first law*, because it underlies everything else. If we think about the system states as nodes in an oriented graph, reversibility imposes that each node has exactly one input edge and one output edge. This fundamental law is also true in quantum mechanics and is called *unitarity*[14], and it assures us that no information is lost. The unitarity law can be expressed as:

Axiom 3. *If two identical isolated systems are in different states, they stay in different states, and they were in different states in the past.*

1.4.2 Time-Development Operator

Considering a system in the state $|\Psi(t)\rangle$, where the t indicates that the state vector evolves over time, quantum motion equations allow us to obtain the state at time t given the initial state:

$$|\Psi(t)\rangle = \mathbf{U}(t)\,|\Psi(0)\rangle\,. \tag{1.12}$$

Determinism

Thanks to the operator $\mathbf{U}(t)$ we can know exactly the state vector $|\Psi(t)\rangle$ at time t, given $|\Psi(0)\rangle$. This assertion can be rephrased as:

Axiom 4. *The temporal evolution of the state vector is deterministic.*

Quantum mechanics is still non-deterministic, because knowing the state vector doesn't mean knowing the result of a measurement.

In order for $\mathbf{U}(t)$ to behave as we want, it has to:

- be a linear operator;
- respect reversibility.

The second constraint allows us to define the mathematical properties of $\mathbf{U}(t)$. Considering two initially different states $|\Psi(0)\rangle$ and $|\Phi(0)\rangle$, since there exists an experiment capable of certainly distinguishing the states, $|\Psi(0)\rangle$ and $|\Phi(0)\rangle$ must be orthogonal:

$$\langle\Psi(0)\,|\,\Phi(0)\rangle = 0.$$

Conservation of Distinctions

The minus first law assures that during the entire temporal evolution of the two systems, the state vectors $|\Psi(t)\rangle$ and $|\Phi(t)\rangle$ will continue to be distinguishable (orthogonal):

$$\langle\Psi(t)\,|\,\Phi(t)\rangle = 0 \qquad\qquad \forall t \geqslant 0.$$

13 For example, if we observe a ball in free fall touching the floor with a certain speed and at a certain time, we can know exactly when and from what height the ball started its fall.

14 We will see in the next paragraph the reason for this name

If we rewrite this equation using Formula 1.12, we obtain:

$$\langle\Psi(0)|\,\mathbf{U}^{\dagger}(t)\mathbf{U}(t)\,|\Phi(0)\rangle = 0.$$

From this we can see that $\mathbf{U}^{\dagger}(t)\mathbf{U}(t)$ must behave as the identity operator, that is:

$$\mathbf{U}^{\dagger}(t)\mathbf{U}(t) = \mathbf{I}. \tag{1.13}$$

An operator that behaves as $\mathbf{U}$ is *unitary*.

Principles 5. *The temporal evolution of state vectors is unitary.*

From the unitarity of $\mathbf{U}$ descends the *conservation of overlaps*: the overlap between two states (their inner product), subjected to the same temporal-development operator, is preserved over time.

1.4.3 The Hamiltonian

Often, in classical physics, a motion law is the result of a differential equation where we have exchanged a finite time interval with an infinite number of infinitesimal intervals.

Continuity

In quantum mechanics we can follow the same path and consider time intervals ϵ close to zero. In this scenario, after an ϵ amount of time, the state vector will change slightly and "smoothly", and the operator $\mathbf{U}(\epsilon)$ will be very similar to the identity. We can rewrite $\mathbf{U}(\epsilon)$ in order to highlight the difference with the identity $\mathbf{I}$ as:

$$\mathbf{U}(\epsilon) = \mathbf{I} - i\epsilon\mathbf{H}. \tag{1.14}$$

For now, i is a mere scale factor that later will help us recognize in $\mathbf{H}$ the quantum version of the classical Hamiltonian.

We can now express the infinitesimal evolution of a quantum system by combining Equations 1.12 and 1.14:

$$|\Psi(\epsilon)\rangle = |\Psi(0)\rangle - i\epsilon\mathbf{H}\,|\Psi(0)\rangle\,.$$

Bringing to the left the time interval:

$$\frac{|\Psi(\epsilon)\rangle - |\Psi(0)\rangle}{\epsilon} = -i\mathbf{H}\,|\Psi(0)\rangle\,.$$

Now considering the limit for $\epsilon \to 0$, we can see in the left member the time derivative of the state vector:

$$\frac{\partial\,|\Psi(t)\rangle}{\partial t} = -i\mathbf{H}\,|\Psi(0)\rangle\,.$$

Before using $\mathbf{H}$ as the quantum Hamiltonian, we have to verify the dimensional correctness. As in classical mechanics, the Hamiltonian is the mathematical construct that represents the energy. In our formula, however, ignoring the state vector, we have the inverse of time on the left and the energy on the right. To resolve this problem, let's introduce an important physical constant: the reduced Planck constant, $\hbar$.

The equation becomes:

$$\hbar \frac{\partial |\Psi\rangle}{\partial t} = -i\mathbf{H}|\Psi\rangle \qquad \text{or} \qquad \frac{\partial |\Psi\rangle}{\partial t} = \frac{-i\mathbf{H}|\Psi\rangle}{\hbar}. \tag{1.15}$$

Time-dependent Schrödinger equation

The constant $\hbar$ has units of $kg \cdot m^2/s$ and resolves the incompatibility between the two members. This equation is fundamental and is called the *generalized Schrödinger equation*, or time-dependent Schrödinger equation. If we know the Hamiltonian of an undisturbed system, we can know the evolution of the state vector.

If $\mathbf{H}$ represents the energy of the system, we should be able to measure it, so $\mathbf{H}$ has to be an observable. If $\mathbf{H}$ is an observable, it must be a Hermitian operator; let's verify it. Starting from Equation 1.13 and substituting $\mathbf{U}$ with Expression 1.14, we obtain:

$$(\mathbf{I} + i\epsilon\mathbf{H}^\dagger)(\mathbf{I} - i\epsilon\mathbf{H}) = \mathbf{I}.$$

Expanding to first order in ϵ, we find:

$$\mathbf{H}^\dagger - \mathbf{H} = 0 \Rightarrow \mathbf{H}^\dagger = \mathbf{H}.$$

We have concluded that $\mathbf{H}$ is an Hermitian operator that represents an observable: the energy of the system. Eigenvalues of $\mathbf{H}$ are the results of all possible direct measurements of the energy of the system.

Quantum Hamiltonian

1.4.4 Commutators

In a system that evolves with time, we expect that the expectation values for a certain observable $\mathbf{L}$ will also change. Thanks to equation 1.11 on page 16, we can write explicitly the time dependence of expectation values:

$$\langle \mathbf{L} \rangle = \langle \Psi(t) | \mathbf{L} | \Psi(t) \rangle .$$

The time derivative[15] is:

$$\frac{d}{dt} \langle \Psi(t) | \mathbf{L} | \Psi(t) \rangle = \left\langle \dot{\Psi}(t) \middle| \mathbf{L} \middle| \Psi(t) \right\rangle + \left\langle \Psi(t) \middle| \mathbf{L} \middle| \dot{\Psi}(t) \right\rangle .$$

Substituting bra and ket with the time-dependent Schrödinger Equation 1.15 (namely $\left|\dot{\Psi}(t)\right\rangle = \frac{-i}{\hbar}\mathbf{H}|\Psi(t)\rangle$), we obtain:

$$\frac{d}{dt} \langle \Psi(t) | \mathbf{L} | \Psi(t) \rangle = \frac{i}{\hbar} \langle \Psi(t) | \mathbf{HL} | \Psi(t) \rangle - \frac{i}{\hbar} \langle \Psi(t) | \mathbf{LH} | \Psi(t) \rangle .$$

That can be rewritten as:

$$\frac{d}{dt} \langle \Psi(t) | \mathbf{L} | \Psi(t) \rangle = \frac{i}{\hbar} \langle \Psi(t) | [\mathbf{H}, \mathbf{L}] | \Psi(t) \rangle .$$

The quantity $\mathbf{HL} - \mathbf{LH}$ is called the *commutator*, and since, in general, the product between operators (matrices) is not commutative, the commutator is not zero (when it is zero, we say that $\mathbf{H}$ and $\mathbf{L}$ commute). Commutators

15 Derivative of a product: $\mathbf{L}$ doesn't depend on time and the dot denotes the time derivative (Newton notation).

are important in physic, and the commutator between two operators, in this case $\mathbf{H}$ and $\mathbf{L}$, is denoted by:

$$\mathbf{HL} - \mathbf{LH} = [\mathbf{H}, \mathbf{L}].$$

With the commutator we can express concisely the derivative of the expectation value for the observable $\mathbf{L}$:

$$\frac{d}{dt} \langle \mathbf{L} \rangle = \frac{i}{\hbar} \langle [\mathbf{H}, \mathbf{L}] \rangle \tag{1.16}$$

or equivalently:

$$\frac{d}{dt} \langle \mathbf{L} \rangle = -\frac{i}{\hbar} \langle [\mathbf{L}, \mathbf{H}] \rangle . \tag{1.17}$$

This equation links variations of the expectation values of an observable ($\mathbf{L}$) to the expectation values of another physical observable ($-\frac{i}{\hbar}[\mathbf{L}, \mathbf{H}]$)[16].

1.4.5 Conservation of Energy

In quantum mechanics, when we say that a quantity is conserved, we mean that the expectation value of that quantity doesn't change. If we look at Equation 1.17, the condition for the expectation value not to change is that the commutator between this quantity and the Hamiltonian is zero. It is possible to demonstrate that:

Theorem 3. *Having an observable $\mathbf{Q}$, if $[\mathbf{Q}, \mathbf{H}] = 0$, then every power satisfies $[\mathbf{Q}^n, \mathbf{H}] = 0$. This means that the expectation value $\langle Q \rangle$ is conserved, and any power of the expectation value $\langle Q^n \rangle$ does not change with time.*

The most obvious quantity that is conserved is the Hamiltonian $\mathbf{H}$ and, since every operator commutes with itself, we always have:

$$[\mathbf{H}, \mathbf{H}] = 0.$$

We can conclude that, under very general conditions, energy is conserved in quantum mechanics.

1.5 CONCLUSIONS

We conclude this chapter with a recap of what we have discovered in these pages, trying to put everything together to answer the question that opened this chapter: what are the physical limits of quantum computing, and why must our algorithms be reversible?

We started the chapter with an experiment that shows that quantum mechanics is not deterministic. We can, however, make some predictions if we consider the expectation value of a measurement instead of a single result.

We have built state vectors and understood their mathematical meaning, focusing on the fact that knowing the state vector doesn't allow us to know the result of a measurement. We have defined the inner product between

16 It is possible to demonstrate that if $\mathbf{L}$ and $\mathbf{H}$ are Hermitian, then $[\mathbf{L}, \mathbf{H}]$ is also Hermitian.

state vectors, observed that it is a measure of the overlap between states, and concluded that two distinguishable states must be orthogonal.

We have linked a state vector to the result of a measurement –to be precise, to the average of the results of multiple measurements– with machines, Hermitian operators that represent observables. We have built the spin operator and used it to predict the result of a simple experiment, showing how the theory we have built so far is in accordance with experimental results.

Our introduction continues with the analysis of the temporal evolution of a quantum system. We have described the evolution of a state vector with an unitary operator; the application of this operator to a state vector produces the new state in which the system will be. We understood that the temporal evolution of the state vector is deterministic and that indeterminacy is caused only by the act of measuring.

Considering infinitesimal time intervals has allowed us to deduce the time-dependent Schrödinger equation and, thanks to this equation, we have shown how to describe the temporal evolution of expectation values for a certain observable. During this analysis, we also introduced the Hamiltonian of the system, a Hermitian operator that describes the energy of the system.

The discussion ends with a comforting result: as in classical physics, the energy of a closed system is conserved. We have obtained this result by presenting the commutator and linking the temporal evolution of an observable with the commutator between the observable and the Hamiltonian (energy) of the system. The commutator of the Hamiltonian with itself is trivially zero, so the expectation value for the energy doesn't change.

All the information that we have learned allows us to understand the constraint of writing only reversible algorithms for quantum-gate-based quantum computers. Quantum gates operate on qubits through physical transformations[17]. These transformations, like all transformations in quantum mechanics, are described by unitary operators that are intrinsically reversible. This means that all quantum gates are reversible.

In other words, we can build only quantum gates that, having as input different (distinguishable) states, return orthogonal states; also, due to the conservation of overlaps, the inner product between input states is conserved during the quantum gate transformation.

Reversibility doesn't mean that we can go forward and backward in time as we please, but that all quantum gates express injective functions: if we know the output, we can know the input, or in more physical terms, if we know the final state of qubits[18] and the transformations applied to this system (i.e., those implemented by the quantum gates), we can determine the initial state.

Since every quantum algorithm has to be implemented as a path through quantum gates, and every quantum gate is reversible, the algorithms as a whole must also be reversible.

17 How depend strongly on the particular physical implementation.

18 This is a complex system (composed of more than one qubit); to fully understand these systems, we should take into account entanglement. Since our discussion is already quite long, and the temporal evolution of an entangled system is still unitary (reversible), we exclude entanglement from our introduction.

2 COMPUTATIONAL COMPLEXITY AND QUANTUM COMPUTING

In this chapter we introduce the basic concepts of quantum computing. We examine the reasons that push us to search for new computational paradigms and why quantum computing appears to be promising.

In order to fully understand the expectations placed on quantum computing, it is first necessary to introduce some elements of computability and complexity. We will, therefore, examine the foundations of computer science starting from computational models. We will give a definition of a computational problem, and see which problems we can solve with a computer. Finally, we will study the computational complexity of problems by classifying them according to the resources required to find a solution.

Having completed this introduction to computer science, we will see how quantum computers fit into this analysis. We will begin by presenting a computational model that allows us to define which problems are solvable on a quantum computer, and we will note how this naturally leads us toward quantum-based quantum computers. We will see which complexity classes can be defined considering this new paradigm, and we will try to understand what relationships exist between the classes of problems identified with the classical model of computation and those derived from the quantum model.

In the second section of this chapter we will present in a more rigorous way quantum gate-based computers. We will see what is meant by a quantum gate, present the most common gates, and construct a universal set of gates. We will then see what it means to program a quantum gate-based computer by presenting the Deutsch algorithm.

The chapter concludes by presenting *Quantum Adiabatic Computing* (QAC): an alternative paradigm to computation based on quantum gates. We will describe the theoretical foundations of this paradigm and show that QAC and quantum gate-based computing are equivalent. We will see that programming a machine that exploits the QAC paradigm means translating the problem we want to solve into a minimization. We present another approach natively implementable with QAC: *quantum annealing* QA. We will describe QA by drawing a parallel between classical simulated annealing and quantum annealing. Finally, we will focus on the formulation of minimization problems naturally solvable with quantum annealing.

This chapter is inspired by the following books and papers: Section 2.1.1 is inspired by [3, chapter 3 and 4], Section 2.2 is taken from [3, chapter 2, 3, 4 and 7], finally Section 2.3 is inspired by [4, chaper 2 and 3] and [5]. In the following pages these references are no longer considered and, when needed, we cite other works.

2.1 INTRODUCTION TO COMPUTER SCIENCE

Computer science deals with studying which problems can be solved through a computer and which resources are necessary to find the solution. We can divide these two objectives into two parallel and complementary lines of research. On one hand, to prove that a problem is solvable, it is sufficient to show an algorithm capable of solving that problem, without worrying too much about its complexity. On the other hand, to understand which resources are necessary, we would like to find a lower bound on the operations that must be performed to obtain the solution.

These lines of research are complementary in the sense that, ideally, once a lower bound has been identified, we would like to be able to construct an algorithm that matches it. In practice, this is not always possible, because constructing algorithms that behave well on all instances of a problem is difficult, and because the lower bound of the required operations is not always known, so one must settle for the best algorithm developed so far.

We have referred to an algorithm intuitively knowing that it is a sequence of operations that allows us to find the answer to a problem. To formally study the characteristics of an algorithm, this intuition is not sufficient; we need something that we can characterize more precisely. We, therefore, introduce computational models.

2.1.1 Computational models

A computational model allows us to mathematically define the concept of an algorithm. These abstractions were first introduced by Turing and Church to find a solution to Hilbert's *entscheidungsproblem*[1]. We now consider two equivalent computational models: the *Turing machine* and the *circuit model*.

Turing machine

This is the most well-known computational model. Introduced by Turing to formally describe the notion of an algorithm that performs a computational task.

Turing machine definition

A Turing machine can be seen as the conceptualization of a computer. It is composed of four main elements:

PROGRAM: which allows the encoding of an algorithm;

FINITE STATE CONTROL: unit that acts as the processor;

TAPE: memory of the computer;

READ-WRITE TAPE-HEAD: interacts with the memory.

It is possible to define various versions of the Turing machine. For our discussion we can consider the simplest version, in which the memory tape is unbounded to the right and bounded to the left. The symbols that can be written on the tape are 0, 1, b (the blank symbol) and the special symbol ▷ that marks the left end of the tape. These symbols form our alphabet

1 For every mathematical problem, does there exist an algorithm capable of solving it?

Γ: everything that can be represented on the tape consists of finite-length strings Γ^* made up of symbols from the alphabet (except $\triangleright$, which appears only at the beginning of the tape). The finite state control consists of a finite set of states $q_1, \ldots, q_m$ plus two special states q_s and q_h that represent, respectively, the initial state of the machine and the final (halt) state; the latter is reached when the machine has finished the computation.

A program for the Turing machine consists of program lines; each line is a quintuple $\langle q, x, q', x', s \rangle$ which is interpreted as follows: if the machine is in state q and reads the symbol $x \in \Gamma$ on the tape, then it transitions to state q', writes the symbol $x' \in \Gamma$ on the tape, and moves as indicated by s (to the right $+1$, to the left -1, or remains stationary 0).

The Turing machine model may appear simple, but it is the formalism of reference to tell what we consider computable or not. The importance of the Turing machine is supported by the *Church–Turing thesis* proposed independently by Church and Turing:

Church–Turing thesis

Thesis 1. *The class of functions computable by a Turing machine corresponds exactly to the class of functions which we would naturally regard as being computable by an algorithm.*

Church–Turing thesis states that the Turing machine is a computational model capable of providing a rigorous definition of the intuitive concept of an algorithm, and is therefore suitable as a foundation for computer science.

The Turing machine we have described can be modified, for example, by imagining having multiple memory tapes; this new machine could be able to solve problems more quickly than the original, but we will always be able to simulate the multi-tape machine with a single-tape machine. This means that considering multiple tapes does not increase the power of the Turing machine.

Another modification, which will be useful later, that we can make to this computational model is to consider a probabilistic version of the Turing machine: if the machine is in state q and the symbol x is read, a coin is flipped; if the result is heads, the machine transitions to state q_H, otherwise to state q_T. This modification can also be simulated with a Turing machine that explores all possible computation paths one at a time (using exponentially more steps), so once again the set of computable functions is not extended.

UNDECIDABILITY: Computational models, and with them computer science, were developed to answer Hilbert's entscheidungsproblem. Church, with the λ-calculus[2], and Turing, with the Turing machine, showed that the answer is no: there cannot exist an algorithm to decide all the problems of mathematics.

This result clearly divides mathematical problems into two classes: those that can be solved with an algorithm and those so difficult as to be computationally unsolvable. We call the problems belonging to this class *undecidable*. Turing shows that the *halting problem* (the problem of determining whether a Turing machine halts given an input) is, in general, undecidable[3]. In ad-

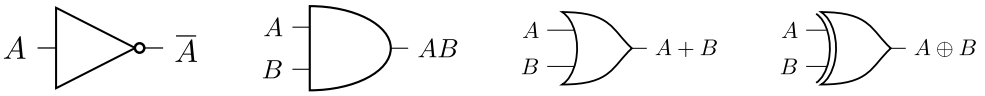
2 Another computational model equivalent to the Turing machine.

3 Church shows that there is no computable function that allows one to determine whether two λ-calculus expressions are equivalent.

dition to the problems used in the proofs of Church and Turing, there exist many other undecidable problems, for example finding a homomorphism between two topological spaces or some problems related to the behavior of dynamical systems.

Circuit model

The circuit model is an alternative model of computation in which we describe an algorithm by implementing it through a circuit made of logic gates. This approach is interesting because it brings us closer to the first quantum model of computation: the gate-based model.



(a) NOT gate **(b)** AND gate **(c)** OR gate **(d)** XOR gate

Figure 2.1: Logic gate

In Figure 2.1 the most well-known logic gates are shown, with which we can build a digital circuit. To these we normally add the possibility of obtaining a copy of a bit (a wire that splits), an operation we call FANOUT, and the possibility of swapping two bits, called CROSSOVER.

It is possible to show that with this limited set of gates one can compute any function $f : \{0,1\}^n \to \{0,1\}^m$. This is not sufficient to describe a useful computational model. Before seeing which characteristics we require, we define the concept of a circuit family.

A *family of circuits* is a set of circuits $\{C_n\}$ indexed by $n \in \mathbb{N}$. The circuit C_n has n input bits and a finite number of extra work bits and output bits. Given a circuit C_n and a bit string x of maximum length n as input of the circuit, we denote by $C_n(x)$ the output of the circuit.

We say that a family of circuits is *consistent* if all circuits in the family compute the same function $C(\cdot)$. Formally, the definition of consistency is: the family of circuits $\{C_n\}$ is consistent if for $m < n$ and x of length at most m bits, then $C_m(x) = C_n(x)$.

Finally, we define a family of circuits to be *uniform* if there exists an algorithm for a Turing machine that, given n as input, is able to generate a description of C_n. That is, this algorithm must provide the list of gates that constitute C_n, and the description of how these gates are connected to each other.

Circuit model definition

We are finally ready to describe the formal characteristics of the circuit model of computation. In this model we consider functions computable by families of circuits $\{C_n\}$ such that:

- the circuits in the family are *acyclic*: no loops must be present in the circuit, this prevents the occurrence of possible instabilities;
- $\{C_n\}$ is a *consistent* family of circuits: all circuits compute the same function (on inputs of increasing size);
- $\{C_n\}$ is a *uniform* family of circuits: the circuits can be generated from a reasonable algorithm.

It is possible to show that the class of functions computable by a consistent, uniform, and acyclic family of circuits is exactly the same as that computable on a Turing machine.

2.1.2 Computational problems

The theoretical models that we have defined in the previous section (and other equivalent ones that have been proposed in the course of studying computer science) allow us to formally study computational problems. This analysis depends on the answer to three fundamental questions:

WHAT IS A COMPUTATIONAL PROBLEM: computational problems are a specific class of problems called *decision problems*. This choice allows us to isolate a restricted class to simplify the analysis and make progress in the development of the theory. We will see that the results obtained extend well beyond decision problems;

HOW TO DEVELOP AN ALGORITHM TO SOLVE IT: after the problem identification, it is necessary to develop an algorithm capable of finding a solution. We are then interested in determining whether there are general algorithms capable of solving large classes of problems. Finally, we would like to find tools to guarantee that an algorithm actually works as expected;

WHAT RESOURCES ARE NECESSARY FOR THE SOLUTION: an algorithm consumes computational resources such as time, space, and energy; we want to know whether the developed algorithm consumes more resources than actually necessary. Moreover, we would like to classify problems according to the resources required to solve them.

Computational complexity

Asymptotic notation

The study of computational complexity aims to quantify the computational resources required to solve a given problem. To study resource consumption independently of the characteristics of the reference architecture[4], *asymptotic notation* is introduced, which captures the essential behavior of a function:

- O ("big O"): used to define the *upper bound* of a function;
- Ω ("big Omega"): used to define the *lower bound* of a function;
- Θ ("big Theta"): is the combination of O and Ω, describing the asymptotic behavior of a function.

Computational complexity considers the spatial and temporal resources required to solve a problem and seeks to demonstrate the existence of a lower bound for these resources whether or not an optimal algorithm exists that meets this lower bound. Clearly, the development of algorithms should

4 We mean independence from specific architectural decisions that, for example, allow a processor to save a few clock cycles whenever a certain operation is performed. We will soon see that different computational models can, instead, have asymptotically different resource consumption.

aim to find an algorithm that does not use more resources than necessary, that is, the theoretical lower bound identified by the study of computational complexity.

In developing the theory of computational complexity, it has been observed that resource consumption depends on the computational model of reference; for example, the three different Turing machines presented in the previous section can solve the same problem with a very different number of operations.

Easy and hard problems

This problem has been clearly addressed by dividing problems into two classes:

- *easy tractable* or *feasible* problems: are problems for which there exists an algorithm capable of finding a solution in *polynomial* time with respect to the size of the problem. An algorithm whose resource consumption grows as $O(p(n))$, where n is the size of the problem and $p(\cdot)$ is a polynomial, is called *efficient*;
- *hard*, *intractable*, or *infeasible* problems: if the best algorithm requires exponential resources[5]. In this case we speak of *inefficient* algorithms.

There are several reasons why the choice of dividing problems into two classes of computational complexity (polynomial and exponential) is appropriate. The first is historical: the polynomials describing most proposed algorithms do not have high degree, which means polynomial algorithms almost always outperform exponential algorithms. The second reason is more foundational and is due to the *strong Church–Turing thesis*:

Strong Church–Turing thesis

Thesis 2. *Any model of computation can be simulated on a probabilistic Turing machine with at most a polynomial increase in the number of elementary operations required.*

This means that probabilistic Turing machines appear to be the strongest "reasonable" model of computation. Formally, we can state that if, considering a certain computational model (different from the probabilistic Turing machine), it is possible to solve a problem with k elementary operations, then it is possible to solve the same problem on a probabilistic Turing machine with at most $p(k)$ elementary operations (where $p(\cdot)$ is a polynomial).

Reversing this description, the strong Church–Turing thesis states that, given a certain problem to be solved, if there is no efficient algorithm for the probabilistic Turing machine, then an efficient algorithm does not exist for any computational device.

Classes of problems

For now we have divided problems into polynomial and exponential, let us now consider finer divisions of computational problems, taking into account not only the time required but also the space needed to find the solution. Before dividing problems into classes, we formally define the decision problems introduced at the beginning of this section.

5 We consider exponential anything that grows faster than a polynomial; $n^{\log n}$, with some abuse of notation, is considered an exponential function.

A decision problem is a problem whose answer is yes or no. We can define them formally using the terminology of *formal languages*. A language L over the alphabet Γ is a subset of the set Γ^* of all (finite) strings of symbols from Γ. A string represents the input of our problem and can be written on the tape of a Turing machine. The machine performs a computation that may terminate by writing on the tape a string equivalent to yes or no depending on whether the input string belongs to the language L or not.

In general, a language L is *decided* by a Turing machine if the machine is able to decide whether an input x on its tape is a member of the language L or not.

P CLASS: We say that a problem can be solved in polynomial time if there exists a Turing machine capable of deciding whether the input x of size n belongs to the language in time $O(n^k)$ for some finite k; we say that this problem belongs to $\mathbf{TIME}(n^k)$. The set of all languages in $\mathbf{TIME}(n^k)$ is denoted by **P**. **P** is our first example of a *complexity class*.

To prove that a language L belongs to **P**, it is sufficient to show an algorithm that decides L in polynomial time. Unfortunately, there exist problems for which such an algorithm has not yet been found. One may then ask whether there exist problems outside **P**.

Conversely, although there exist non-constructive proofs showing that there must exist problems that require exponential resources, it seems very difficult to prove that a specific problem does not belong to **P**.

NP CLASS: There exist problems for which a polynomial algorithm has not been found, but if the solution to the problem is somehow provided, it is possible to apply an efficient algorithm to verify such a solution. Problems with this property belong to the class **NP**. Formally, the set **NP** contains the languages L such that there exists a Turing machine M such that:

- if $x \in L$, there exists a witness string w such that M halts answering "yes" after a time polynomial in the size of x when the machine starts from the state xbw (b is the blank space);
- if $x \notin L$ for any witness string w, M halts answering "no" after a time polynomial in the size of x when the machine starts from the state xbw.

$P \neq NP$ open question

The most famous open problem in computer science is whether or not there are problems in **NP** which are not in **P**, often abbreviated as the $\mathbf{P} \neq \mathbf{NP}$ problem.

Within the class of problems **NP** we find the **NP**-*complete* problems. A language L is **NP**-complete if any other language in **NP** can be *reduced* to L. A language B is said to be reducible to a language A if there exists a Turing machine that, in polynomial time, starting from x produces $R(x)$ and $x \in B$ if and only if $R(x) \in A$. We can view reduction as a translation of one problem into another equivalent one: finding the solution to one means solving the other as well. **NP**-*complete* problems represent the hardest problems in the class **NP**, in the sense that finding a lower bound for an **NP**-*complete* problem means finding a lower bound for all problems in **NP**.

An **NP**-*complete* problem is the following: given a circuit with n input bits and one output bit, determine whether there exists an assignment of the

input bits such that the output of the circuit is 1. This problem is called the *circuit satisfiability* problem or CSAT. The proof of **NP**-*completeness* of CSAT is due to the *Cook–Levin theorem*

Theorem 4. *CSAT is NP-*complete.

Once a language L that is **NP**-*complete* has been found, to prove that another language L′ is **NP**-*complete* it is sufficient to find a reduction from L to L′.

PSPACE CLASS: If we stop considering the time required to solve a problem and instead focus on space consumption, we can define the class of problems **PSPACE**. This class contains the languages L that can be recognized by a Turing machine using a polynomial number of working bits without time limitations. The classes of problems **P** and **NP** are subsets of **PSPACE**, but there are no proofs that these subsets are strict or not.

A CHAIN OF INCLUSIONS: We can organize the main classes of computational problems in the following chain of inclusions:

$$\mathbf{L} \subseteq \mathbf{P} \subseteq \mathbf{NP} \subseteq \mathbf{PSPACE} \subseteq \mathbf{EXP}. \tag{2.1}$$

Where **L** are the problems solvable in logarithmic space and **EXP** the problems solvable in exponential time.

Although most computer scientists believe that each of these inclusions is strict, there are no proofs of this. However, there exist results that allow us to state with certainty that there is not a single class of problems. The *time hierarchy theorem* and the *space hierarchy theorem* ensure respectively that $\mathbf{L} \subset \mathbf{PSPACE}$ and $\mathbf{P} \subset \mathbf{EXP}$. This means that in Equation 2.1 there is at least one strict inclusion, even if we do not know which one.

Approximate solutions

If we need to solve an **NP**-*complete* problem, today we can rely on many algorithms that perform well on most instances of these difficult problems. If we happen to be particularly unlucky and the instance we are trying to solve does not appear to have favorable properties for our algorithms, we can change approach and search for an *approximate solution*.

There exist **NP**-*complete* problems for which approximate algorithms are known that guarantee a certain maximum error with respect to the correct solution. There is an entire branch of computer science devoted to approximation algorithms. It is possible to define the class of languages **MAXSNP**, analogous to the class **NP**, for which it is possible to efficiently verify an approximate solution to the problem.

BPP CLASS: This is the last class of problems that we introduce considering classical computational models, and it will be useful in the next section where we will introduce quantum models of computation. To define this class of languages we change the computational model; we have already introduced the probabilistic Turing machine and have seen that, following the

strong Church–Turing thesis, it is assumed to be the most powerful classical model of computation.

With this model we can define the class **BPP**, that is *bounded-error probabilistic time*, which contains the languages L for which there exists a probabilistic Turing machine M such that if $x \in L$ then M accepts x with probability $\geqslant \frac{3}{4}$, and if $x \notin L$ then M rejects x with probability $\geqslant \frac{3}{4}$[6].

The *Chernoff bound* implies that with a limited number of repetitions of the algorithm, the probability of success is amplified to the point of being essentially one. For this reason, languages in **BPP**, even more than those in **P**, are considered those that can be solved efficiently on a classical computer.

2.1.3 Quantum computational complexity

We now extend the classification of computational problems by considering a new computational model: *quantum circuits*. In Section 2.2.2 we will precisely define what a *quantum gate* is and how they can be assembled to create quantum circuits. For now, it is sufficient to know that a quantum gate applies a unitary transformation (introduced in Section 1.4.2) to one or more quantum bits (*qubits*). The size of a quantum circuit is defined by the number of gates that are in the circuit and by the depth of the circuit[7]. The latter is the number of distinct timesteps at which gates are applied.

BPQ Class

Thanks to the concept of quantum circuits we can define the quantum alternative to the class **BBP**: class of problems **BQP**. This set contains the decision problems that can be solved with bounded probability of error using a polynomial size quantum circuit (both the number of gate and the depth). To be precise a circuit is a fixed entity that can found the solution of a problem up to a certain input dimension. To better formalize the **BQP** class we recall the concept of circuit family. Considering a language L and an input string x of size n we say that $L \in$ **BPQ** if:

Quantum circuit model definition

- the number of gates of the circuits in the family that solve the problem grows, at most, polynomially with respect to n;
- the family of circuits is *uniform*, that is, there exists a set of clear instructions, our algorithm, that generates the quantum circuit.

Relation with classical classes

One of the most significant results in quantum computational complexity is that **BQP** $\in$ **PSPACE**. It is clear that **BPP** $\in$ **BQP**. The question if **BPP** $\neq$ **BQP** remains open; in other words, a proof that a quantum computer is more powerful than a classical computer has not yet been found.

The inequality **BPP** $\neq$ **BQP** would violate the strong Church–Turing thesis and does not only affect the comparison between classical and quantum

6 The value $\frac{3}{4}$ is arbitrary; it is sufficient that the probability is greater than $\frac{1}{2}$.

7 In Section 2.2.3 we will use another way of measuring the complexity: the number of query to an oracle.

models of computation, but would also have repercussions on the hierarchy of classical inclusions. If we rewrite Equation 2.1 in light of **BQP** $\in$ **PSPACE** we obtain:

$$\mathbf{L} \subseteq \mathbf{P} \subseteq \mathbf{NP} \subseteq \mathbf{BQP} \subseteq \mathbf{PSPACE} \subseteq \mathbf{EXP}. \tag{2.2}$$

Assuming that **BPP** $\neq$ **BQP**, we could deduce that **BPP** $\neq$ **PSPACE**.

There are some hints that quantum computer might be more powerful than classic ones. The two most famous example of quantum superiority are the Grover's algorithm and the Shor's algorithm.

Grover's algorithm shows a quadratic speedup in finding an element from an unordered list respect to the lower bound of a classical algorithm. Grover's search algorithm is optimal: his complexity is the same of the theoretical lower bound in the quantum computational model.

The Shor's algorithm shows an exponential speedup in factorizing a number respect to the best know classical algorithm. The key to the efficient quantum mechanical solution of the factoring problem was the exploitation of a structure hidden within the problem.

Limits of quantum computation

Many computer scientists believe that the essential reason for the difficulty of **NP**-complete problems is that their search space has no structure, and that the best possible method for solving such problems (in the most general case) is to adopt a search method.

Taking this point of view means that quantum computer can not solve efficiently an **NP**-complete problem because the maximum speedup achievable is quadratic. Researchers in quantum computation and quantum information focus, then, on another class of problems to show the superiority of quantum computation over classical computation. This class of problems is called **NPI** (**NP**-*intermediate*) and contains problem in **NP** that are both outside **P** (or **BPP**) and **NP**-complete.

The factoring problem is believed to be in **NPI** and the Shor's algorithm would seem to demonstrate the computational superiority of the quantum model; however, unfortunately, there is no proof that **P** $\neq$ **NPI**. These considerations are deeper explained in [3, pages 150, 151, 270 and 271]

2.2 QUANTUM GATE BASED PARADIGM

The quantum gate-based computation paradigm is the direct implementation of the computational model that we have just described. This approach to quantum computation is founded on extensive theoretical work, part of which we have mentioned in the previous section, and for this reason most of the theoretical results on quantum computation derive from this paradigm.

As suggested by the name, this model is based on quantum gates; quantum gates manipulate *qubits* by applying physical transformations to them. The purpose of Chapter 1 was to convince us that these gates must have certain characteristics in order to comply with the laws of physics.

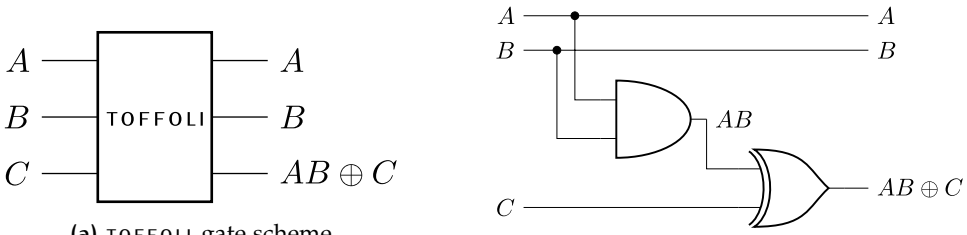
Quantum gates modify *qubits* by applying unitary operators. We focused on the physical meaning of a unitary operator in Section 1.4.2 where we saw how, knowing the state of the system after the application of the operator, we could infer the initial state.

We now focus on the implications that a unitary operator has on a computational model. The first consequence, the most obvious one, is that we are allowed to construct only reversible gates, that is, gates that allow us to determine the state of the input *qubits* if we know the state of the output *qubits*.

2.2.1 Classical reversible gates

Reversible gates do not exist only when dealing with quantum computers. There are numerous examples of classical reversible gates and there are studies that start from a completely reversible classical computation to drastically reduce the energy consumption of computers. It is also useful for us to examine some classical reversible gates because some considerations we can make about them will be useful in the study of quantum gates.

The simplest reversible gate is the NOT gate (Figure 2.1a). Another important classical reversible gate is the TOFFOLI gate shown in Figure 2.2[8].



(a) TOFFOLI gate scheme

(b) TOFFOLI gate Implementation

Figure 2.2: Circuit sampling

The TOFFOLI gate is also called *controlled controlled* NOT. From Figure 2.2 we see that the value of bit C is negated if both the control bit A and the control bit B are set.

If we observe the truth table of the TOFFOLI gate (Figure 2.3) we can notice how the inputs are uniquely identified by the outputs (definition of a reversible gate). The TOFFOLI gate has three input bits, therefore the truth table has $2^3 = 8$ rows; since the outputs must be unique keys there are no repetitions in the outputs. This means that all possible combinations of three bits also appear in the outputs. We can then deduce that the TOFFOLI gate acts by *permuting* the input rows of the truth table. This is true in general:

A	B	C	A	B	$AB \oplus C$
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	1	0
0	1	1	0	1	1
1	0	0	1	0	0
1	0	1	1	0	1
1	1	0	1	1	1
1	1	1	1	1	0

Figure 2.3: TOFFOLI gate truth table

Theorem 5. *The output multi-column of the truth table of a classical reversible gate is a permutation of the input multi-column.*

8 Figure 2.2b shows the implementation of the TOFFOLI gate using the gates shown in Figure 2.1.

This concept of permutation of the input will also reappear when we discuss quantum gates.

2.2.2 Quantum gate

In Chapter 1 we saw that spin is an example of a binary property, and we classified as *qubit* all those physical properties that, when measured, return only two values. We measured spin with respect to the coordinate axes and identified different spin states (*up* and *down*, *left* and *right*, *in* and *out*).

These states slightly change notation when we move from the study of physics to computer science. Before describing quantum gates we introduce this alternative notation for the states of a *qubit* and show a graphical representation that will be useful to understand the effect of quantum gates.

qubits

Spin and qubits

In computer science the states *up* and *down* (Equations 1.4 and 1.5) represent respectively a *qubit* off $|0\rangle$ and on $|1\rangle$. We denote with $|+\rangle$ and $|-\rangle$ the *qubits* in the states *right* and *left* (Equations 1.6 and 1.7); finally the states *in* and *out* (Equations 1.8 and 1.9) are represented by $|i\rangle$ and $|-i\rangle$. We can rewrite the cited equations considering the new notation:

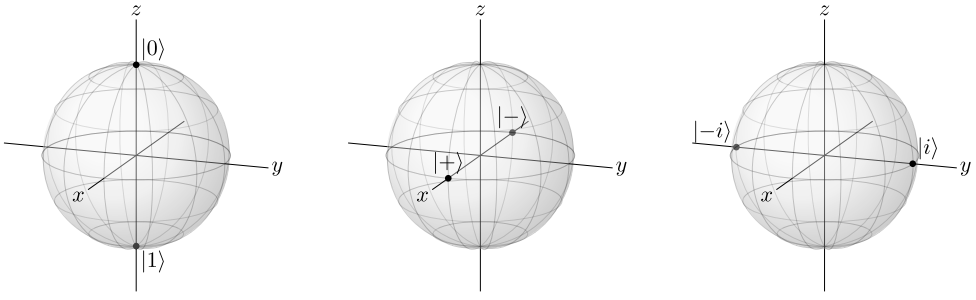
$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \qquad |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \tag{2.3}$$

and:

$$|i\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i\,|1\rangle) \qquad |-i\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i\,|1\rangle) \tag{2.4}$$

Bloch sphere

To represent *qubits* in a graphical way that respects the mathematical properties of quantum states, we can use a sphere called the *Bloch sphere*. The sphere has radius 1 and every point on the sphere represents a *qubit*[9]. Figure 2.4 shows, on the Bloch sphere, the *qubits* described above.



(a) States $|0\rangle$ and $|1\rangle$ **(b)** States $|+\rangle$ and $|-\rangle$ **(c)** States $|i\rangle$ and $|-i\rangle$

Figure 2.4: Bloch Sphere [6]

A state represented as a point on the surface of the Bloch sphere is called a *pure state*. It is also possible to consider the points inside the sphere, in which case one speaks of *mixed states*. To continue our discussion, we can consider only pure states.

9 Up to a global phase, which is however irrelevant.

Quantum gate

Quantum gates properties

Recalling the results achieved in Chapter 1 and also considering what we have said about classical reversible gates in Section 2.2.1, we can make the following statements regarding quantum gates:

- quantum gates are linear maps preserving the total probability equal to 1[10];
- we can represent a quantum gate as a matrix that preserves the total probability equal to 1;
- quantum gates are unitary matrices and unitary matrices are quantum gates;
- a quantum gate $\mathbf{U}$ is reversible and $\mathbf{U}^\dagger$ is the inverse gate;
- classical reversible gates are valid quantum gates[11].

Now that we have described the main properties of quantum gates we can examine the effect that these gates have on *qubits*; we begin our analysis with gates that operate on a single *qubit*.

SINGLE QUBIT QUANTUM GATE: Quantum gates that operate on a single *qubit* can be visualized as *rotations* on the Bloch sphere. Any rotation can be traversed both forward and backward, which makes rotations reversible and, consequently, valid quantum gates. We now examine some rotations that define particularly interesting gates.

Pauli gates

Rotations of 180° with respect to the axes define the *Pauli gates*:

- Pauli gate X: rotation with respect to the x axis. This gate is the quantum version of the classical NOT gate:
$$X|0\rangle = |1\rangle \qquad X|1\rangle = |0\rangle\,;$$
- Pauli gate Y: rotation with respect to the y axis. This gate has no classical counterpart and, applied to the *qubits* $|0\rangle$ and $|1\rangle$, modifies them as follows:
$$Y|0\rangle = i|1\rangle \qquad Y|1\rangle = -i|0\rangle\,;$$
- Pauli gate Z: rotation with respect to the z axis. This is also a purely quantum gate, which acts as follows:
$$Z|0\rangle = |0\rangle \qquad Z|1\rangle = -|1\rangle\,;$$
- Pauli gate I: zero rotation. This gate does not modify the state of the *qubits*:
$$I|0\rangle = |0\rangle \qquad I|1\rangle = |1\rangle\,.$$

10 Recall that the total probability of the state $|s\rangle = c_1|0\rangle + c_2|1\rangle$ formed by a *qubit* is $\alpha^2 + \beta^2$.

11 In particular, it is possible to show that a classical reversible gate permutes probability amplitudes, thus preserving the total probability equal to 1.

It is also possible to define these gates as matrices; the definition is the same as that of the operators that represented the observables σ_x, σ_y and σ_z (respectively Equations 1.3.3, 1.3.3 and 1.3.3) plus the identity matrix for the I gate. In fact, if we observe the matrices that describe the observables we note that, in addition to being Hermitian operators (which describe observables), they are also unitary, therefore valid quantum gates.

We can then consider rotations smaller than 180°; we define the S gate or *phase gate* as a rotation of 90° with respect to the z axis and by halving the rotation angle again we obtain the T gate (rotation of 45° with respect to z). The action of T on $|0\rangle$ and $|1\rangle$ is:

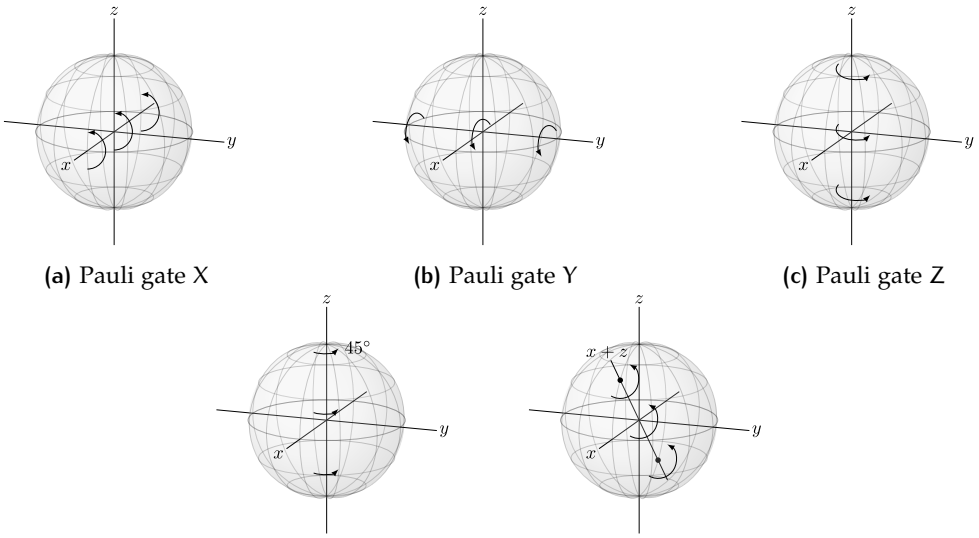
$$T|0\rangle = |0\rangle \qquad T|1\rangle = e^{i\frac{\pi}{4}}|1\rangle .$$

The last gate that acts on a single *qubit* that we consider is the *Hadamard gate* (H). This gate can be interpreted as a rotation of 180° with respect to the $x+z$ axis. H allows the creation of superposition of states[12], in particular:

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle \qquad H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |-\rangle .$$

Important gates on the Bloch sphere

Figure 2.5 shows the described gates as rotations on the Bloch sphere.



(a) Pauli gate X **(b)** Pauli gate Y **(c)** Pauli gate Z

(d) Gate T **(e)** Hadamard gate H

Figure 2.5: Gate as rotation [6]

MULTIPLE QUBITS QUANTUM GATE: When we describe states composed of multiple *qubits* we must represent them as a *tensor product* $\otimes$[13]. For example, a state in which two *qubits* are both in the state $|0\rangle$ is written:

$$|0\rangle \otimes |0\rangle \text{ abbreviated as } |0\rangle|0\rangle \text{ abbreviated as } |00\rangle .$$

With two *qubits* the z basis is $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$. A generic state $|s\rangle$ will be the superposition of the basis states

$$|s\rangle = c_1|00\rangle + c_2|01\rangle + c_3|10\rangle + c_4|11\rangle ,$$

12 We will return to this in Section 2.2.2.

13 Describing mathematically the tensor product is beyond the goal of this work.

and the total probability will be $|c_1|^2 + |c_2|^2 + |c_3|^2 + |c_4|^2$ which, if the state is normalized, will be equal to 1.

Some composite states can be written as a tensor product of single *qubit* states, for example:

$$\frac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle) = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |+\rangle\,|-\rangle\,.$$

Entangled states

These states are called *product states* or *simply separable states*. There exist states that cannot be decomposed into simple states. In this case we speak of *entangled states*, states in which complete knowledge of the state vector of the entire system does not provide any information about the states of the elements composing the system[14].

We study multiple qubits quantum gates precisely for their ability to create entangled states. The first gate that enables entanglement between *qubits* is the CNOT or CX.

The CNOT gate (*controlled* NOT) is a two-*qubit* gate; the first is called the *control*, the second is the *target qubit*. If the control *qubit* is set, the target *qubit* is flipped (hence the name controlled NOT). The behavior of the CNOT gate can be described as:

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

Figure 2.6: CNOT gate

$$\text{CNOT}\,|a\rangle\,|b\rangle = |a\rangle\,|a \oplus b\rangle\,.$$

Figure 2.6 reports the matrix definition of the CNOT. This gate is able to create entangled states. In particular:

$$\begin{aligned} \text{CNOT}\,|+\rangle\,|0\rangle &= \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \left|\Psi^+\right\rangle \\ \text{CNOT}\,|-\rangle\,|0\rangle &= \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) = \left|\Psi^-\right\rangle \\ \text{CNOT}\,|+\rangle\,|1\rangle &= \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) = \left|\Phi^+\right\rangle \\ \text{CNOT}\,|+\rangle\,|1\rangle &= \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) = \left|\Phi^-\right\rangle\,. \end{aligned} \tag{2.5}$$

Bell states

The states $|\Psi^+\rangle$, $|\Psi^-\rangle$, $|\Phi^+\rangle$ and $|\Phi^-\rangle$ cannot be obtained as tensor products two of single *qubit* states and are, therefore, entangled states. These four states are called *Bell states*; they are orthogonal and form a basis called the *Bell basis*.

Other important quantum gates are those we have already encountered when discussing classical gates; we have already seen that a classical reversible gate is a valid quantum gate. The gates of most interest to us are the TOFFOLI gate and the CROSSOVER gate.

It is not possible, however, to construct a quantum analogue of the FANOUT gate; therefore we cannot obtain a copy of a *qubit* without knowing its state. Measuring the state, however, would inevitably modify it, so in general we

14 Entanglement is one of the most interesting aspects of quantum mechanics. A detailed discussion of its physical implications is beyond the scope of this work. For some introductory material about entanglement the reader is referred to [2].

cannot duplicate a *qubit*. This is an important theoretical result and is known as the *non-cloning theorem*:

Non-cloning theorem

Theorem 6. *It is impossible to make a copy of an unknown quantum state. Qubits cannot be copied.*

Universal quantum gates

Just as there exist sets of classical gates that allow the construction of any other gate[15], we can identify sets of universal quantum gates. A set of universal quantum gates must be able to:

- create superpositions of states such as the Hadamard gate H;
- create entangled states such as the CNOT gate;
- create complex probability amplitudes; H and CNOT contain only real values, while the phase gate S allows obtaining complex probability amplitudes.

Although the set of gates we have implicitly defined $\{ \text{CNOT}, H, S \}$ has all the required properties, it can be shown that it is not a universal set. In particular, the *Gottesman-Knill theorem* states that:

Theorem 7. *A quantum circuit containing only* CNOT, H *and* S *gates is efficiently simulated by a classical computer.*

We call the circuits generated by this set of gates the *Clifford group*, and the set $\{ \text{CNOT}, H, S \}$ consequently takes the name of *Clifford group generator*. A set of universal gates must, therefore, generate more than the Clifford group, even though it has not been proven that this is a sufficient condition to be a universal set.

Universal quantum gates sets

Examples of universal gate sets are:

- $\{ \text{CNOT}, H, T \}$: replacing the phase gate with the T gate in the Clifford group generator allows us to obtain a universal set of gates;
- $\{ \text{TOFFOLI}, H, S \}$: starting again from the Clifford group generator, we obtain a universal set by replacing the CNOT with the TOFFOLI gate.

The concept of a universal set of gates in quantum computing is different from what we are used to with classical gates. In the classical case we can construct any truth table starting from a set of universal classical gates. The meaning of a universal set in quantum computation is instead described by the *Solovay-Kitaev theorem*:

Theorem 8. *Any universal gate set can approximate a quantum gate on* n *qubits to precision* ϵ *using* $\Theta(2^n \log^c(\frac{1}{\epsilon}))$ *gates for some constant* c.

The term 2^n should not be surprising because a unitary operator acting on n *qubits* (that is, the gate we are trying to approximate) is a matrix of dimension $2^n \times 2^n$. What is really relevant is the term $\log^c(\frac{1}{\epsilon})$, which tells us that we can reduce the approximation error at a cost that grows as a polynomial of a logarithm —this function is often called *polylog*— which is efficient.

15 Examples of these sets are: $\{ \text{AND}, \text{NOT} \}$, $\{ \text{OR}, \text{NOT} \}$, $\{ \text{NAND} \}$ and $\{ \text{TOFFOLI} \}$.

2.2.3 Algorithms

We have discussed what quantum gates are and how they can be assembled to build a quantum circuit. We now see how this circuit must be constructed in order to be useful for solving a computational problem. In this section we introduce quantum oracle-based circuits and present the simplest oracle-based algorithm.

Quantum Oracles

An oracle is a Boolean function, that is, a function that acts on *qubits* and that can be composed with gates. In order to be translated into quantum gates we must ensure that the resulting gate is reversible. In general, it is possible to make a circuit reversible by using an *extra qubit* and applying the XOR gate between the output of the circuit and the extra *qubit*.

Single qubit oracle

For example, if the function $f(x)$ acts on a single *qubit* we can construct a reversible gate $\mathbf{U}_f$ as shown in Figure 2.7. It is easy to see that regardless of the action of $f(x)$, which may or may not be reversible, knowing the outputs of the gate it is possible to reconstruct the inputs. We can rewrite the application of $\mathbf{U}_f$ to the *qubits* as:

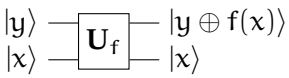


Figure 2.7: Quantum oracle gate

$$|x\rangle\,|y\rangle \xrightarrow{\mathbf{U}_f} |x\rangle\,|y \oplus f(x)\rangle$$

The extra *qubit* —the one we used to make $\mathbf{U}_f$ reversible— is called the *answer qubit* or *target qubit*, while $|x\rangle$ is called the *input qubit*. When we set $|y\rangle = |0\rangle$ we can read the result of $f(x)$ applied to $|x\rangle$ in the target *qubit*.

In reality, this is not the standard way of using a quantum oracle. What is done instead is to prepare the target *qubit* in the state $|-\rangle$ and then apply the gate $\mathbf{U}_f$. Quantum gates can, indeed, also be applied to superposed states, and in this case the result is a phase change of the *qubit* $|x\rangle$ while the target *qubit* remains unchanged. We can write the application of this gate as:

$$|x\rangle\,|-\rangle \xrightarrow{\mathbf{U}_f} \begin{cases} |x\rangle\,|-\rangle & \text{if } f(x) = 0 \\ -|x\rangle\,|-\rangle & \text{if } f(x) = 0 \end{cases} = (-1)^{f(x)}\,|x\rangle\,|-\rangle = (-1)^{f(x)}\,|x\rangle\,,$$

where, in the last step, we omit the value of the target *qubit* because it is not modified by the gate.

Phase oracle

Since, used in this way, the oracle applies a phase change to $|x\rangle$, it is called a *phase oracle*. Figure 2.8 shows the application of the phase oracle; in particular, it can be seen how it is possible to prepare $|y\rangle$ in the state $|-\rangle$ in order to then use $\mathbf{U}_f$ as a phase oracle[16].

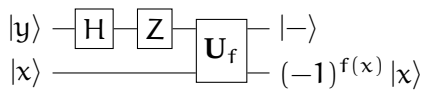


Figure 2.8: Phase oracle

16 It is possible to obtain the same result by first applying the Pauli gate X and then the Hadamard gate H.

Deutsch' algorithm

The Deutsch algorithm shows how to use the phase oracle to solve a computational problem. We will see that already with this first algorithm we obtain better performances compared to what we could achieve with a classical architecture. In this case, however, the performance gain is polynomial; therefore this is not one of those algorithms that suggest that quantum computers can efficiently solve a broader class of problems than classical computers[17].

The computational problem we want to solve is to determine the *parity* of two unknown bits b_1 and b_2. Determining parity means computing $b_1 \oplus b_2$. To obtain the result we are given access to an oracle $f(x)$ with $x \in \{0, 1\}$ which, when queried with the index of the unknown bit, returns the value of the bit, that is:

$$f(0) = b_0 \qquad f(1) = b_1 .$$

In a classical computation model we must query the oracle twice, obtain the values of the bits, and compute the result of the XOR between the two values.

Deutsch's algorithm allows us to obtain the parity value by querying the oracle only once. Figure 2.9 shows how the quantum circuit is implemented.

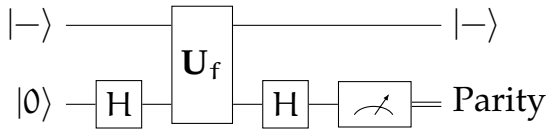


Figure 2.9: Deutsch' algorithm

Without developing all the algebraic steps, let us see what happens to the input qubit $|0\rangle$ as it passes through the quantum gates. The first quantum gate is the H gate; its application transforms $|0\rangle$ into

$$|0\rangle \xrightarrow{H} \frac{1}{\sqrt{2}} |0\rangle + |1\rangle = |+\rangle .$$

Phase oracle and superposition

We now query the oracle with an input in a superposition of states. The result of the query is[18]

$$\left[\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)\right] |-\rangle \xrightarrow{U_f} \frac{1}{\sqrt{2}}\left[(-1)^{f_0} |0\rangle + (-1)^{f_1} |1\rangle\right].$$

Ignoring some steps we can simplify this expression as:

$$\begin{cases} (-1)^{b_0} |+\rangle & \text{if } b_0 = b_1 \\ (-1)^{b_0} |-\rangle & \text{if } b_0 \neq b_1 \end{cases} .$$

After the oracle, if $b_0 = b_1$, we obtain the state $|+\rangle$ up to a global phase; if $b_0 \neq b_1$ we obtain the state $|-\rangle$ up to a global phase. This phase does not modify the position of the qubit on the Bloch sphere, and we can already distinguish the two states with a measurement parallel to the x axis. Since,

17 Showing these algorithms, including Shor's algorithm, is beyond the scope of this work.
18 Where, in the result, we omitted the answer qubit $|-\rangle$.

conventionally, measurements are performed with respect to the z axis, we apply the H gate once more obtaining

$$\begin{cases} (-1)^{b_0} |0\rangle & \text{if } b_0 = b_1 \\ (-1)^{b_0} |1\rangle & \text{if } b_0 \neq b_1 \end{cases}.$$

Global phases are physically irrelevant

The act of measurement discards the global phase and allows us to distinguish the two cases with certainty: we obtain $|0\rangle$ if the unknown bits were equal (even parity) and, $|1\rangle$ if the unknown bits were different (odd parity). It is interesting to note that throughout the entire algorithm we never learn the value of the individual bits, but only whether they are equal or not.

Deutsch's algorithm allows us to halve the number of oracle queries, and it can be extended to an arbitrary number of bits whose parity we want to determine. This example concludes our discussion of the quantum gate-based paradigm. We will encounter this paradigm again in Chapter 7, where we will combine it with the concepts of quantum annealing that we now illustrate.

2.3 ADIABATIC QUANTUM COMPUTING

Adiabatic Quantum Computing (AQC) is a paradigm of quantum computation alternative to the gate-based quantum model. The two models are computationally equivalent, but deeply different in the nature of computation. AQC has an analog nature that contrasts with the digital one of the gate-based paradigm. When we adopt AQC, we study the evolution of a physical system over time.

Before describing this approach to quantum computation in more detail, it is necessary to give some definitions that will be useful in the following discussion. We call a *quantum dynamical system* a system formed by n *qubits* that evolves over time according to certain forces. The time evolution is determined by the Hamiltonian $\mathbf{H}(t)$[19] of the system and described by the Schrödinger equation (Equation 1.15). As we observed in Section 1.4.3, $\mathbf{H}(t)$ is a Hermitian operator, and the observable associated with it is the energy of the system. The eigenvalues of $\mathbf{H}(t)$ at a given instant represent all the possible energy levels of the system that we can measure. We call this set of values the *energy spectrum* of the system. We are interested in two particular energy levels: the lowest energy one, called the *ground state*, and the *first excited state*, the energy level immediately above the ground state.

2.3.1 AQC algorithm

AQC algorithms are designed to solve optimization problems formulated as follows: given an objective function $f : \mathbb{D}^n \to \mathbb{R}$ defined on n variables $x = x_1, \ldots x_n$ from some discrete domain $\mathbb{D}$, find an assignment of values to x that minimizes $f(x)$.

An AQC algorithm to solve optimization problem P with objective function $f(x)$ is described by a time-varying Hamiltonian $\mathbf{H}(t)$ that defines the

19 We write it in this way to emphasize that, in general, $\mathbf{H}$ also depends on time.

Components of AQC

evolution of the system Q composed of n *qubits*. $\mathbf{H}(t)$ is specified by three components:

- an *initial Hamiltonian* $\mathbf{H}_i$ chosen so that its ground state is easy to find;
- a *final* or *problem Hamiltonian* $\mathbf{H}_f$ that encodes the objective function so that the ground state of the system is an eigenstate of $\mathbf{H}_f$ having minimum eigenvalue. That is, the ground state of the system corresponds to an optimal solution to P;
- an *adiabatic evolution path*, a function $s(t)$ that decreases from 1 to 0 as time goes from 0 to t_f.

The AQC algorithm is defined by $\mathbf{H}(t)$ which creates a gradual (adiabatic) transition from $\mathbf{H}_i$ to $\mathbf{H}_f$ according to

$$\mathbf{H}(t) = s(t) \cdot \mathbf{H}_i + (1 - s(t)) \cdot \mathbf{H}_f. \tag{2.6}$$

To create the initial Hamiltonian we apply to all *qubits* the Pauli operator X (σ_x in Equation 1.10) —this operator describes the observable spin parallel to the x axis, consequently— the associated eigenvectors are $|+\rangle$ and $|-\rangle$. In particular, $\mathbf{H}_i$ is defined as

$$\mathbf{H}_i = \sum_{i=1}^{n} \frac{1}{2}(1 - X\, q_i)$$

where $X\, q_i$ indicates the application of the Pauli gate X to the *qubit* i. Then the ground state is formed by all n *qubits* in the state $|+\rangle$. This means that a state has been created in which the eigenstates in the standard basis ($\{|0\rangle, |1\rangle\}$) are equiprobably observable.

Initial state as maximum superposition

In general, $\mathbf{H}_i$ is specified in a basis orthogonal to that of the final Hamiltonian in order to create a state of maximum superposition.

Starting from the known ground state of $\mathbf{H}_i$ and evolving thanks to $s(t)$ up to $\mathbf{H}_f$, the *adiabatic theorem* ensures that, under certain conditions, Q will always remain in the ground state of $\mathbf{H}(t)$, and at time t_s we will be able to read in the configuration of the *qubits* the solution of problem P.

At first glance it might seem that this formulation limits the capabilities of AQC; in reality, we can naturally rewrite many problems in **NP** as optimization problems. In Section 2.1.3 we mentioned the problem of prime factorization as a candidate to demonstrate the superiority of quantum computing. This problem can be translated into an optimization problem in which, given an integer N, one must find two integers $x, y > 1$ such that they minimize $f(x, y) = (N - x \cdot y)^2$.

2.3.2 Theoretical results

Let us see some results on which AQC is based, which define the classes of problems that can be addressed and establish the relationship between AQC and the gate-based model.

Adiabatic theorem

The adiabatic theorem guarantees, under fairly strict conditions, that the computation carried out by QAC actually ends in the ground state of $\mathbf{H}_f$. Let us consider the algorithm $\mathbf{H}(t)$ that evolves over time $t : 0 \to t_f$; we remap the flow of time with $s(t)$ strictly decreasing from 1 to 0 while $t : 0 \to t_f$. The Hamiltonian associated with $s(t)$ is $\tilde{\mathbf{H}}(s)$. Finally, we call $\tau(s)$ the rate at which $\tilde{\mathbf{H}}(s)$ evolves as a function of $s(t)$. Denoting by $|\Phi_s\rangle$ the state of the system as a function of $s(t)$, we can rewrite the Schrödinger equation 1.15 for our system as[20]:

$$\frac{\partial}{\partial s}|\Phi_s\rangle = -i\tau(s)\tilde{\mathbf{H}}(s)|\Phi_s\rangle .$$

Assuming that throughout the entire evolution of the system the ground state is always unique, we call $\delta(s)$ the *spectral gap*: the difference between the eigenvalues of the ground state and of the first excited state, at any time s. We call δ_m the *minimum spectral gap* $\delta_m = \min_s \delta(s)$. We assume that

$$\tau(s) \ll \frac{\left|\left|\frac{\partial}{\partial s}\tilde{\mathbf{H}}(s)\right|\right|}{\delta_m^2} \tag{2.7}$$

where $\|\cdot\|$ is the matrix norm induced by the L_2 metric.

Denoting by $|\Phi_g\rangle$ the ground state of the final Hamiltonian, the *adiabatic theorem* states that:

Theorem 9.

Adiabatic theorem

- *If the system is in the ground state at* $s = 0$, *and*
- *if* δ_s *is strictly greater than* 0 *throughout time* s, *and*
- *if the process evolves slowly enough to obey Equation 2.7*

then the process will finish in the ground state $|\Phi_{t_f}\rangle = |\Phi_g\rangle$ *with sufficiently high probability.*

Temporal complexity

To obtain an upper bound on the computational time t_f for a problem P of size n, it is necessary to find an upper bound for the numerator $\left|\left|\frac{\partial}{\partial s}\tilde{\mathbf{H}}(s)\right|\right|$ and a lower bound for the denominator δ_m^2 in Equation 2.7.

The numerator is bounded by the maximum difference between the eigenvalues obtainable from the Hamiltonians $\mathbf{H}_i$ and $\mathbf{H}_f$. It is possible to show that this difference is polynomial in the size n of the problem.

Spectral gap and $\mathbf{P} \neq \mathbf{NP}$

In general, the analysis of the minimum spectral gap δ_m^2 for a given algorithm $\mathbf{H}(s)$ and a given instance I of a problem P is difficult. In particular, when trying to develop an algorithm $\mathbf{H}_L(s)$ to decide a language $L \in \mathbf{NP}$, it is not known whether δ_m^2 is polynomial or exponential for $\mathbf{H}_L(s)$. Establishing this would mean proving $\mathbf{P} \neq \mathbf{NP}$ or $\mathbf{P} = \mathbf{NP}$ in this computational model.

20 We omitted the term $\hbar$ because it is only a multiplicative constant and we set it equal to 1. This is an approach often adopted in quantum mechanics to simplify the arithmetic steps.

Relation with gate based model

Equivalence between AQC and quantum gates model

By studying the relationships between AQC and quantum gate computing, it has been shown that a universal version of AQC can simulate the gate-based model with, at most, polynomial slowdown. It has also been shown that the converse is true: the quantum gate model can simulate any AQC computation efficiently. These two theorems show that adiabatic quantum computing and quantum gate computing are polynomially equivalent computational models. The class of problems solved efficiently by AQC algorithms is therefore **BQP**.

2.3.3 Quantum annealing

Quantum annealing (QA) is an approach based on heuristics that allows solving combinatorial optimization problems. It is a system inspired by *simulated annealing*. Simulated annealing is a stochastic algorithm for finding the minimum energy of a system inspired by thermodynamics.

There is a close relationship between AQC and QA; in particular, it is established that an algorithm for quantum annealing can be implemented natively with the instruction set provided by AQC. The two approaches are so closely connected that the two terms are often used interchangeably, and in this work we can also consider them as synonyms. However, there are differences, not so much in the algorithm, but in the reference computational model:

- An AQC algorithm is designed to be implemented in the universal AQC model. This is the computational model discussed in the previous section and is polynomially equivalent to the quantum gate-based model;
- A QA algorithm aims to solve a certain type of problems: combinatorial optimization problems. This allows imposing restrictions on the final Hamiltonian and relaxing the conditions of the AQC model.

difference between AQC and QA

In a QA algorithm, the final Hamiltonian $\mathbf{H}_f$ represents a classical ground state, that is, without superposition at the time of reading the results. The condition that is relaxed with respect to AQC, instead, is the adiabaticity of the evolution of the physical system. In particular, it is no longer required that the entire computation occurs in the ground state of $\mathbf{H}(t)$.

Annealing

Let us now examine the ideas underlying annealing-based systems, starting from simulated annealing developed for classical computation models, and then seeing how quantum annealing fits into the discussion and what advantages the latter has over the classical model.

SIMULATED ANNEALING: This stochastic approach allows finding the minimum of a multivariate function. The method is inspired by thermodynamics and aims to find the most stable configuration of the system, that is, the one with the lowest energy. The function to be minimized therefore represents

the energy of the system and is generally composed of many local minima separated by energy barriers. To overcome these barriers, energy is supplied to the system. In simulated annealing, this energy is thermal energy.

The physical system starts in a certain (non-optimal) configuration and with a certain (simulated) temperature T. The system changes configuration according to a certain heuristic and in the new configuration it will have a new energy; let ΔE be the difference between the energy in the second configuration and the energy in the first. If $\Delta E < 0$ the new configuration is accepted. If $\Delta E > 0$ the transition is not immediately discarded but is accepted with a certain probability equal to the Boltzmann factor $e^{-\frac{\Delta E}{k_b T}}$ where k_b is the Boltzmann constant. We observe that the higher the temperature, the higher the probability of overcoming high energy barriers. As the temperature decreases —thus assuming we are approaching the global minimum— transitions to higher-energy configurations become rarer. When we reach $T = 0$, the total energy is given only by the internal energy of the system, and, at the end of the annealing, the system is in the state of minimum internal energy.

The time τ required to evolve from $T = T_i$ to $T = 0$ is called the *thermal relaxation time of the system*. If the system is *ergodic*[21], the relaxation time increases linearly with the size of the problem (while the possible configurations increase exponentially); in this case simulated annealing is an efficient algorithm for finding the ground state of the system. However, if the system is not ergodic, any thermal fluctuation at finite temperature would require an infinite relaxation time to bring the system to the minimum energy configuration; that is, if the system is not ergodic $\tau \to \infty$.

Limits of simulated annealing

QUANTUM ANNEALING: Quantum annealing, like its classical version, can be seen as a heuristic that moves within a solution space searching for the best one. The way the space is explored by a QA algorithm is described by a parameterized Markov chain. To move from one configuration to another, temperature T is no longer used but a *transverse field coefficient* Γ (also called the *tunneling coefficient*). Like T, Γ also starts with a high value and decreases to zero following a certain scheduling.

Quantum annealing introduces a *transverse field* or *disordering Hamiltonian* $\mathbf{H}_d$ that does not commute with $\mathbf{H}_f$ and is scaled with the tunneling coefficient. The Hamiltonian that describes the time evolution of the system is therefore:

Disordering Hamiltonian

$$\mathbf{H}(t) = \mathbf{H}_f + \Gamma(t)\mathbf{H}_d. \tag{2.8}$$

The disordering Hamiltonian $\mathbf{H}_d$ introduces kinetic energy to the annealing process in the form of quantum fluctuations of the solution space. Decreasing $\Gamma(t)$ moves the system closer to $\mathbf{H}_f$ while damping the quantum fluctuations. Figure 2.10 graphically shows the advantage of QA compared to the classical counterpart: high values of $\Gamma(t)$ allow the computation to escape local minima by *tunneling* through hills instead of climbing over them incrementally. This allows the algorithm to move faster and further across the landscape early in the process.

21 For more information about ergodic systems see [7].

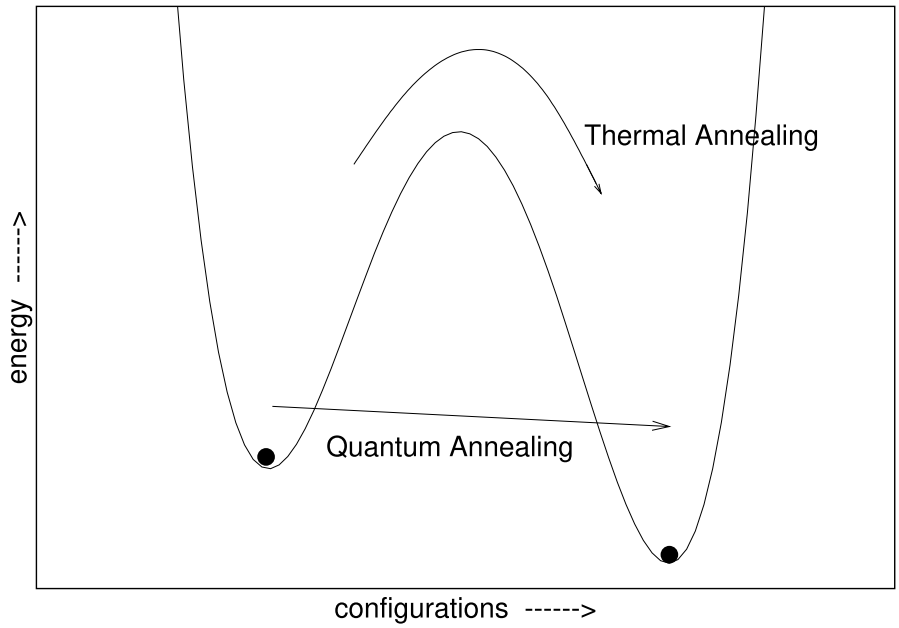


Figure 2.10: Simulated and quantum annealing [5]

Simulated quantum annealing

Another advantage of QA is that, during the evolution, the system is not in a single state, but in a superposition of states whose probability is defined by $\mathbf{H}(t)$. This idea is also adopted by algorithms that implement QA on classical architectures. In this case we refer to *simulated quantum annealing* and the objective is to try to capture the superposition property by representing many states in parallel[22].

Formulating a problem for annealing

Let us see how to formulate a problem so that it can be solved with a *quantum annealer* (a machine capable of implementing QA). We will obtain two equivalent formulations that describe problems solvable with quantum, simulated, or simulated quantum annealing. For now we focus on the theoretical description of the formulation. For a practical implementation compatible with a QA architecture, see Section 4.3.

ISING MODEL: it is a mathematical model that describes phase transitions and other properties of a physical system that evolves over time. The problem is defined by n particles arranged on a graph $G = (V, E)$ that can be described as a d-dimensional grid. Each particle can be in a spin state $\in \{+1, -1\}$. We call *spin configuration* $s = s_1, \ldots, s_n$ an assignment of spins to the particles.

Forces on particles

Two types of forces can act on the described system:

- *external forces* h_i applied to individual particles;
- *interaction forces* J_{ij} between grid neighbors (two vertices v_i and v_j connected by an edge e_{ij}). If there are no edges between v_i and v_j we assume $J_{ij} = 0$.

22 A more in-depth discussion on simulated quantum annealing is beyond the scope of this work, but can be found in [8].

The energy of a given configuration is defined by

$$\mathbf{H}(s) = \sum_i h_i \cdot s_i + \sum_{j<i} J_{ij} \cdot s_i s_j \ . \tag{2.9}$$

Let us see how to encode this energy function in a quantum annealer. Considering a collection of n *qubits* $Q = q_1, \ldots, q_n$ arranged in the grid defined by the graph $G = (V, E)$, the state of Q is $|\Phi\rangle = |\phi_1, \ldots, \phi_n\rangle$. We can translate this energy into the problem Hamiltonian H_f:

$$\mathbf{H}_f = \sum_{i \in V} h_i \cdot Z\, q_i + \sum_{(i,j) \in E} J_{ij} \cdot Z\, q_i\ Z\, q_j. \tag{2.10}$$

Where $Z\, q_i$ indicates the application of the Pauli gate Z to the *qubit* i. This is a valid Hamiltonian for a QA algorithm and a quantum annealer is able to retrieve the solution at minimum energy.

QUBO PROBLEM: The Ising Model is defined on spins $s_1 \in \{+1. - 1\}$. An equivalent formulation, but closer to the usual one in computer science, is the *Quadratic Unconstrained Binary Optimization* problem (QUBO form). In this formulation we consider n Boolean variables $x_i \in \{0, 1\}$ and, considering an upper triangular matrix Q_{ij} of weights of dimension $n \times n$, we want to minimize the function

$$f_{QUBO}(x) = \sum_{i,j} Q_{ij} \cdot x_i x_j \ . \tag{2.11}$$

from QUBO form to Ising form and vice-versa

It is possible to show that a problem in QUBO form always has a corresponding version in Ising model form, and to move from one formulation to the other one imposes $s_i = 1 - 2x_i$ [9]. It is also possible to show that the ground state of the problem in QUBO form is the same as that of the problem in Ising form up to a finite offset δ_e

$$\min_{s \in \{-1,+1\}} \sum_i h_i \cdot s_i + \sum_{j<i} J_{ij} \cdot s_i s_j = \min_{x \in \{0,1\}} \sum_{i,j} Q_{ij} \cdot x_i x_j + \delta_e \ . \tag{2.12}$$

2.4 CONCLUSION

In this work we have justified the interest in quantum computing by showing how this new paradigm is positioned within the study of computational complexity and in the definition of classes of computational problems.

We started by defining theoretical models that describe classical computation such as the Turing machine and the circuit model. Starting from these, we classified computational problems based on the resources required to solve them.

We then introduced quantum computation theoretically (through quantum circuits) and showed how the classes of computational problems change when considering this new paradigm. In particular, we observed that proving the computational advantages of the quantum paradigm would lead to significant results also in classical computation, making it possible to prove that $\mathbf{P} \neq \mathbf{PSACE}$.

The chapter then continues by defining the quantum computation paradigm based on gates, this time in a less theoretical way, observing how a *qubit* is defined in a computational model and what it means to apply a quantum gate to a system of *qubits* to modify its state. The section on quantum gates concludes with the Deutsch algorithm, which shows a first speedup (even if only polynomial) of quantum computers compared to classical architectures.

We concluded the chapter by examining another approach to quantum computation, this time an analog approach: AQC. We described what an AQC algorithm consists of and which conditions must be satisfied for this algorithm to allow us to obtain the solution to our problem. Finally, we moved from the universal AQC model to QA algorithms, losing generality but relaxing the constraints imposed on computation. We derived two forms for formulating a problem suitable for QA (QUBO form and Ising form) and highlighted their equivalence.

These two formulations that we have obtained are precisely the objective of the rest of our work: transforming structured knowledge, and a query on this knowledge, into a formulation suitable for QA.

3 ONTOLOGY

In this chapter we explain what kind of knowledge base (KB) an ontology is, how to build an ontology, and why this knowledge representation is important. To clarify and demonstrate why ontologies are useful, we present an example of a foundational ontology[1], briefly discussing its utility.

The rest of the chapter is about reasoning on ontologies; we discuss the semantics of the formal language used to represent knowledge, what we mean when saying interpretation of a KB, and the complexity of finding an interpretation.

3.1 KNOWLEDGE BASE

In the field of information technologies, an ontology is a structured representation of knowledge about a certain domain of interest; however, the study of knowledge began much before informatics. To better understand what an ontology is, let's start with the philosophical definition and then point out the differences between this vision and the IT one.

3.1.1 Ontology in philosophy

Ontology was born as a branch of philosophy. In this context it is the science of what is, of the kinds and structures of objects, properties, events, processes, and relations in every area of reality [10].

The goal of an ontology is to give a definitive and exhaustive classification of entities in all spheres of being. With the term "definitive" we mean that an ontology should answer questions such as: "What classes of entities are needed for a complete description and explanation of all the goings-on in the universe?" With the term "exhaustive", instead, we mean that all types of entities and relations between these entities are included in our ontology [10].

3.1.2 Ontology in computer science

Thanks to the advent of the internet and the development of bigger and bigger software used by bigger and bigger groups of users, what we might call the Tower of Babel problem emerged. Each research group develops its KB with terms and concepts shared and accepted only inside the group. For example, different databases may use identical labels but with different meanings, and the same meaning may be expressed with different names [10].

1 a very general template the can be used as base (foundation) to build an ontology about a specific domain (more about that in Section 3.2.2).

To address the incompatibility problem between software, databases, and research groups, ontologies have become an important research topic in computer science where the goal is to define standards for data exchange, information integration, and interoperability [11].

In this field the term ontology gains a new meaning:

Ontology definition

Definition 1. *Ontologies represent a formal and explicit specification of a shared conceptualization [12].*

In this definition the keywords are:

CONCEPTUALIZATION: an ontology creates an abstract model identifying and defining only the relevant concepts;

EXPLICIT: the types of concepts and constraints on their use are explicitly defined;

FORMAL: an ontology should be machine-readable;

SHARED: the knowledge represented by the ontology has to be accepted by a group of people, ideally by everyone.

A-Box and T-Box

When we use an ontology to represent knowledge we are describing a graph where entities are bound together through relationships, and classified according to a formal description of the world [13]. Knowledge bases expressed with this formalism are divided into two components [14]:

T-BOX: stores a set of universally quantified assertions (inclusion assertions) stating general properties of concepts and roles;

A-BOX: contains assertions on individual objects (instance assertions).

The T-Box is the conceptualization of the world, while the A-Box is a certain instance of the world we have modelled in the T-Box.

We can see some similarities between an ontology and a database: the T-Box can be seen as the Entity-Relation schema and the A-Box as the set of all entries of the database. There is, however, a logical difference between the world represented by an ontology and the world represented by a database.

Databases make the *closed world assumption*: everything that is not present in the database is automatically false; for example, if a person does not appear in a bank registry it means that that person is not a client of the bank.

Ontologies, on the other hand, make the *open world assumption* [15], which means, for example, that we can assert that a certain person is a parent even if we have not specified any son or daughter.

3.1.3 OWL Language

OWL 2 Web Ontology Language is an ontology language for the Semantic Web with a formally defined meaning [16]. Thanks to OWL we can model classes and relations between classes (T-Box) and individuals with their specific properties and relations between individuals (A-Box).

OWL is a declarative language and defines the state of the world in a logical way. In particular, we are interested in OWL DL where the meaning of ontologies expressed with this language is assigned in a Description Logic style. OWL DL is, therefore, decidable and an appropriate tool (so-called reasoner) can then be used to infer further information about that state of the world [16].

OWL DL

OWL per se does not specify any syntax, it states only what can or cannot be expressed in an ontology. The World Wide Web Consortium (W3C) standardizes various syntaxes, some inspired by functional languages, others more suitable for storing on web pages. The only syntax that must be implemented by all tools to be compliant with the OWL standard is the RDF/XML syntax [16] (examples of this syntax are provided in Section 3.2.1).

3.1.4 Importance of ontologies

Ontologies are important in various fields, from interoperability to machine learning.

In the Semantic Web context, ontologies are a main vehicle for data integration, sharing, and discovery [17]. Different research groups can use the same ontology to share a unified vocabulary that helps build common knowledge and helps to better integrate the results obtained by each group.

In a more commercial scenario, an ontology can be used as a translation layer between different databases or software that are built by different teams and use different vocabularies.

In the machine learning field an ontology could be used to support the sharing and reuse of formally represented knowledge among neuro-symbolic AI systems [12].

3.2 EXAMPLE ONTOLOGIES

To help understanding the structure of ontologies and to show a practical example of ontology, we present two ontologies: a simple ontology about family relationships and DOLCE, a foundational ontology.

3.2.1 Simple ontology

This simple ontology about parental relationships shows the basic structure of an ontology, helping to understand the graph structure of these KBs and the relations between the T-Box and A-Box.

In Figure 3.1, we can see the T-Box of the ontology: this structure specifies what our domain of interest is, and what entities could possibly populate our world. This ontology is about people, so the main class/concept is `People`: this class has several subclasses that represent parents, children, and married people. We can assert that a person belongs to the married class without specifying the partner (open world assumption) but we can also infer that a person belongs to the parents class because we have created a relationship of type `parent_of` between that person and another person.

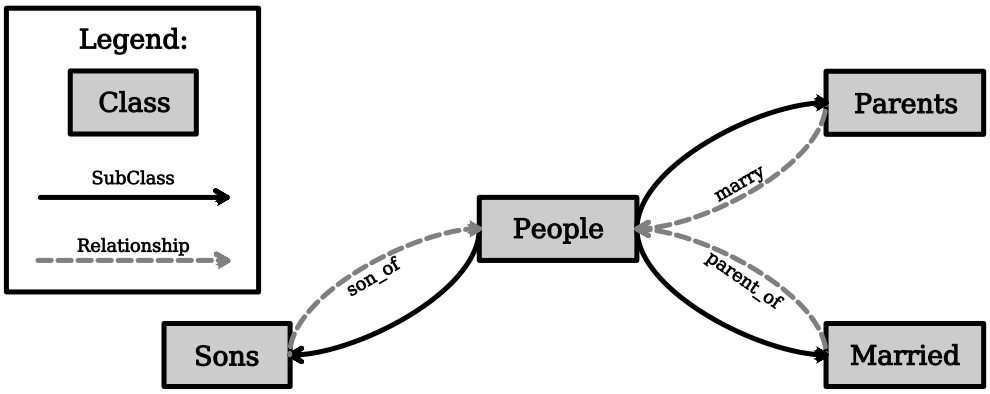


Figure 3.1: Graph for T-Box

OWL allows us to express rules to infer when a member of a class belongs also to another class. The following code shows (in the RDF/XML syntax) the definition of the class `Parent`[2]:

```
1 <owl:Class rdf:about="http://people#Parent">
2   <owl:equivalentClass>
3     <owl:Restriction>
4       <owl:onProperty rdf:resource="http://people#parent_of"/>
5       <owl:someValuesFrom rdf:resource="http://people#Person"/>
6     </owl:Restriction>
7   </owl:equivalentClass>
8   <rdfs:subClassOf rdf:resource="http://people#Person"/>
9 </owl:Class>
```

Listing 3.1: Definition of parents

At line 8 we can see that `Parent` is a subclass of `People`, and at lines 4 and 5 it is specified that a parent is a person that is `parent_of` another person. Other interesting properties that we can specify thanks to OWL constructs are:

- relation `marry` is symmetric;
- relation `parent_of` is the inverse of `son_of`;
- we can specify a domain and a range for relations.

Now we can populate the ontology by adding individuals and relations between individuals. For this small example we take inspiration from the Simpson family, and in the family tree (Figure 3.2 on the right) we can see the small portion of the family represented. To show what we mean by open world assumption we have asserted that Jackie is a married person even if in our representation there is no husband.

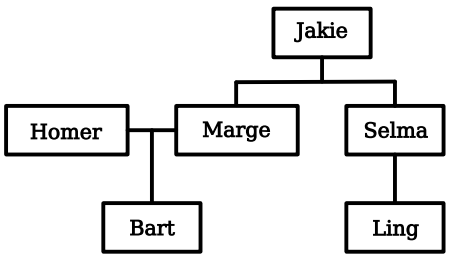


Figure 3.2: Simpson family tree

2 The complete code of the ontology can be seen at https://github.com/DavideCamino/Thesis/blob/main/Code/Ontology/people.rdf.

Our ontology covers a small domain, the types of entities that populate our model are very limited; the next example shows the commitment of engineering an ontology to represent virtually anything in the universe.

3.2.2 DOLCE ontology

DOLCE (Descriptive Ontology for Linguistic and Cognitive Engineering) is a top-level (foundational) ontology [18]; this means that this ontology describes fundamental aspects of reality and should be used as a base for constructing an ontology about a particular domain of interest. For this reason DOLCE defines only the T-Box; the user will then expand the T-Box with specific classes and relations of interest, and lastly will populate the A-Box.

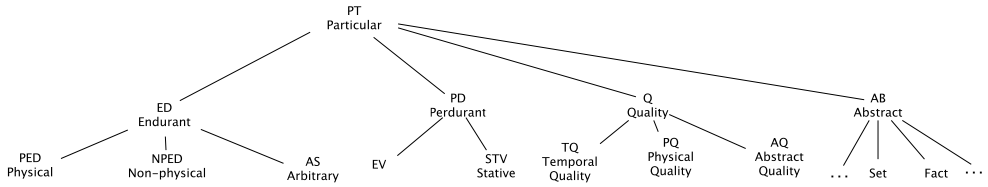


Figure 3.3: First layer of DOLCE taxonomy [18]

STRUCTURE OF DOLCE: in DOLCE we can model the modification of objects during time; for this reason DOLCE distinguishes between *endurants* and *perdurants*. Endurants may acquire or lose properties and parts through time, perdurants are fixed in time [18]. With a simplification we can see endurants as the physical entities that are modified by the passing of time (like objects, animals, and people) and perdurants as events that, once they have passed, cannot be changed anymore (like a tennis match or a conference).

endurants and perdurants

The relation connecting endurants and perdurants is called participation. A physical entity can be in time by participating in a perdurant, and perdurants happen in time by having endurants as participants [18].

Another important aspect of DOLCE is the way we attribute a property to an entity; this is done by using qualities, which are what can be perceived and measured. To do so we can assert that a certain entity has a specific quality and then, when it is possible, quantify that quality.

IMPORTANCE OF DOLCE: foundational ontologies can be useful in several fields, from conceptual modeling to natural language processing. DOLCE, today, is used in a variety of domains where it provides the general categories and relations needed to give a coherent view of reality [18].

3.3 REASONING ON ONTOLOGY

In Section 3.1.3 we have introduced the standard language to encode an ontology; in order to infer new information, starting from the one we already have, we need to better specify the semantics of OWL DL.

3.3.1 $\mathcal{SROIQ}$ DL

The semantics of OWL DL extends the semantics of the description logic (DL) $\mathcal{SROIQ}$ to provide support for datatypes and punning [19]. For constructs available both in OWL DL and in $\mathcal{SROIQ}$ the semantics correspond exactly.

Description logics allow the modeling of the domain of interest with three kinds of entity: concepts, roles, and individual names. These entities correspond to unary predicates, binary predicates, and constants in first-order logic [15]. From the point of view of ontology and OWL, concepts are classes, roles are relationships, and individual names are the individuals that can belong to one or more classes.

Construct in $\mathcal{SROIQ}$ DL

$\mathcal{SROIQ}$ is one of the most expressive description logics where we have constructors for:

- transitive roles: $\mathcal{S}$
- role inclusions, local reflexivity, universal role, symmetry, asymmetry, role disjointness, reflexivity, and irreflexivity: $\mathcal{R}$;
- nominals: $\mathcal{O}$;
- inverse roles: $\mathcal{I}$;
- qualified number restrictions: $\mathcal{Q}$.

For example, we can construct the ontology shown in Figures 3.1 and 3.2 with a set of assertions like:

```
person(selma)        married(jackie)        parent_of(marge, bart)
```

Each of these statements is called an axiom and the set of all axioms constitutes our KB.

3.3.2 Interpretation of a knowledge base

An interpretation I consists of a domain Δ^I and an interpretation function $\cdot^I$ that maps:

$$\begin{gathered}\text{concept } A \to A^I \subseteq \Delta^I \\ \text{role } R \to R^I \subseteq \Delta^I \times \Delta^I \\ \text{named individual } a \to a^I \in \Delta^I\end{gathered}$$

Meaning of interpretation

In other words I assigns a fixed meaning to all entities in the KB [15]. By having a fixed meaning, we can say if an axiom α holds in I or not; in the first case we say that I satisfies α and we write $I \models \alpha$.

If all axioms in an ontology are satisfied by I we say that I is a *model* of the ontology. An ontology is consistent if it accepts at least one model.

A reasoner should at least be capable of saying if an ontology is consistent, but we are also interested in querying knowledge to retrieve new information.

QUERY INTERPRETATION: Considering a KB K, a query q consists of axiom templates where $\mathcal{SROIQ}$ axioms are composed of concept names, role names, and individual names, but also of concept variables, role variables, and individual variables. A solution for the query is an interpretation μ that allows us to rewrite all variables in q with names; we denote with $\mu(q)$ the result of the substitution.

The evaluation of q over K is a set of solutions μ with: [20]

$$\{\ \mu | K \cup \mu(q) \text{ is a } \mathcal{SROIQ} \text{ knowledge base and } K \models \mu(q)\}$$

In other words μ binds all free variables of q to names present in K [20].

A naive approach to find the solution to a query is to simply test for each possible solution mapping μ, if $K \models \mu(q)$; however, in the worst case, the number of mappings that have to be tested is exponential in the number of variables in the query [20].

3.3.3 Complexity of reasoning

Since presenting an actual algorithm for reasoning on ontologies is out of the scope of this work, we only give some hints about the reasons for the complexity and then present the theoretical results that prove the problem of reasoning on ontology is at least PSpace-hard.

It is easy to convince oneself that the more axioms there are in an ontology, the fewer interpretations exist that satisfy all axioms. On the other hand, if an ontology has fewer models, the more axioms hold in all of them and the more logical consequences follow from the ontology.

In other words the semantics of description logics are *monotonic*: the more knowledge we embed in an ontology, the more results it returns [15]. A more formal view is given in [21], where two *sources of complexity* are identified:

- OR-branching: the presence of disjunctive constructors;
- AND-branching: the presence of qualified existential and universal quantifiers.

The AND-branching is responsible for the exponential size of a single interpretation, and the OR-branching is responsible for the exponential number of different interpretations.

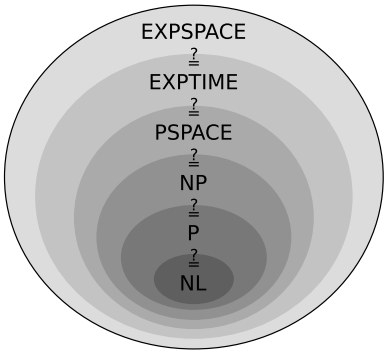


Figure 3.4: Complexity classes

To discuss the complexity of reasoning we take into account the description logic $\mathcal{ALC}$; this DL is a restriction of $\mathcal{SROIQ}$ [15], so its complexity is a lower bound for $\mathcal{SROIQ}$. It is possible to prove the **PSPACE**-hardness of satisfiability in $\mathcal{ALC}$ [21], therefore also $\mathcal{SROIQ}$ DL is at least **PSPACE**-hard. This result shows that, unless **PSPACE** = **P**, the exponential time complexity of any algorithm that makes inference on an ontology cannot be improved .

$\mathcal{SROIQ} \in$ ***PSPACE***

For those interested in some numerical examples to better understand what this class of complexity means in a real context, [22] presents the reasoner HermiT and evaluates its performance on some real ontologies.

3.4 CONCLUSIONS

In this chapter we have explained what an ontology is and we have motivated the interest in this field.

We showed that an ontology can be useful both in academic research and in industry. Moreover, being a formal and machine-readable structure, it can be queried and used to perform logically demonstrable reasoning whose subject is precisely the knowledge represented within the ontology.

We have shown both theoretically and with examples what can be expressed in an ontology and what cannot. We have formally defined what the interpretation of a KB is and showed what a query and its results are.

Lastly, we have characterized the complexity of reasoning on ontologies. This complexity is what motivated us to search for other paradigms to infer new knowledge starting from an ontology. In the next chapters we will build the tools necessary to achieve this goal.

Part II

TOOLS

4 ENVIRONMENT SETUP

In this chapter we describe the environment, libraries and tools we use to execute our tests.

In the following sections we install the SDKs to develop and interact with quantum computers from IBM and D-Wave. We also present two other useful tools to easily write optimization problems.

4.1 PYTHON ENVIRONMENT

The language used to interface with quantum computers is usually Python. In this section we create a virtual environment in Python in order to communicate with the IBM quantum computer and the D-Wave quantum computer.

For our tests we manage Python environments with `conda`. Let's start by creating the virtual environment named `quantum` and activating it with:

```
$ conda create --name quantum python=3.12 pip
$ conda activate quantum
```

For our tests and to follow the various examples presented both by IBM and D-Wave, it is also useful to be able to run a Jupyter notebook. We can install Jupyter with:

```
$ pip install jupyter
```

4.2 IBM QISKIT

To program a gate-based architecture and to access IBM quantum computers we use the *Qiskit* software stack. The name Qiskit is a general term referring to a collection of software for executing programs on quantum computers.

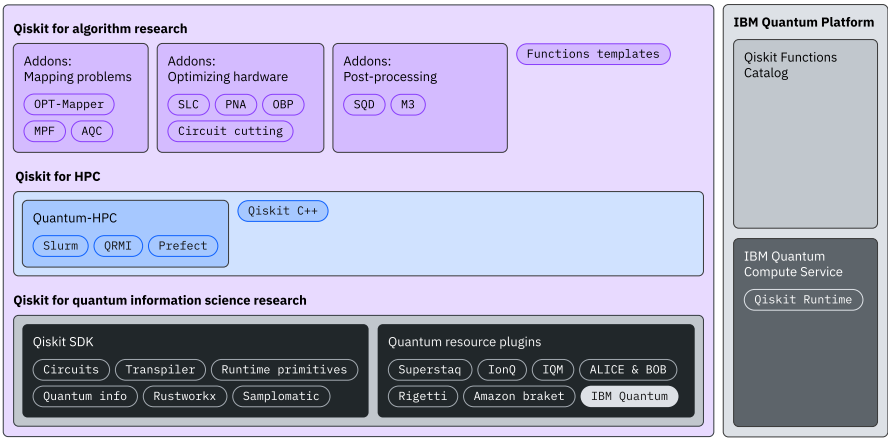


Figure 4.1: Qiskit software stack

The core components are *Qiskit SDK* and *Qiskit Runtime*. The first one is completely open source and allows the developer to define his circuit; the second one is a cloud-based service for executing quantum computations on IBM quantum computers.

4.2.1 Hello World

Following the IBM documentation[1] we can install the SDK and the Runtime with:

```
$ pip install qiskit matplotlib qiskit[visualization]
$ pip install qiskit-ibm-runtime
$ pip install qiskit-aer
```

Last command installs Aer, which is a high-performance simulator for quantum circuits written in Qiskit. Aer includes realistic noise models, and we will use it later to test our circuit.

Sometimes the Qiskit stack suffers from incompatibilities between the various software components that compose the environment. At the moment of writing, the latest packages seem to work without any problem. For our tests we will use `qiskit: 2.2.3`, `qiskit-ibm-runtime: 0.43.1` and `qiskit-aer: 0.17.2`.

If the setup is successful we are now able to run a small test to build a Bell state (two entangled qubits)[2]. The following code assembles the gates, shows the final circuit and uses a sampler to simulate on the CPU the result of 1024 runs of the program.

```
from qiskit import QuantumCircuit
from qiskit.primitives import StatevectorSampler

qc = QuantumCircuit(2)
qc.h(0)
qc.cx(0, 1)
qc.measure_all()

sampler = StatevectorSampler()
result = sampler.run([qc], shots=1024).result()
print(result[0].data.meas.get_counts())
qc.draw("mpl")
```

Listing 4.1: Building Bell state

4.2.2 Transpilation

Each Quantum Processing Unit (QPU) has a specific topology. We need to rewrite our quantum circuit in order to match the topology of the selected device on which we want to run our program. This phase of rewriting, followed by an optimization, is called transpilation.

Considering a fake hardware (so we do not need an API key) we can transpile the quantum circuit `qc`, from the code above, to match the topology

1 https://quantum.cloud.ibm.com/docs/en/guides/install-qiskit
2 We will obtain state $|\Psi^+\rangle$ (Equation 2.5)

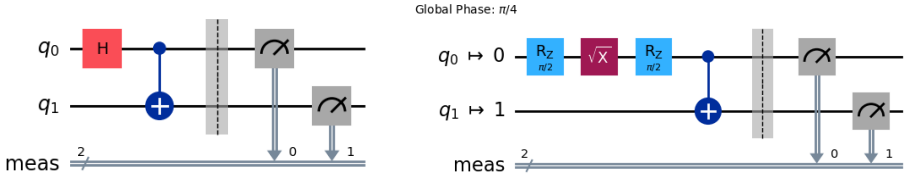
of a specific QPU. Listing 4.2 implement the transpilantion, and Figure 4.2 shows the circuits generated: (a) is the circuit whe have defined in Listing 4.1 and (b) is the transpiled version where the Hadamard gate is replaced to match the actual topology of the QPU.

```python
from qiskit_ibm_runtime.fake_provider import FakeWashingtonV2
from qiskit.transpiler import generate_preset_pass_manager

backend = FakeWashingtonV2()
pass_manager = generate_preset_pass_manager(backend=backend)

transpiled = pass_manager.run(qc)
transpiled.draw("mpl")
```

Listing 4.2: Transpilation



(a) Original circuit **(b)** Transpiled circuit

Figure 4.2: Transpilation example

4.2.3 Execution

To test our transpiled circuit we use Aer, which allows us to simulate also the noise of real quantum hardware. Listing 4.3 shows how to simulate the execution.

```python
from qiskit_aer.primitives import SamplerV2

sampler = SamplerV2.from_backend(backend)
job = sampler.run([transpiled], shots=1024)
result = job.result()
print(f"counts Bell circuit: {result[0].data.meas.get_counts()}")
```

Listing 4.3: Simulated execution

Simulated noise results

If we look at the results of the execution we can observe that some answers present non-entangled qubits; this is caused by the (simulated) noise of the quantum device. A typical output of the execution could be:

```
> counts Bell circuit: {'00': 504, '11': 503, '01': 10, '10': 7}
```

Where states `01` and `10` should not be present in an ideal execution with no errors.

4.3 D-WAVE OCEAN

To define an optimization problem that can be solved on a D-Wave quantum computer we use the Ocean software stack. Ocean also allows us to interact with D-Wave hardware, submit a problem, and simulate the execution on a classical CPU.

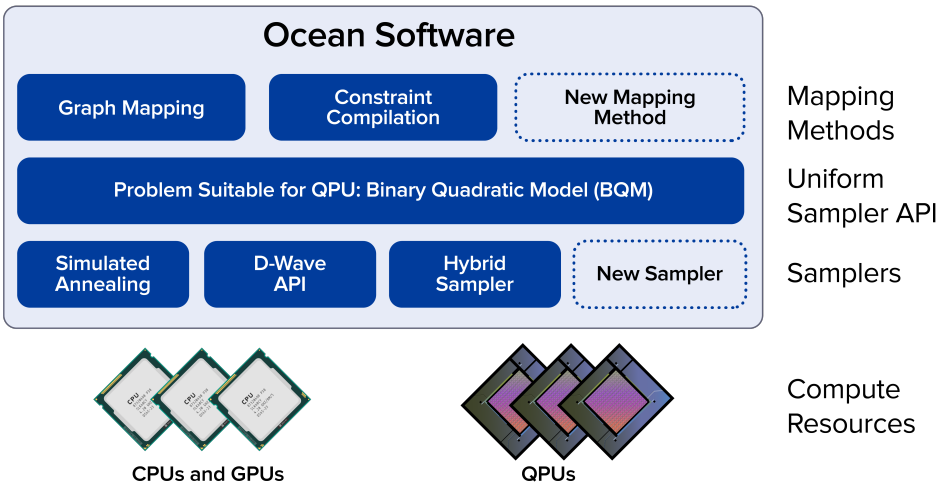


Figure 4.3: Ocean software stack

All tools that implement the steps needed to solve your problem on a CPU, a D-Wave quantum computer, or a quantum-classical hybrid solver can be installed with:

```
$ pip install dwave-ocean-sdk
```

After the installation, running the command `dwave setup` will start an interactive prompt that guides us through a full setup. During the setup it is also possible to add an API token or connect to the D-Wave account to import a key directly to use the quantum hardware.

4.3.1 Hello World

To present a simple optimization program we consider the minimum vertex cover (MVC) problem. Given a graph $G = (V, E)$, the problem asks to find a subset $V' \subseteq V$ such that, for each edge $\{u, v\} \in E$, at least one of u or v belongs to V', and the number of nodes in V' ($|V'|$) is the lowest possible.

MVC as Ising problem

The reduction from MVC to an Ising formulation is well known. The cost function that we want to minimize can be expressed by:

$$\text{cost} = \sum_{i=1}^{|V|} v_i + 2 \cdot \sum_{\{i,j\} \in E} (1 - v_i - v_j + v_i v_j)$$

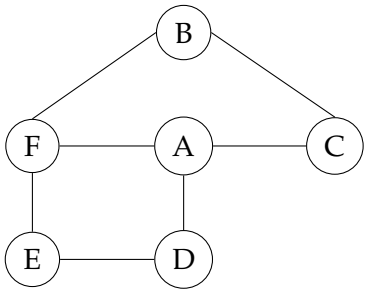
where $v_i \in \{-1, 1\}$ and $v_i = 1$ means that $v_i \in V'$, otherwise $v_i = -1$.

Like all problems in Ising form we can express the cost as a symmetric matrix, so our function becomes

$$\text{cost} = v^T \times \mathbf{M} \times v$$

where v is the vector containing the binary variables v_i.

The figure shows an example graph (4.4a) and the corresponding matrix (4.4b) expressing the cost function.



(a) Original graph

$$\mathbf{M} = \begin{pmatrix} 1 & 0 & 2 & 2 & 0 & 2 \\ 0 & 1 & 2 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

(b) Corresponding matrix

Figure 4.4: Ising formulation

Simulation of execution

The following code presents a possible implementation of the Ising model described above. We have defined two dictionaries to store the matrix coefficients. The last line of code finds ten possible answers to the problem using the simulated annealing function implemented by D-Wave.

```
from dwave.samplers import SimulatedAnnealingSampler
linear = {'A': 1, 'B': 1, 'C': 1, 'D': 1, 'E': 1, 'F': 1}
quadratic = {('B', 'C'): 2, ('B', 'F'): 2, ('C', 'A'): 2, ('D',
↪  'A'): 2, ('E', 'D'): 2, ('E', 'F'): 2, ('F', 'A'): 2}
sampler = SimulatedAnnealingSampler()
result = sampler.sample_ising(linear, quadratic, num_reads=10)
```

Listing 4.4: Ising example

If we print the results with `print(result.aggregate())` we can observe something similar to this:

```
  A  B  C  D  E  F energy num_oc.
0 -1 -1 +1 +1 -1 +1  -14.0       6
1 +1 +1 -1 -1 +1 -1  -14.0       4
['SPIN', 2 rows, 10 samples, 6 variables]
```

The two different results represent the two correct answers to our particular instance of the MVC problem.

4.4 PYQUBO AND QUBOVERT

In Listing 4.4 we have manually built the matrix representing the function that we want to minimize. It can be useful to have some tools that allow us to work at a higher level, defining cost functions like Equations 2.9 and 2.11 that we defined in the section about quantum annealing (Section 2.3.3).

Considering again the MVC problem, the objective function tends to minimize the number of nodes in our subset, while the penalty increases the cost if we leave out some edges. This interpretation allows us to transform the Ising model into the more familiar —from the point of view of a computer scientist— QUBO model, where all variables $x_i \in \{0, 1\}$. Let's see how PyQUBO and qubovert help us in this task.

4.4.1 PyQUBO

Reading from the documentation on the PyQUBO site[3], PyQUBO allows us to create QUBOs or Ising models from flexible mathematical expressions easily. Some of the features of PyQUBO are:

- Python based (C++ backend);
- Fully integrated with Ocean SDK;
- Automatic validation of constraints;
- Placeholder for parameter tuning.

We can install PyQUBO with `$ pip install pyqubo` and rewrite our MVC problem by defining the Hamiltonian that we want to minimize.

```
1  from pyqubo import Binary, Placeholder, Constraint
2  from dwave.samplers import SimulatedAnnealingSampler
3
4  A, B, C, D, E, F  = Binary('A'), Binary('B'), Binary('C'),
   ↪  Binary('D'), Binary('E'), Binary('F')
5
6  H_objective = (A + B + C + D + E + F)
7  H_penalty = Constraint(((1 - A - C + A*C) +\
8  (1 - A - D + A*D) +\
9  (1 - A - F + A*F) +\
10 (1 - B - C + B*C) +\
11 (1 - B - F + B*F) +\
12 (1 - D - E + D*E) +\
13 (1 - E - F + E*F)) ,label='cnstr0')
14
15 L = Placeholder('L')
16 H = H_objective + L*H_penalty
17 H_internal = H.compile()
18 bqm = H_internal.to_bqm(feed_dict={'L': 2})
19
20 sampler = SimulatedAnnealingSampler()
21 result = sampler.sample(bqm, num_reads=10)
```

Listing 4.5: Rewriting MVC with pyQUBO

Listing 4.5 presents a possible re-implementation of Listing 4.4, where we also see how PyQUBO interfaces with the Ocean SDK (line 17), and how to create (lines 14–16) and instantiate (line 17) a parametric Hamiltonian.

4.4.2 qubovert

As written in the documentation[4], qubovert is the one-stop package for formulating, simulating, and solving problems in boolean and spin form. Using our nomenclature, boolean and spin form are respectively QUBO and Ising form.

3 https://pyqubo.readthedocs.io/en/latest/
4 https://qubovert.readthedocs.io/en/latest/index.html

Qubovert allows us to define various types of optimization problems that can be solved by brute force, with qubovert's simulated annealing, or with D-Wave's solver. Models defined in qubovert are:

QUBO: Quadratic Unconstrained Boolean Optimization;

QUSO: Quadratic Unconstrained Spin Optimization (Ising model);

PUBO: Polynomial Unconstrained Boolean Optimization;

PUSO: Polynomial Unconstrained Spin Optimization;

PCBO: Polynomial Constrained Boolean Optimization;

PCSO: Polynomial Constrained Spin Optimization.

In addition to generic models, qubovert has a library of famous NP-complete problems mapped to QUBO and Ising forms.

```
1  from qubovert import boolean_var
2  from dwave.samplers import SimulatedAnnealingSampler
3
4  A, B, C, D, E, F  = boolean_var('A'), boolean_var('B'),
   ↪  boolean_var('C'), boolean_var('D'), boolean_var('E'),
   ↪  boolean_var('F')
5
6  model = A + B + C + D + E + F
7  model.add_constraint_OR(A, C, lam=2)
8  model.add_constraint_OR(A, D, lam=2)
9  model.add_constraint_OR(A, F, lam=2)
10 model.add_constraint_OR(B, C, lam=2)
11 model.add_constraint_OR(B, F, lam=2)
12 model.add_constraint_OR(D, E, lam=2)
13 model.add_constraint_OR(E, F, lam=2)
14
15 qubo = model.to_qubo()
16 dwave_qubo = qubo.Q
17
18 sampler = SimulatedAnnealingSampler()
19 result = sampler.sample_qubo(dwave_qubo, num_reads=10)
```

Listing 4.6: Rewriting MVC with qubovert

Listing 4.6 shows a possible implementation of the MVC problem using the tools provided by qubovert. Qubovert allows us to express our problem as a PCBO; we use this formulation to express constraints in a more natural way. In our example we ensure that each edge is covered simply by enforcing that at least one of the nodes linked by the edge is present in the solution. This constraint is repeated for each edge in the graph (lines 7–13). Constrain are encoded in the cost function using a penalty function multiplied for a scalar: *Lagrange multiplier*. To specify the Lagrange multiplier we use the keyword `lam`.

Qubovert, like PyQUBO, can interface with the Ocean SDK, transforming a PCBO problem into a QUBO problem (line 15) and then rewriting it in the format accepted by the D-Wave solver (or sampler).

4.5 CONCLUSION

In this chapter we have set up an environment to run our future experiments and tests. We have also shown some small examples to present the main characteristics and test the tools we will use in our work.

Following this setup allows anyone to recreate exactly the same configuration we use, avoiding (for what we know and test) incompatibilities between Python packages.

5 QA-PROLOG

QA-Prolog is a tool that allows one to write a program in a logic programming language and execute it on a quantum annealer. QA-Prolog also retrieves the results returned by the quantum annealer and presents them in a natural and comprehensible way.

In this chapter we introduce the project and give some pointers to other related works. We describe extensively the pipeline of transformations starting from a Prolog code and ending with a Hamiltonian $\mathbf{H}_f$, as we have described in Section 2.3.3. Next we present our contribution to the project, discussing the changes we have made to the original QA-Prolog code to restore compatibility with the modern framework used to interface with the D-Wave quantum annealer and to support the latest version of the library used in the project.

The chapter ends with an installation guide that ensures a working and reproducible environment to run experiments, both for this chapter and for the following ones.

5.1 THE PROJECT

QA-Prolog is a project developed by Scott Pakin[1] in 2017 – 2019. It starts from the question: "Can one express constraint logic programming in the form accepted by quantum-annealing hardware?" [1].

The hope is that even if today we live in the *NISQ*[2] era of quantum computing, quantum annealers are more easily scalable than quantum gate-based computers [23], and QA-Prolog could improve Prolog program execution by replacing backtracking with fully parallel annealing into a solution state [24].

5.1.1 Reason

As we have shown in Chapter 2, programming a quantum computer is not an easy task. We express our algorithm in a very low-level way.

On a quantum annealer we have to define a cost function, without constraints (which must be transformed into a penalty function). Even if there are libraries that allow us to express these functions in an easier way, we need at least to find a QUBO representation of our problem.

Quantum gates and FPGAs

Even worse is the situation on quantum gate-based computers. The programmer has to build a quantum circuit gate by gate, an approach similar to what is done with FPGAs (Field Programmable Gate Arrays) [25]. We can indeed see a strong analogy summed up in Table 5.1.

1 Los Alamos National Laboratory: pakin@lanl.gov.
2 Noisy intermediate-scale quantum computing.

FPGA	Quantum gates computer
programmable logic blocks which implement logic functions	quantum gates
programmable routing that connects logic functions	possibility to define order of gates
I/O blocks connected to logic	input *qubits* and output *qubits* that can be measured

Table 5.1: Parallelism between FPGA and quantum gate

FPGAs are components that the majority of computer scientists are not used to and are probably out of the interest of a programmer. In the same way, the hardness of programming a quantum computer could be a significant deterrent to attracting new researchers in the field.

Today some "high-level gates" are available that wrap multiple low-level gates into useful patterns. There also exist, both for quantum gate-based computers and quantum annealers, some templates of well-known problems that need only fine-tuning to be useful for a specific problem.

Despite these sorts of abstractions, programming a quantum computer is difficult and very close to machine language.

5.1.2 Prolog

We can see QA-Prolog as a compiler from Prolog to $\mathbf{H}_f$, where the ground state of $\mathbf{H}_f$ is the solution of our Prolog program. Before starting with the compilation process, it is useful to understand the main characteristics of Prolog, because it is not an imperative programming language like C or Java, but a *declarative* one like SQL, moreover, Prolog is a *logic programming language*. This means that the core of programming is not to tell the computer what to do, but to tell it what is true and ask it to try to draw conclusions [26].

Logic programming language

Discussing in detail the characteristics of Prolog and logic programming languages is out of scope of this work. More information about Prolog can be found in *Programming in PROLOG* [26] and *Learn Prolog Now!* [27]. For the ones interested in logic programming language *Logic for problem solving*[28] and *Introduction to logic programming*[29] are recommended.

In Prolog we do not specify step by step an algorithm that resolves our problem, we describe instead the formal relationship between the objects in our problem and which relations have to be true in our solution [26].

Elements of a Prolog Program

Programming in Prolog consists of:

- listing *facts* about objects and relationships between objects;
- specifying *rules* to derive new facts from the ones already asserted;
- asking questions (*queries*) about objects and their relationships.

From these characteristics we can understand what "declarative" means: the program is a list of statements about our problem (our domain of interest); the relation between a Prolog program and an ontology is very strict.

In Prolog we encode a KB[3] made of facts and rules; in Capter 6 we can see a complete example of rewriting from an OWL ontology into a Prolog KB.

ARITHMETIC PREDICATES: To better understand some QA-Prolog features that differ from basic Prolog we explain here how to assert equality between arguments when referring to integer. Predicates offered by Prolog are:

- `+Expr1 =:= +Expr2`: true if expression `Expr1` evaluates to a number equal to `Expr2`;
- `-Number is +Expr`: true when `Number` is the value to which `Expr` evaluates.

Arguments mode indicator

Where signs before arguments are *arguments mode indicators*. Sign `+` specifies that the following argument must be instantiated, `-` indicates that the argument is a output [30]. This mean that we cannot have variables when using `=:=` predicate because everything must be known at the time of evaluation. Using `is` the only variable can be the value of `Number`; `Expr`, on the other hand, has to be known.

EXAMPLE: Now we present a basic Prolog program in order to show the syntax and the usage of queries. We encode a KB about a party, we describe two people, Yolanda and Mia, asserting what they are doing, and giving some rules to derive other information about our small scenario.

```
1 sings(mia).
2 listens2Music(yolanda).
3 party.
4
5 dance(yolanda):- listens2Music(yolanda).
6 happy(yolanda):- dance(yolanda).
7 happy(mia):- sings(mia).
8
9 smile(X) :- happy(X).
```

Listing 5.1: Basic KB

Interpretation of Prolog lines

In Listing 5.1, adapted from [27], lines 1 to 3 are facts: we are asserting that Yolanda is listening to music, Mia is singing, and there is a party. The other lines are rules: we can identify rules by the `:-` sign that divides the *head* of the rule on the left, from the *body* on the right. The head of a rule is true if the body is true.

For example, the rule at line 5 can be read as: "If Yolanda is listening to music, then Yolanda is dancing". Line 9 shows the usage of a variable: variables start with an uppercase letter and are placeholders for information. We can read this rule as "If someone is happy, they smile".

We can query our KB, asking for example if Yolanda is happy. In SWI-Prolog [31] we can interact with the interpreter, and the query we evaluate is `?- happy(yolanda).` (the full stop is part of the syntax and tells the interpreter that the query is complete). Prolog will answer `yes.`; this is because Yolanda

3 Knowledge Base

is listening to music, and if she is listening to music she dances, and if she dances she is happy.

By analogous reasoning it should be clear why the result of `?- smile(X).` is `X = mia; X = yolanda.`, where `;` means logical disjunction: *or*.

5.1.3 Features of QA-Prolog

QA-Prolog does not support all the features of Prolog, but enough to make basic logic programming possible [1].

QA-Prolog supports atoms and positive integers but not floating-point numbers, strings, or lists. It supports arithmetic and relational operations, and rules can reference other rules but not recursively. QA-Prolog supports unification, backtracking, and predicates comprising multiple clauses [1].

QA-Prolog also supports some features not present in the basic version of Prolog. In particular, operations can be performed on variables even before they are ground [1]; this means that QA-Prolog is more powerful in manipulating free variables. We can understand this feature looking back to Paragraph 5.1.2 where we have shown that aritmetica expression can have a free variable at most (`is` predicate) on the left operand.

Hints of reversible computation

In QA-Prolog we can have variables on the left and on the right of predicate[4], that way we can pass the input to the formula to obtain the result, but also assign a value to the result to retreive the input. We can already see that QA-Prolog allows a reversible computation: from input to output but also from output to input.

5.1.4 Related works

There are some other attempts to develop a high-level programming language for quantum computers, both for the quantum gate model and for quantum annealers.

Quantum Prolog [35] demonstrates that one can express the equivalent of a pure version of Prolog over finite relations in terms of a model of discrete quantum computing. This work targets quantum gate computers and focuses on the mathematical equivalence of relational programming and discrete quantum computing over the field of Booleans [1]. Quantum Prolog, however, remains a theoretical work that has the goal of proves some mathematical properties, there is no implementation and therefore is not suitable to run experiments.

C to D-Wave [36] reuses the pipeline we will discuss, replacing the starting step. The paper addresses the difficulty of programming quantum annealers by presenting a compilation framework that compiles a subset of C code into quantum machine instructions to be executed on a quantum annealer.

4 In Prolog is possible to obtain the same result with library like CLP(FD) [32, 33] and CLP($\mathbb{Z}$) [34]

5.2 PIPELINE

We are now ready to describe the pipeline of transformations that brings us from a Prolog program to a Hamiltonian $\mathbf{H}_f$.

The chain of transformations is shown in Figure 5.1, where the last step (in orange) is the quantum annealer capable of finding the ground state of $\mathbf{H}_f$. In purple we can see the various file formats throughout the pipeline, and in yellow the software that performs the rewriting. Most of this software is made ad hoc for the QA-Prolog pipeline on the other hand, we will see that Yosys is a tool used in digital circuits design.

From a high-level point of view, the pipeline rewrites an initial KB expressed in Prolog into different objects. The logical meaning of the entities we build during the pipeline is:

1. Prolog program (KB);
2. High-level digital circuit in Verilog;
3. Low-level digital circuit in EDIF format;
4. Symbolic Hamiltonian in qmasm formalism;
5. Physical Hamiltonian.
6. Execution un a *quantum processing unit*

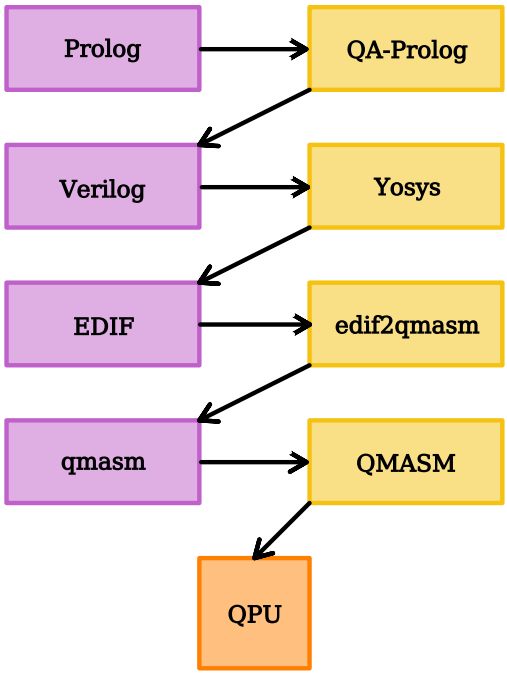


Figure 5.1: QA-Prolog pipeline

Let us start analyzing the pipeline from the last step, the one nearest to the QPU.

5.2.1 QMASM

QMASM is a *quantum macro assembler* [37]; it processes a symbolic Hamiltonian and assembles a physical Hamiltonian that can be embedded on a D-Wave quantum annealer. It was developed in Python by Scott Pakin with the goal of filling a gap in tools for the creation of D-Wave programs. QMASM is an abstraction layer that allows the programmer not to care about manually setting specific point weights and coupler strengths on the physical topology.

This software can be considered an assembler in the sense that it maps a symbolic representation of the operations (assembly language) to the machine language. QMASM is not only an assembler, but extends its functionality by including macros: parameterized named blocks of assembly language that a program can instantiate multiple times [37].

QMASM as macro assembler

FEATURE: QMASM provsingides a number of features to simplify low-level D-Wave programming [37]. Moreover, we can also use QMASM to assemble a physical Hamiltonian that we can later manipulate or solve using classical methods or other quantum systems.

Some of the most useful and interesting features of QMASM are:

- *qubits* are referenced symbolically, not numerically, both in the source code and when QMASM reports execution results;
- *qubits* can be pinned to `true` or `false`;
- *qubit* patterns can be encapsulated into macros and instantiated repeatedly;
- groups of macros can be encapsulated into libraries and reused across multiple programs;
- QMASM can automatically exclude from the results the solutions known to be incorrect and shows only "interesting" *qubits*.

Thanks to thiings set of features, QMASM is already a useful abstraction layer that simplifies the development of programs that target an annealer, classical or quantum.

Example of satisfiability problem

EXAMPLE: Let us consider an example that shows the potential of QMASM and that is also useful for the following steps of the pipeline. The satisfiability problem is well known to be an NP-complete problem [38], so it is a good candidate for our example. We take into account the simple formula:

$$y = x_1 \wedge \neg(x_2 \vee x_3) \tag{5.1}$$

QMASM allows us to define a macro for every logical operator needed to represent the formula and then assemble the final formula by calling these macros. In Listings 5.2 we call `A` and `B` the input *qubits* and `Y` the output *qubit*. Weights are specified as an Insing problem (Section 2.3.3), which is the default format for QMASM source files[5].

There are multiple interesting details about these macros:

- the symbol `#` identifies a comment line not processed;
- the symbol `$` is used to tag a *qubit* as ancillary: unless explicitly requested by the programmer, the intermediate results are not reported in the solutions;
- thanks to the directive `!assert` we can inform QMASM about constraints on the solution. This directive does not change the weights, but allows the programmer to exclude from the solutions those that are surely incorrect.

Compared with what we have done in Listing 4.4, defining weights in QMASM is much easier and the result is more readable;

```
# Y = A AND B
!begin_macro AND
    !assert $Y=$A&$B
    $A -0.5
    $B -0.5
    $Y  1

    $A $B  0.5
    $A $Y -1
    $B $Y -1
!end_macro AND
```

(a) AND gate

```
# Y = NOT A
!begin_macro NOT
    !assert $Y=!$A
    $A $Y 1.0
!end_macro NOT
```

(b) NOT gate

```
# Y = A OR B
!begin_macro OR
    !assert $Y=$A|$B
    $A  0.5
    $B  0.5
    $Y -1

    $A $B  0.5
    $A $Y -1
    $B $Y -1
!end_macro OR
```

(c) OR gate

Listing 5.2: T-Box definition

It is possible to verify, for each macro in Listings 5.2, that given a configuration of the input *qubits*, the value of `Y` that minimizes the energy corresponds to the output of the logic gate we are modeling. For example, the explicit polynomial that describe AND operator (where we have removed the $ symbol for simplicity of reading) is:

Macro interpretation as a polynomial

$$\mathbf{H} = -\frac{1}{2} \cdot (A + B) + Y + \frac{1}{2} \cdot (AB) - (AY + BY)$$

Where $A, B, Y \in \{-1, 1\}$ because we are describing an Ising problem.

Listings 5.2, also, show the usage of directive `!assert` (line 3 of each listing): considering the OR gate, we impose that in a valid solution $Y = A \vee B$, solutions that not verify this constraint are automatically discarded by QMASM.

We can save these macros in a file named `gates.qmasm` and use it to solve our problem. To compute the formula $y = x_1 \wedge \neg(x_2 \vee x_3)$ we will use some intermediate results: we start with $x_4 = (x_2 \vee x_3)$, then we apply the negation $x_5 = \neg x_4$, and lastly we compute the result as $y = x_1 \wedge x_5$.

Assemble macros

```
# Solve circuit-satisfiability problem.

!include <gates>

!use_macro OR x2_or_x3
  x2_or_x3.$A = x2
  x2_or_x3.$B = x3
  x2_or_x3.$Y = $x4

!use_macro NOT not_x4
  not_x4.$A = $x4
  not_x4.$Y = $x5

!use_macro AND x1_and_x5
  x1_and_x5.$A = x1
  x1_and_x5.$B = $x5
  x1_and_x5.$Y = y
```

Listing 5.3: Circuit satisfiability

5 as specified in https://github.com/lanl/qmasm/wiki/File-format.

The QMASM code implementing this procedure is reported in Listing 5.3. This example shows how to use macros: we instantiate a macro and give it a name with `!use_macro <macro_name> <instance_name>` (e.g. line 5), and then instantiate the *qubits* defined in the macro with our actual *qubits*. For example, considering the `!use_macro OR` at line 5, we can see that we use x_2 and x_3 as input and an ancillary *qubit* $\$x_4$ as output. Another detail to point out is the directive `!include` (line 3), which imports all gates used in this source file.

Run a QMASM "program"

Finally, we can query the quantum annealer (or in this case a classical solver) to find a solution for our satisfiability problem. To do so we pin the output variable `y` to be sure that in the solution its value will be `true`. We can query the solver with:

```
qmasm --run --pin="y := true" --solver="sim_anneal" circ_sat.qmasm
```

where `circ_sat.qmasm` is the file name of our source code.

QMASM interfaces directly with the Ocean framework and can query one of the solvers made available by D-Wave. Retrieving the solutions results in a call to Ocean's library functions very similar to the one shown in Listing 4.4 on line 5. QMASM also offers the possibility to set the annealing time and the number of reads directly from the command line; in this case, the default values were used, which are the default annealing time stetted by D-Wave for the specific solver and 1000 samples, respectively.

Results are reported in Listing 5.4. Here we can see that the solver has correctly found the solution, but only 642 times out of the default 1000 samples. This is caused by the stochastic nature of annealing, simulated or quantum. QMASM automatically removed the solutions that do not respect the `!assert` directive or do not have the minimum energy.

```
# x1 -->   12
# x2 -->    3
# x3 -->    4
# y --> [True]
Solution #1 (energy = -20.0000, tally = 642):

Variable  Value
--------  -----
x1        True
x2        False
x3        False
y         True
```

Listing 5.4: Circuit satisfiability results

QMASM already offers some powerful abstractions to work with quantum annealers. Now we add two additional layers that allow a programming style more similar to the paradigms computer scientists are used to, while preserving strong control over variable dimensions and therefore over the number of *qubits* used.

5.2.2 Yosys and edif2qmasm

These steps of the pipeline take as input a *Verilog HDL* (Hardware Description Language) program and transform it into a symbolic Hamiltonian.

We aggregate two steps because Yosys is not a tool developed for this pipeline and is used mostly to optimize the intermediate results of the transformations. Also, Yosys and edif2qmasm work on the same logical entity: a digital circuit that, in these steps of the pipeline, is converted into a symbolic Hamiltonian.

Verilog

Verilog is a hardware description language that offers different levels of design abstraction. The highest level is *behavioural* and uses programming constructs such as assignments, conditionals, and while-loops. The lowest level is a connection of gates (*netlists*) [39].

Hardware description languages are not something that the majority of programmers are familiar with, but they offer some advantages when targeting a quantum annealer [40]:

- they provide precise control over bit widths in order not to waste any *qubit*;
- they can be compiled to a small set of primitives: the gates we can define with QMASM.

In these steps of the pipeline we start using behavioural constructs, then we automatically generate a netlist as the output of *synthesisers*.

The synthesiser used is Yosys [41], a free and open-source software for Verilog HDL synthesis.

The two most useful features of Yosys are the possibility of specifying a *cell library*, the set of gates used to synthesize the netlist, and the use of the external tool *Berkeley ABC* [42] (incorporated in Yosys), providing additional code optimizations.

Gates selection

Yosys lowers a Verilog program to an EDIF netlist that uses only a specified set of gates; then edif2qmasm lowers the netlist into a symbolic Hamiltonian. This Hamiltonian uses a set of macros defining the energy for each type of gate Yosys can use[6]. What we end up with is a symbolic Hamiltonian that can be used as input for QMASM.

EXAMPLE: Considering again the simple formula $y = x_1 \wedge \neg(x_2 \vee x_3)$ (Equation 5.1).

Figure 5.2 shows a possible Verilog implementation of the formula (a), the netlist Yosys has produced from the Verilog code (b), and, only for clarification purposes, the corresponding digital circuit implemented with logic gates (c).

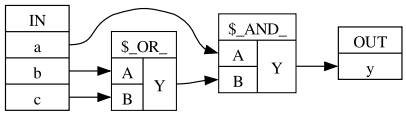
Now we can feed the obtained netlist to edif2qmasm to obtain the code reported in Listing 5.5.

Comparing the automatically generated Listing 5.5 and the manually written 5.3, we can observe some differences:

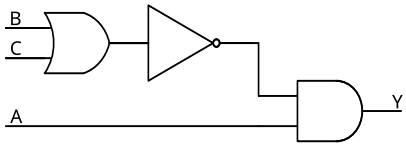
6 The actual set of gates extends the logical ports *and*, *or*, and *not* and is available at: https://github.com/lanl/edif2qmasm/blob/master/stdcell.qmasm.

```
1 module sat (A, B, C, Y);
2   input A, B, C;
3   output Y;
4
5   assign  Y = A & ~(B | C);
6 endmodule
```

(a) Verilog Code



(b) Netlist synthesized



(c) Corresponding circuit

Figure 5.2: Yosys processing

```
1  !include <stdcell>
2
3  !begin_macro sat
4    !use_macro AND $id00006
5    !use_macro NOT $id00004
6    !use_macro OR $id00005
7    $id00004.A = $id00005.Y  # $or$sat.v:5$1_Y
8    $id00005.A = b
9    $id00005.B = c
10   $id00006.A = a
11   $id00006.B = $id00004.Y  # $not$sat.v:5$2_Y
12   $id00006.Y = y
13 !end_macro sat
14
15 !use_macro sat sat
```

Listing 5.5: Simbolic Hamiltonian

- in Listing 5.5 we define the new macro `sat`: each module in the Prolog program is converted into a macro;
- the generated code is a little less readable but more compact. We do not need to generate new ancillary variables, but instead use directly as input for macros the output of other macros.

Even if it is less readable, we can comment on the code in Listing 5.5. At lines 4, 5, and 6 we declare that we need the macros `AND`, `NOT`, and `OR`, giving to each macro a name like `$id<number>`[7]. At line 7 we assign as input `A` of the macro `NOT` the output `Y` of the macro `OR`. Given these hints, the rest of the code should be clear.

5.2.3 QA-Prolog

This is the last step of the pipeline, the one that provides the maximum level of abstraction from the hardware that finds the solution to our problem.

QA-Prolog, like edif2qmasm, is written in GO [1, 40] and compiles a Prolog program to a Verilog one; each fact or rule is converted into a module.

7 The symbol `$` tells QMASM that we are not interested in the value of *qubits* inside the macro.

Once QA-Prolog has compiled the Prolog source code, including the query that can be specified on the command line, into Verilog, QA-Prolog invokes Yosys, edif2qmasm, and QMASM, as illustrated in Figure 5.1 [1].

Results in human-readable format

QMASM executes the user's program on a D-Wave system or on a simulator and reports the value of each symbol appearing in the QMASM source file. QA-Prolog maps these lists of Booleans back to integers and named atoms, associates those values with variables named in the user's query, and reports all variables and their values to the user just as a typical Prolog environment would [1].

EXAMPLE 1: The first example we present shows a small KB that can be summarized as "the enemy of my enemy is my friend".

```
1 hates(alice, bob).
2 hates(bob, charlie).
3
4 enemy(X, Y) :- hates(X, Y).
5 enemy(X, Y) :- hates(Y, X).
6
7 friend(X, Y) :-
8   enemy(X, Z),
9   enemy(Z, Y).
```

Listing 5.6: KB enemy of my enemy is my friend

The code is reported in Listing 5.6. Thanks to the declarative nature of Prolog we can immediately understand the meaning of our KB: we describe relations between three people, assert that if a person hates another one they are enemies (lines 4 and 5), and that if two people share an enemy they are friends (lines 7, 8, and 9).

We can now query the KB to find if there are two people who are friends with:

```
QA-Prolog --qmasm-args="--solver=tabu" --query='friends(P1, P2).'
↪  --work-dir="~/work" friends.pl
```

In this command we can see that QA-Prolog allows us to query the KB with the exact same syntax we would use to interact with a Prolog interpreter. As expected, QA-Prolog *binds* the variables `P1` and `P2` to Alice and Charlie.

Another detail to point out is the parameter `--work-dir="<path>"`, which specifies the folder in which QA-Prolog outputs all files and the solver: we use a *tabu search algorithm* because the simulated annealing implemented in Ocean is not capable of finding a solution even in this small scenario[8]. We say small because, from the study reported in [43], we can observe that real-world ontologies define a number of classes, individuals, and properties[9] that can be on the order of hundreds of thousands.

By inspecting the working directory, we can check the output of each step of the pipeline. In Appendix A.1 is shown the rewritten of a Prolog KB into a Verilog program: it is possible to observe in detail how each fact or rule has been transformed. If we inspect the working directory we can see the

8 More considerations about that in Chapter 8.
9 That we will transform in facts and rules in Chapter 6.

output of every step of the pipeline; in particular, to see how exactly the KB is rewritten into a Verilog program, in Appendix A.1 there is the result of compilation, and it is possible to observe in detail how each fact or rule has been transformed.

Satisfiability problem in QA-Prolog

EXAMPLE 2: The second example concludes our transformation of the satisfiability problem (Equation 5.1); unfortunately, this is not very didactic because the Prolog program suitable for QA-Prolog processing introduces a consistent amount of overhead. This is due to the fact that in Prolog we can simply import `library(clpb)` [44] and then formulate the query `?- sat(A * (~(B + C))).`; to use QA-Prolog, instead, we have to implement logical operators by defining their truth tables, then assemble the operators in our logical formula. This leads to the overhead. Listing 5.7 shows the implementation we use in our test.

```
logicAnd(false, false, false).
logicAnd(false, true,  false).
logicAnd(true,  false, false).
logicAnd(true,  true,  true).

logicNot(false, true).
logicNot(true,  false).

logicOr(false, false, false).
logicOr(false, true,  true).
logicOr(true,  false, true).
logicOr(true,  true,  true).

satFormula(A, B, C, Y) :-
  logicOr(B, C, X),
  logicNot(X, Z),
  logicAnd(A, Z, Y).
```

Listing 5.7: Satisfiability in Prolog

We can now run QA-Prolog with the query `sat(A, B, C, true)` to find the correct assignment of Boolean variables that satisfy our logic formula. Again, we are forced to use the tabu search algorithm. If we now explore the working directory of QA-Prolog we can see that the symbolic matrix generated by the pipeline is a lot more complex that the one reported in Listing 5.5, we can interpret that as the cost of abstraction.

5.2.4 Overview

We have described a pipeline of rewritings that, layer after layer, makes it possible to abstract away from the hardware that actually solves our problem, moving toward a language, and thus a way of reasoning, that is at a higher level. We speak of rewritings because the object we are manipulating remains logically unchanged, just as it is essential that the solutions we are searching for remain unchanged; the problem and its solutions are simply expressed in different languages as one moves down the pipeline.

Chain of rewritings

The advantages of abstraction are essentially the same as those obtained when moving from assembly language to a high-level language: greater ease of expression, improved readability, and the possibility of working with already familiar paradigms. The drawbacks are the overheads that naturally arise during the translation process. QA-Prolog is still a prototype, and if today compilers produce better machine code than a programmer could write manually, it is because significant effort and many years of research have been invested to achieve excellent results. The hope is that QA-Prolog and related works represent a first step in the same direction within the field of quantum computing.

5.3 UPDATE TO THE PROJECT

In this section we briefly describe the updates we have made to the project in order to restore compatibility with the Ocean framework and with the other libraries used. We also provide a small guide to install the tools to ensure that all packages work properly and our experiments are reproducible.

5.3.1 Restoring QMASM

The last commit to QA-Prolog was made in 2019, and the last one to QMASM in 2021. Since then, packages and libraries have changed and the compatibility with the QA-Prolog pipeline has broken.

Obsolete code

The most fragile component of the pipeline is QMASM, because it is the one that interacts with external frameworks that are likely to change. In particular, the development of Ocean is very active, and some functions used in QMASM have now been removed, deprecated, or moved to other packages.

The incompatibilities encountered between imports used in QMASM and external libraries are:

- in the `dwave.cloud` and `hybrid` libraries;
- in the libraries defining D-Wave samplers (classical solvers);
- in the `scipy.stats` library.

The documentation available for Ocean[10] and for scipy[11] helped us resolve these incompatibilities: most of the time the solution consisted simply in correcting the name of the library.

Another small bug, caused by a different name of a solver in the command line helper and in the actual code, was fixed by restoring the correspondence.

5.3.2 Fixing interaction edif2qmasm-QMASM

During the execution of the complete pipeline we encountered an error caused by undefined macros in the QMASM source file.

10 Available at https://docs.dwavequantum.com/en/latest/index.html.
11 Available at https://docs.scipy.org/doc//scipy/index.html.

```
// Define hates(atom, atom).
module \hates/2 (A, B, Valid);
  ...
endmodule
```

(a) Verilog Code

```
# hates/2
!begin_macro id00011
  ...
!end_macro id00011
```

(b) Macro definitions

```
# enemies/2
!begin_macro id00010
  !use_macro hates/2 $id00032  # hates_PWYwG/2
  ...
!end_macro id00010
```

(c) Macro usage

Listing 5.8: Undefined macros error

The error can be seen in Listings 5.8: edif2qmasm rewrites the netlist generated from (a) into the macro (b); here `hates/2` is just a comment, the actual name of the macro is `id00011` (specified at line 2). In (c), however, we can see that the macro previously defined is called symbolically and not with its actual name (lines 3). This obviously produces an error: the name `hates/2` in the QMASM file has no meaning, it is only a comment.

```
with open(file, 'r') as file_in:
  first = file_in.readline()
  second = file_in.readline()
  while(second_row != ""):
    if first.startswith("#") and second.startswith("!begin_macro"):
      self.name[first[2:-1]] = second[len("!begin_macro "):-1]
    first_row = second_row
    second_row = file_in.readline()

with open(file, 'r') as file_in:
  doc = file_in.read()
  lines = doc.splitlines()
  for line in lines:
    for word in line.split():
      if word in self.name.keys():
        line = line.replace(word, self.name[word])
      self.new_lines.append(line)
```

Listing 5.9: Preprocessor's core

Fixing macros' names

To solve the problem we added a preprocessing step before parsing with QMASM. During the preprocessing all symbolic names are substituted with the actual name of the corresponding macro.

The core code of the preprocessor is reported in Listing 5.9. The program scans the source file two times: during the first reading a Python dictionary *symbolic_name-actual_name* is built, then, during the second reading, all instances of the symbolic name are replaced with the actual name.

Now all components should work properly, and we can install all the software needed and run some experiments.

5.3.3 Installation guide

In order to have a working environment where we can run our experiments, we need to install all the software required by the QA-Prolog pipeline. In Chapter 4 we have set up a Python environment with all the essential tools; to start the installation of the QA-Prolog pipeline we need at least the Ocean SDK installed as described in Section 4.3.

The first component we need is QMASM: we can install the updated and fixed version from `https://github.com/DavideCamino/qmasm.git`. The installation procedure consists of the following three commands:

```
$ git clone https://github.com/DavideCamino/qmasm.git
$ cd qmasm
$ python setup.py install
```

Next we need Yosys and GO: Yosys is part of the pipeline, and GO allows us to compile edif2qmasm and QA-Prolog. Both Yosys and GO are available from their official website `https://yosyshq.net/` and `https://go.dev/`, also on most of GNU/Linux distributions Yosys and GO can be installed directly from the package manager of the distribution.

The `go install` command installs Go executables in the default directory `$HOME/go/bin`. It is useful to add the bin subdirectory to `PATH`. This can be done by editing the `.bashrc` file (or the corresponding one for the specific shell), adding:

```
PATH=$PATH:$HOME/go/bin
export PATH
```

Finally we can install edif2qmasm and QA-Prolog with:

```
$ go install github.com/lanl/edif2qmasm@latest
$ go install github.com/lanl/QA-Prolog@latest
```

5.4 CONCLUSION

In this chapter, we presented QA-Prolog, a rewriting pipeline that enables the transformation of a Verilog program into a Hamiltonian whose ground state represents the solution of the original program. We discussed the various steps of the pipeline in detail, also showing how some parts were modified to make them compatible again with current frameworks. Finally, we described how to install a working version of the pipeline.

From our discussion, it emerges that QA-Prolog is a modular software stack in which it is possible to replace a component of the pipeline with another to modify the result; for example, we presented work that starts from C instead of Prolog.

This work has the potential to provide a foundation for many other experiments, that can rely on an already implemented and functioning infrastructure, in order to develop new extensions, both upstream and downstream, of the pipeline.

Part III

EXPERIMENTS

6 A QUANTUM ONTOLOGY

In this chapter, we present a complete example of rewriting, starting from an ontology expressed in OWL (Section 3.1.3), and arriving at a Prolog formulation suitable for the pipeline presented in the previous chapter.

Following the literature, we begin with the translation of the ontology into Prolog, discussing how to translate concepts such as classes, subclasses, individuals, and relations. We then focus on the incompatibility of this first version with QA-Prolog, highlight the reasons for this incompatibility, and propose possible solutions.

We present some examples of queries to demonstrate the equivalence of the two transformations. Finally we provide observations on the final result obtained after the transformations performed by QA-Prolog.

6.1 STARTING ONTOLOGY

The domain of interest of our ontology is the animal kingdom; this domain serves as a pretext for a deeper description than the one shown in the family tree example (Section 3.2.1). We will see how to describe animal species, their feeding habits, and the relationships between individuals.

6.1.1 Ontology structure

In this ontology, we describe a small part of the animal kingdom. The more complex class hierarchy represents the taxonomy of animals: it characterizes living beings with increasing precision until identifying their species.

The other classes focus, instead, on feeding habits of animals. We distinguish between prey and predators, and between carnivorous and herbivorous animals (in this small example, for simplicity, we ignore omnivores. The reason will be clear in the next section).

Finally, we consider relationships between animals. We define two types of relationships: parental relation that connect two animals in parent–son roles, and relation related to feeding habits that link two animals in the roles of predator and prey.

The T-Box of our ontology can be seen in Figure 6.1, we define also some individuals and their relations, to populate our ontology[1].

1 The code for this ontology is available at https://github.com/DavideCamino/Thesis/blob/main/Code/Ontology/animals.rdf.

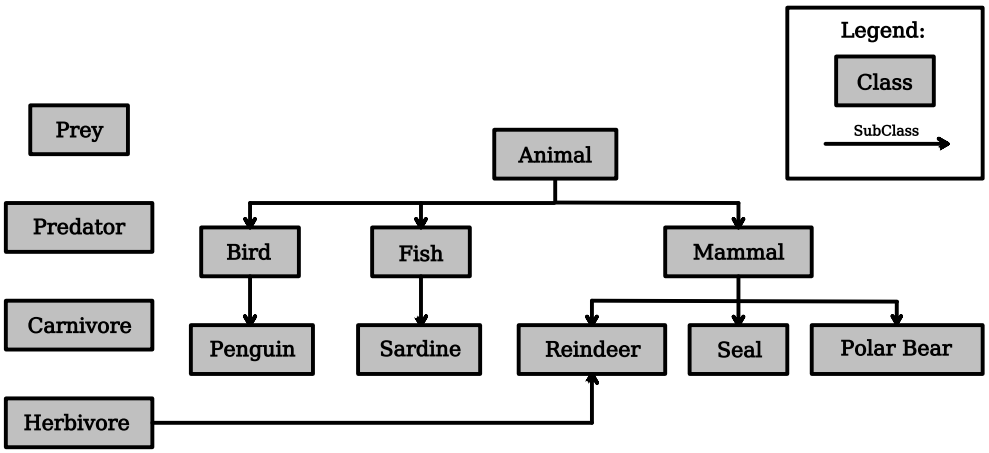


Figure 6.1: Ontology T-Box

6.1.2 Ontology constraints

In this section, we present the constraints on classes and individuals represented in our ontology. These constraints allow us to make inference on the ontology and derive new information from what is already present. Thanks to these constraints we can determine if all the information contained in the ontology is consistent with the theoretical model we are describing.

Ontology constrains

Constrains defined in animal ontology are:

- Constrain on classes:
  - carnivore and herbivore are disjoint classes;
- Constrain on individual:
  - If animal A hunts animal B, A is a predator and carnivore and B is a prey.

From defined constraints we can see that a prey and predator are not disjoint classes and being a prey does not automatically mean that the animal is a herbivore. This is coherent with the animal kingdom where a certain animal could be both a prey and a predator.

It should also be clear why we have omitted the omnivores from our description: this simplification allows us to describe herbivore and carnivore as disjoint and to infer that an animal that hunt is automatically a carnivore.

6.1.3 Ontology contents

We now examine some facts represented in the ontology; these will serve as examples to show how integrity checking of the information collected in the KB, and the reasoning processes work.

Let us begin by considering the classes. As shown in the T-Box of Figure 6.1, the animals we can describe are: sardines, penguins, seals, polar bears, and reindeers. From the T-Box we also see that we have only specified reindeer as herbivorous; feeding habits of other spices can be inferred with the hunt relationship.

We now examine some facts about the individuals that populate our KB:

- Beary is a polar bear and hunts Sealy and Penguy;
- Sealy is a seal and hunts Penguy;
- Penguy is a penguin and hunts Sardy;
- Sardy is a sardine;
- Reiny is a reindeer and hunts Sardy;
- Reiny_b and Reiny_c are reindeers, Reiny_b is son of Reiny and Reiny_c is son of Reiny_b.

From this information, we can deduce that Beary and Reiny are predators, Sealy and Penguy are both predators and prey; and Sardy is a prey.

Inconsistency source

CONTRADICTION: A contradiction has also been deliberately introduced into the ontology. From the fact that Reiny hunts, it follows that she is a carnivorous animal; however, her species is labeled as herbivorous. Reiny is therefore both herbivorous and carnivorous. Moreover, herbivores and carnivores are disjoint classes. This makes the KB inconsistent.

If we used a reasoner to verify the consistency of the KB, for example HermiT [22], we would obtain an error message informing us that the ontology is inconsistent and that the reasoner's results cannot therefore be considered reliable.

The reasoner is also able to provide an explanation of the inconsistency (shown in Figure 6.2[2]), from which it is clear that the problem is that Reiny is simultaneously herbivorous and carnivorous.

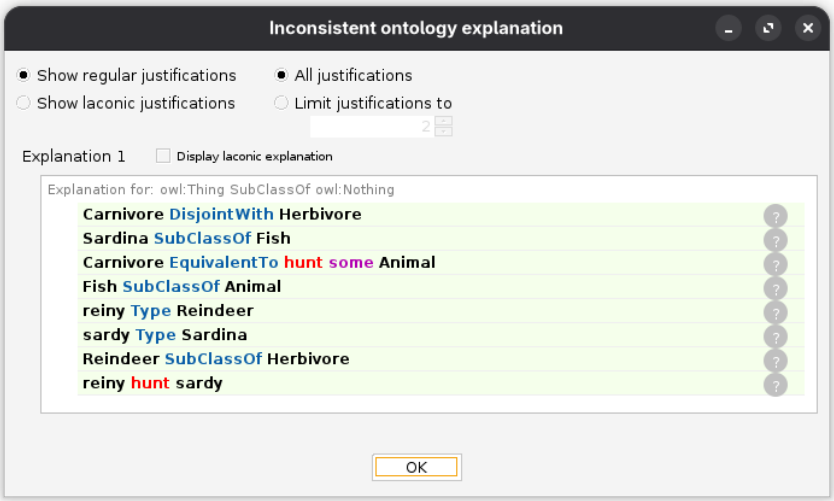


Figure 6.2: Error dialog in Protégé

We introduced this inconsistency in order to show —in the following sections— how it is also possible in Prolog to detect this type of error and identify contradictions in the KB. If we remove the fact "Reiny hunts Sardy" the ontology will return consistent.

2 The figure shows the dialog window of Protégé[45], a widely used ontology editor.

6.2 PROLOG VERSION

To translate the ontology into a Prolog KB, we follow the work presented in [46]. This article discusses in depth how to replicate the various constructs provided by OWL in order to obtain a logically equivalent KB in Prolog.

In this section, we will see how to translate the concepts of class and subclass, disjoint classes, and relationships between individuals; we will present rules to infer new information, as was done in the ontology. Lastly, we will show how to detect errors and inconsistencies in the KB.

Our goal is not to go through the entire translation of an ontology into Prolog, but rather to convince ourselves that such translation is possible.

The original ontology includes only a small subset of the constructs available in OWL. This is because the translation into Prolog is only the first step; in the following sections, we will modify the Prolog program to make it compatible with QA-Prolog and obtain a representation suitable for a quantum annealer.

Advanced topics such as the open world assumption, enumerated classes, existential quantification, cardinality rules, cyclic hierarchies, and more are covered in [46].

6.2.1 Classes and subclasses

We start our KB describing the T-Box of the ontology: here we define the classes' hierarchy. Listing 6.1 shows how to build the hierarchy: (a) is a list of facts that defines the classes (for example, line 1 can be read as "animal is a class."), (b) defines the structure of the classes (line 1 is read as "mammal is a subclass of animal.").

```
1  class(animal).
2  class(mammal).
3  class(polar_bear).
4  class(seal).
5  class(reindeer).
6  class(fish).
7  class(sardine).
8  class(bird).
9  class(penguin).
10 class(herbivore).
11 class(carnivore).
12 class(prey).
13 class(predator).
```

(a) Classes definitions

```
1 subclass(mammal, animal).
2 subclass(bird, animal).
3 subclass(fish, animal).
4 subclass(polar_bear, mammal).
5 subclass(seal, mammal).
6 subclass(reindeer, mammal).
7 subclass(sardine, fish).
8 subclass(penguin, bird).
9 subclass(reindeer, herbivore).
```

(b) Subclasses definitions

Listing 6.1: T-Box definition

Transitivity property of subclass relationship

We must now create a rule to ensure that the subclass relation satisfies the transitive property:

$$\text{X subclass of Y} \wedge \text{Y subclass of Z} \Rightarrow \text{X subclass of Z.}$$

In our ontology, for example, this rule is needed to establish that a penguin is an animal, but not a mammal. The most natural way, in a logic

programming language, to enforce the transitive property is through a recursive definition. Listing 6.2 shows how the rule is defined: lines 1 and 2 implement the base case of the recursion, while lines 4, 5, and 6 implement the recursive call.

Recursive rule

```
1 subClass(C1, C2) :-
2     subclass(C1, C2).
3 subClass(C1, C2) :-
4     subclass(C1, C3),
5     subClass(C3, C2).
```

Listing 6.2: Recursive definition of subclass predicate

It is important to point out that the predicate we are defining in Listing 6.2 is different from the one used in Listing 6.1. The predicate `subclass(C1, C2)` (where `C1` and `C2` are classes) is used to construct the subclass hierarchy with a list of facts. The predicate `subClass(C1, C2)` captures the transitive property we are interested in. This one is the predicate that must be used in queries and in any subsequent rules whenever it is necessary to reason over the class hierarchy.

6.2.2 Individuals and their relationships

We define individuals by assigning them to a class; to do so, we use the predicate `isa(I, C)` where `I` is an individual, `C` is a class. Relationships, on the other hand, are defined through the predicate `hasproperty(I1, R, I2)` that means individual `I1` is in the relation `R` with individual `I2`. Listing 6.3 is a collection of facts that populate our KB, and that define the relationships between individuals.

```
1 isa(beary, polar_bear).
2 isa(sealy, seal).
3 isa(reiny, reindeer).
4 isa(reiny_b, reindeer).
5 isa(reiny_c, reindeer).
6 isa(sardy, sardine).
7 isa(penguy, penguin).
```

(a) Individuals definitions

```
1 hasproperty(beary, hunt, sealy).
2 hasproperty(beary, hunt, penguy).
3 hasproperty(sealy, hunt, penguy).
4 hasproperty(penguy, hunt, sardy).
5 hasproperty(reiny, hunt, sardy).
6 hasproperty(reiny_b, sonOf, reiny).
7 hasproperty(reiny_c, sonOf, reiny_b).
```

(b) Relationships definitions

Listing 6.3: A-Box definition

The predicate `isa(I, C)` allows us to define the class of an individual, but it does not take the class hierarchy into account. For example, we know that Beary is a polar bear, but we cannot infer that he is a mammal or an animal.

To also consider the superclasses of an individual's class, we define the predicate `isA(I, C)`. This predicate, thanks to the recursive definition of `subClass(C1, C2)`, permits to assign to an individual all the classes inherited from the class defined through the predicate `isa(I, C)`. Listing 6.4 shows the implementation of rule `isA(I, C)`.

`isA` predicate definition

As for the predicates `subclass(C1, C2)` and `subClass(C1, C2)`, also `isa(I, C)` must be used only in the definition of an individual's class, while `isA(I, C)` is to be used in queries and future rules.

```
isA(I, C) :-
    isa(I, C).
isA(I, C) :-
    isa(I, C2),
    subClass(C, C2).
```

Listing 6.4: isA predicate

6.2.3 Other rules

The KB that we have built so far contains classes, individuals, and relationships between individuals; we now want to add rules that allow us to extract new knowledge from the facts contained in the ontology.

In particular, we now present the rules to: assign a class to an individual, identify the ancestors of an individual, define two classes as disjoint, and find inconsistencies in the KB.

ASSIGNING A CLASS: The rules defined in Listing 6.5 allow us to assign the class carnivore and predator/prey to an individual. These rules are an example of how logical programming languages are easy to interpret: we can read Listing 6.5 as "If an animal hunts, it is a carnivore and it is a predator. If it is hunted, it is a prey."

```
isA(I, carnivore) :-
    hasproperty(I, hunt, _).
isA(I, predator) :-
    hasproperty(I, hunt, _).
isA(I, prey) :-
    hasproperty(_, hunt, I).
```

Listing 6.5: Class attribution rules

ANCESTORS: Similarly to what was done for the definition of subclasses `subClass(C1, C2)`, we can define the predicate `hasproperty(I1, ancestor, I2)`. Listing 6.6 shows the basic implementation.

```
hasproperty(I1, ancestor, I2) :-
    hasproperty(I2, sonOf, I1).
hasproperty(I1, ancestor, I2) :-
    hasproperty(I2, sonOf, I3),
    hasproperty(I1, ancestor, I3).
```

Listing 6.6: Ancestor definition

Ancestor predicate characteristic

Although in an ontology that deals with the animal kingdom this predicate may seem of little interest, it provides us with a pretext to:

- define a predicate that can be expanded with other rules: in Listing 6.6 we define only the base step and the recursive call, but by imagining building an ontology that deals with people, we can enrich the predicate with numerous rules that reflect family relationships between humans;

- have a predicate that can trigger many recursive calls[3]: the number of individuals populating an ontology can be considerable and can increase over time; moreover, if complex relationships are also considered —such as those suggested in the previous point— the number of ancestors of an individual can be very high and not easily known a priori.

DISJOINT CLASSES AND INCONSISTENCY: In our ontology, logical inconsistency is due to the fact that the reindeer Reiny is both herbivorous and carnivorous, but the two classes are disjoint. Let us therefore see how to express this disjointness in Prolog and how to capture the inconsistency in our KB.

Listing 6.7 shows how to define two disjoint classes and the consequences this has for individuals. At line 1 we see that to state that carnivores and herbivores are disjoint classes, a single fact is sufficient. The other lines show the consequences that disjointness has on individuals: if the classes `C` and `C2` are disjoint (line 4) and `I` belongs to class `C2` (line 5), then `I` does not belong to class `C` (line 3). `logicNot(P)` is a predicate that indicates that the proposition `P` is false.

Providing a detailed explanation of `logicNot(P)` predicate is beyond the scope of this work; in [46] the introduction of `logicNot(P)` is justified in order to bring the open world assumption into Prolog.

```
1 disjointClasses(carnivore, herbivore).
2
3 logicNot(isA(I, C)) :-
4     disjointClasses(C, C2),
5     isA(I, C2).
```

Listing 6.7: Disjoint classes

Identifying the inconsistency of our KB means finding a statement that is both true and false. In particular, Reiny is carnivorous because she hunts (line 5 Listing 6.3), but she cannot be so because she is a reindeer and reindeers are herbivores (line 9 Listing 6.1).

Inconsistency finding rule

It is possible to identify a proposition that is simultaneously true and false with the rule shown in Listing 6.8. The head of the rule is a list composed of a string (which serves only to print an informational message) and the proposition `P`. For the predicate to be true, the proposition `P` must be both false and true.

```
error(['A term cannot be true and false.', P]) :- logicNot(P), P.
```

Listing 6.8: Error rule

We can now test the consistency of the KB by asking the Prolog interpreter whether there exists a `P` that is simultaneously false and true. The result of the query is shown in the following listing:

```
?- error(X).
    X = ['A term cannot be true and false.', isA(reiny, carnivore)].
```

3 It will be useful in Section 6.3.2.

6.2.4 Final result

Thanks to this translation process, we have obtained a KB expressed in Prolog equivalent to the original ontology expressed in OWL. Although with some simplifications, we were able to implement in Prolog the most important constructs offered by OWL, and we built a set of rules to perform inference on the newly created KB. The translation process can be further generalized to more advanced OWL constructs by following [46].

6.3 QA-PROLOG VERSION

The KB we have just obtained[4] is not compatible with the QA-Prolog pipeline. As presented in Section 5.1.3, QA-Prolog does not implement all the features of Prolog. In particular, lists and complex terms are not supported, and rules cannot be recursive.

In this section we will see our contribution on how to modify unsupported rules in order to make them compatible with the pipeline described in Chapter 5.

6.3.1 Lists and complex terms

In our KB we introduced a single list, in Listing 6.8, exclusively to print an additional message to explain the inconsistent proposition. The list can simply be removed, or we can use an additional variable that will be bound to the message we want to print.

Complex terms are predicates that have one or more predicates as arguments. The complex term present in our KB is `logicNot(isA(I, C))` introduced in Listing 6.7 line 3. We used this predicate to state that if an individual belongs to a class `C1`, disjoint from class `C2`, then the individual cannot belong to `C2`.

We can replace the complex term with a new predicate that represents the same concept: `isNotA(I, C)`. This new predicate forces us to modify also the rule for capturing inconsistencies in the KB. Listing 6.9 shows the final result.

```
1 isNotA(I, C) :-
2     disjointClasses(C, C2),
3     isA(I, C2).
4
5 error(I, C, M) :-
6     isA(I, C),
7     isNotA(I, C),
8     M = 'Individual X cannot belong and not belong to class Y'.
```

Listing 6.9: List and complex term removal

Eliminating the predicate `logicNot(P)` where `P` is a proposition, therefore possibly a predicate, forces us to define a new predicate for each statement

4 Available at https://github.com/DavideCamino/Thesis/blob/main/Code/2_QA-prolog/3_QA-prolog/KB1/animal.pl.

we want to negate, and to test each *predicate-predicate_negation* pair to check for possible inconsistencies.

In our KB, only class membership can generate inconsistency, so a single rule is sufficient. In general, other definitions of the predicate `error` may be necessary to account for all possible causes of inconsistency[5].

6.3.2 Unfolding

In our KB there are two rules that are defined recursively: `subClass(C1, C2)` and `hasproperty(I1, R, I2)`.

Unfolding and loop unrolling

To eliminate recursion, we can *unfold* the recursive calls. The procedure is analogous to the *loop unrolling* performed by modern compilers to speed up the execution of machine code. Loop unrolling consists of transforming a loop of instructions into a repeated sequence of instructions; this increases the length of the code but reduces the number of loop condition checks.

Unfolding consists of developing the recursion by explicitly repeating the function call; in this way the recursive call is replaced by many calls to the function `A` in the function `B`.

SUBCLASSES: To eliminate recursion in the rule `subClass(C1, C2)` we can remove the rule with the recursive call, replacing it with a sufficient number of other rules to cover the entire class hierarchy; each new rule adds one call and allows moving one level higher in the hierarchy.

Our ontology has a maximum depth of three, therefore at most two recursive calls are needed to start from any node in the class hierarchy and reach the root. Listing 6.10 shows the updated version of the rule `subClass(C1, C2)` (to clarify the procedure, unfolding is developed up to three calls, even though only two were necessary to cover the entire class hierarchy).

```
1 subClass(C1, C2) :-
2     subclass(C1, C2).
3 subClass(C1, C3) :-
4     subclass(C1, C2),
5     subclass(C2, C3).
6 subClass(C1, C4) :-
7     subclass(C1, C2),
8     subclass(C2, C3),
9     subclass(C3, C4).
```

Listing 6.10: Subclasses unfolding

ANCESTOR: The issue with the predicate `hasproperty(I1, R, I2)` concerns the ancestor-descendant relationship. Indeed, its definition was very similar to the rule `subClass(C1, C2)`. In this case, however, recursion is also present in the base step, as can be seen in lines 1 and 2 of Listing 6.6. Listing 6.11 shows a possible solution. About this code it is important to note:

5 However, this also gives us the possibility to define more precise error messages.

- unlike Listing 6.10, we define many new predicates: this is useful if there were multiple ways to define new ancestors starting from those already identified[6];
- the predicate `hasProperty(I1, R, I2)` has been added: this must be used in queries, while `hasproperty(I1, R, I2)` is used only to define relationships between individuals;
- we stopped, arbitrarily, after building a family tree of depth ten; in general there is no way to know how many recursive calls should be unfolded to be sure of capturing all the ancestors of an individual.

```
ancestor1(I1, I2) :-
    hasproperty(I2, sonOf, I1).
ancestor2(I1, I2) :-
    ancestor1(I1, I3),
    hasproperty(I2, sonOf, I3).
ancestor3(I1, I2) :-
    ancestor2(I1, I3),
    hasproperty(I2, sonOf, I3).
    ...
ancestor9(I1, I2) :-
    ancestor8(I1, I3),
    hasproperty(I2, sonOf, I3).

ancestor_unfold(I1, I2) :-
    ancestor1(I1, I2);
    ancestor2(I1, I2);
    ...
    ancestor9(I1, I2).

hasProperty(I1, P, I2) :-
    hasproperty(I1, P, I2).
hasProperty(I1, ancestor,I2) :-
    ancestor_unfold(I1, I2).
```

Listing 6.11: Ancestor unfolding

6.3.3 Stopping criterion

As mentioned above, it is not always possible to know in advance how many recursive calls will be performed to capture all the aspects we are interested in. Here we discuss some stopping criteria looking only at the KB[7]. In general, we can distinguish two cases: T-Box and A-Box.

T-BOX: The T-Box describes our domain of interest and should be stable. Once a description of the world suitable for our purpose has been developed, it is rare for it to change; in our case, a modification of the T-Box could occur only if new species were discovered, which is not frequent.

6 Listing A.2 in the appendix shows an example for clarification.
7 For a performance analysis of the impact of unfolding see Section 6.4.3.

If the rules with recursive calls involve the T-Box, it can be possible to know the maximum number of recursive calls needed to capture all relevant information and perform sufficient unfolding to allow an adequate number of function calls.

An example of this stopping criterion was shown in the rewriting of the rule `subClass(C1, C2)`. Once the class hierarchy is known, it is possible to compute the maximum depth and use it as a criterion to stop the unfolding process.

A-Box is more challenging in unfolding process

A-BOX: The A-Box contains the individuals populating the ontology. For this reason, the A-Box is much more subject to change than the T-Box and, as discussed when presenting the ancestor-descendant relationship, relationships between individuals can also be very complex.

Each time an individual is added to the KB or its relationships are modified, the depth of the recursive calls may change.

With regard to the A-Box, the stopping criterion for unfolding must be decided in advance or estimated through heuristics where possible. It is not possible to know in advance the maximum number of recursive calls without performing reasoning, but this would be counterproductive since the objective of the translation is to perform reasoning on a quantum annealer.

6.3.4 Equivalence between the KBs

Before replacing the KB with unsupported construct and recursive rules with the modified one, we must ensure that there is equivalence between the two KBs.

To show equivalence between recursive calls and the unfolded version, it would be possible to construct a proof based on the principle of induction, possibly with the aid of a proof assistant.

Such a constructive proof is not the objective of this work. To convince ourselves of the equivalence and to take into account also other modifications, we test the behavior of queries that use rewritten rules to ensure their correctness.

As far as possible, the unfolding process was carried out so as to make it transparent to the end user. The result was achieved with regard to queries related to the subclass hierarchy and the retrieval of an individual's ancestors.

As for error detection, however, the various rules that define the predicate `error` may have a different number of parameters. For each number of parameters it will be necessary to test the predicate in order to ensure that all inconsistencies are identified. We have not to test for each definition of the predicate; for example, if there are two definitions with 3 parameters and one with 4, it will be necessary to test `error` twice.

Listing 6.12 shows the queries, that use the modified rules to make them compatible with the QA-Prolog pipeline, and their results:

- In (a) we retrieve all superclasses of reindeer, using the unfolded version of `subClass(C1, C2)` predicate;

- In (b) we construct the family tree of the individual Reiny_c to test the new version of `subClass(C1, C2)`;

- In (c) we use the predicate `error(X, Y, Z)` with the definition that avoids list and complex terms.

```
?- subClass(reindeer, X).
    X = mammal ;
    X = herbivore ;
    X = animal ;
    false.
```

(a) Subclasses

```
?- hasProperty(X, ancestor, reiny_c).
    X = reiny_b ;
    X = reiny ;
    false.
```

(b) Ancestor

```
?- error(X, Y, Z).
    X = reiny,
    Y = carnivore,
    Z = 'Individual X cannot belong and not belong to class Y'
```

(c) Error

Listing 6.12: Query results

The result of the queries is as expected: we obtained all and only the results that we would also have obtained by querying the original KB. This intuitively convinces us of the equivalence of the two KBs.

6.4 FINAL RESULT

The obtained KB[8] is compatible with the QA-Prolog pipeline. In this section we go through the entire transformation pipeline until obtaining the Hamiltonian that represents the KB together with the associated query, and we discuss the results obtained.

Formulating a query is essential because it defines the outputs of our program and, therefore, allows us to build the digital circuit, which is the first step of the pipeline.

To evaluate the result of the translation, we do not consider the outcome of the query evaluation because, as already shown in the chapter on the pipeline in Section 5.2.3, the simulators offered by the Ocean framework are able to find the solution only for very small problem instances and we will see, in Section 6.4.2, that we will obtain instances of problems with a very large number of *qubits*.

What we consider instead is the dimension of the minimization problem obtained at the end of the pipeline, extracting the number of qubits composing the polynomial to be minimized, and attempting to provide a justification for the different dimensions we obtain.

8 Available at https://github.com/DavideCamino/Thesis/blob/main/Code/2_QA-prolog/3_QA-prolog/KB1/animal_unfold.pl.

6.4.1 Query

We construct different queries that test different facts and rules defined in our KB. We consider both simple queries, whose answer can be found simply by scanning the facts explicitly listed in the KB, and increasingly complex queries that require applying one or more inference rules.

The queries we test are:

1. `?- class(animal).`
2. `?- class(X).`
3. `?- subClass(seal, X).`
4. `?- isA(sardy, X).`
5. `?- error(X, Y, Z).`
6. `?- hasProperty(I1, ancestor, reiny_c).`

In the results we will refer to the queries directly by the number associated with them in this list.

6.4.2 Results

As can be seen from Figure 6.3, the number of *qubits* required to represent the problem varies significantly, from a few dozen, for the first queries to almost ten thousand *qubits*, for the query related to the ancestors of an individual.

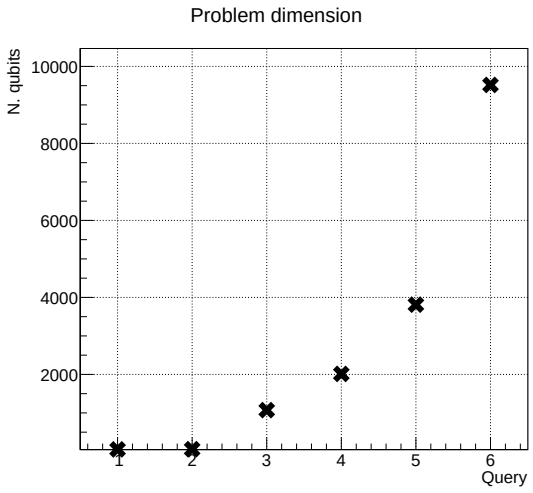


Figure 6.3: Problem size

The large difference in the number of *qubits* used is due to the complexity of the query, to how many facts and rules must be used in order to find its solution.

For simple queries such as the first two, it is sufficient to scan the list of facts. Query 3, which generates a problem with 1072 Boolean variables, requires the construction of the entire class hierarchy. Query 4 (2013 *qubits*) also requires the class hierarchy; moreover, all the rules that allow us to define an individual as belonging to a certain class must be considered (rules of Listing 6.5).

Query 5 adds, to all the rules necessary to solve query 4, also the disjointness between classes and the rule that defines inconsistencies. The number of variables thus increases to 3812. Finally, query 6 uses relationships between individuals and in particular requires the relation that defines ancestors, which was unfolded in Listing 6.11. Precisely because of the unfolding,

the number of rules increases significantly and, consequently, the dimension of the problem grows up to 9519 *qubits*.

From the obtained results, it is clear that, depending on the query, we obtain Hamiltonians of very different sizes. This is advantageous because it allows us not to consider the entire KB for each query, but to extract only the part necessary to answer our specific question.

One may then ask which step of the pipeline is responsible for this extraction. By observing the files in the QA-Prolog working directory, it can be noted that the Verilog file is the same for all queries, whereas the EDIF file, synthesized by Yosys (Section 5.2.2), is radically different from one query to another.

Yosys optimization

The optimization therefore occurs at the level of the logic circuit: once the output of our circuit has been defined through the query, Yosys is able to discard all those parts of the circuit described in Verilog that are not necessary to define the output. Yosys generates a EDIF file using only the Verilog module connected, in circuit term, to the output. This step, as highlighted by the graph in Figure 6.3, is able to significantly reduce the dimension of the problem.

6.4.3 Impact of unfolding on performance

If we observe Listing 6.10 and Listing 6.11, we can see that the unfolding of the two predicates, `subClass(C1, C2)` and `hasProperty(I1, ancestor, I2)`, is very similar. However, the dimension of the problem generated by the queries that use these predicates (queries 3 and 6 in Figure 6.3) is significantly different, i.e. typically smaller, from one query to another. The main difference between the two predicates is the number of times the unfolding has been carried out.

In this section we study the dimension of the problem generated by the QA-Prolog pipeline as a function of the number of recursive calls unfolded. In order to isolate as much as possible the impact of unfolding, we consider the minimal KB and the query shown in Listing 6.13.

```
1 class(animal).
2 class(mammal).
3
4 subclass(mammal, animal).
```

(a) Prolog File (KB)

```
1 ?- subClass(mammal, animal)
```

(b) Query

Listing 6.13: Input unfold

QA-Prolog imposes a maximum number of clauses in the body of a rule, in order to exceed this limit, we are forced to develop the unfolding as we did for the relation `hasProperty(I1, ancestor, I2)` (Listing 6.11).

Since we know the only recursive rule of the KB, we can automate the unfolding process. Listing 6.14 shows the core of the rule generation: for `n_unfold` times (line 2) we generate a new rule (line 3) using the rule defined in the previous iteration[9] (line 4), then adding one level (line 5).

9 Or at line 1 during the first iteration.

```
rule_0 = 'subClass0(C0,C1) :- \n\tsubclass(C0, C1).\n'
for i in range(1, n_unfold):
        rule = 'subClass'+ str(i) + '(C0,C2) :-\n'      +\
               '\tsubClass' + str(i-1) + '(C0,C3),\n'   +\
               '\tsubclass(C3, C2).\n'
```

Listing 6.14: Unfolding script's core

The problem's size, in terms of the number of Boolean variables, was evaluated for trees of depth ranging from two to twenty. To perform these experiments, not only the generation of the rules was automated but also the execution of the experiment itself[10].

QUALITATIVE RESULTS: In Figure 6.4, it is possible to observe the impact of unfolding on the problem dimensions. As can be noted, as the number of rules increases, the number of *qubits* required to represent the problem increases as well.

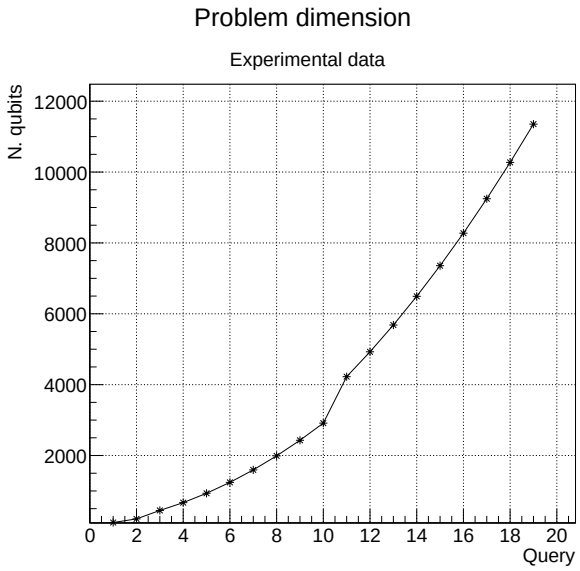


Figure 6.4: Unfold impact on size

Another qualitative observation that can be made is that the number of variables is correlated with the number of rules independently of the class hierarchy. In fact, the KB (Listing 6.13) we are considering has a maximum depth of two, having only two classes (one subclass of the other). We can therefore deduce that Yosys is able to optimize the circuit by discarding all components not connected to the output, but it is not able to select the rules actually necessary to reach the answer to the query.

QUANTITATIVE RESULTS: To quantitatively understand the impact of unfolding on the dimension of the final Hamiltonian, we attempt to fit the experimentally obtained points. To perform the fitting, we use the least squares method by minimizing the statistical variable χ^2. The classes of functions tested were:

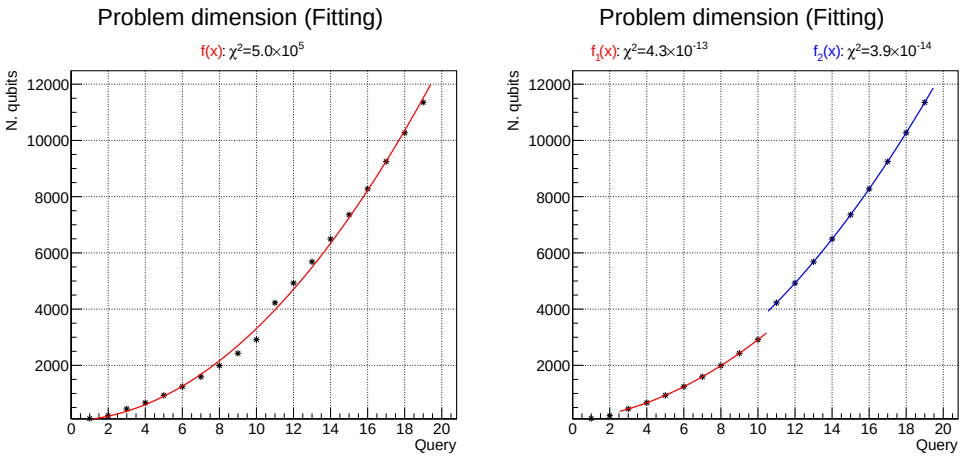
- first-degree polynomial: linear function;
- second-degree polynomial: parabola;
- exponential function.

10 The code is available at https://github.com/DavideCamino/Thesis/blob/main/Code/2_QA-prolog/3_QA-prolog/KB2/animal.ipynb.

Quadratic growing of problem dimension

The function that best interpolates the experimental points is the second-degree polynomial. Figure 6.5a shows the result of the fitting. A very high value of χ^2 can be observed; this would lead us to reject the hypothesis that a parabola correctly interpolates the experimental data. However, it should be recalled that the χ^2 test assumes that a reasonable estimate of the error associated with each measurement is available [47]: in this case we cannot associate any error to the measurement even by repeating the test many times. The conclusion we can nevertheless reach is that, among the classes of functions studied, the parabola is the one that best interpolates the data.

By observing more closely the graph in Figure 6.4, we can notice what appears to be a jump in the distribution of the points (between point 10 and 11). We can therefore try to consider two different quadratic functions that approximate respectively the first set of points and the second. Figure 6.5b shows the result of this hypothesis.



(a) Single fitting

(b) Double fitting

Figure 6.5: Fitted data

In this case we observe a very low χ^2 for both interpolating functions. Again, such a result should surprise us because obtaining such a good fit starting from an experimental data distribution is highly unlikely.

In our case, such a low value indicates that the collected data lie perfectly on a parabola, and the absence of experimental error makes this agreement exact. The "jump" in the data distribution could be due to the internal representation of the rules. Below a certain number of rules, they are identified with a certain number of *qubits*; once this limit is exceeded, more *qubits* are required to identify each rule.

If this hypothesis is correct, we should expect further jumps in the trend of the number of variables, but the growth will always remain quadratic. Moreover, the continuous segments will each be twice as long as the previous one, because each extra *qubit* allows doubling the number of unique identifiers for the rules.

6.5 CONCLUSION

In this chapter we have shown how it is possible to move from an ontology expressed in OWL to a KB expressed in Prolog on which it is possible to highlight inconsistencies in the facts and perform reasoning as on the original ontology.

We have recalled some of the limitations of QA-Prolog, showing how they impacted our translation; we proposed solutions that led us to rewrite some rules, and, above all, to unfold recursive calls.

We showed that the obtained KB is effectively compatible with the QA-Prolog pipeline by going through all the steps until obtaining a Hamiltonian, whose dimensions we discussed by relating them to the query and consequently to the facts and rules necessary to answer that query.

Finally, we have studied the impact of unfolding over the problem dimension. We have obtained a quadratic growing of the Boolean variable used to represent our problem as a function of the number of recursive calls unfolded.

At a high level, this chapter can be seen as an additional upstream step of the pipeline: QA-Prolog, through successive rewritings into different formats that logically always represent the same problem with the same solution, starts from a KB in Prolog and reaches a representation suitable for a quantum annealer. In this chapter we added the rewriting from OWL to Prolog, increasing by one step the chain of rewritings.

7 QAOA

In the previous chapters we focused on obtaining a Hamiltonian whose ground state represented the solution to our problem. Implicitly or explicitly we referred to quantum annealers as machines for finding the configuration that minimizes the energy.

In this chapter we will show how to move from a problem formulated for quantum annealers to a problem formulated for quantum gate based computers through the *Quantum Approximation Optimization Algorithm* (QAOA), proposed for the first time by Farhi et al. in [48].

We begin by describing the QAOA algorithm, highlighting theoretical advantages and disadvantages with respect to QAC[1]. We will then show how it is possible to translate a problem formulated in QUBO form (Section 2.3.3), obtained at the end of the QA-Prolog pipeline, into a problem suitable for IBM quantum gate based computers.

Finally, we will test the proposed translation and present some experimental results, discussing them.

7.1 QAOA

The QAOA algorithm is inspired by QAC; both attempt to find the minimum eigenvalue of a matrix that represents the energy of a physical system [49]. This matrix describes an observable, consequently it is a Hermitian operator and in particular it is the Hamiltonian $\mathbf{H}$ of the system. The smallest eigenvalue of $\mathbf{H}$ is the minimum energy of the system that can be measured; for this reason it is called the ground state energy.

As described in Section 2.3.1 QAC starts from a Hamiltonian whose ground state is easy to know and evolves, adiabatically (to remain in the ground state), towards the Hamiltonian with unknown ground state [49]. The evolution time from one Hamiltonian to another can be exponential[48]; moreover, generally the success probability of QAC does not increase as the annealing time increases.

QAOA levels

QAOA is a modular algorithm that targets quantum gate based machines and, once its parameters are optimized, the quality of the result increases as the number of modules, called *levels*, increases.

In the gate model the algorithm is described by the circuit and vice versa a circuit define an algorithm. From now on we use the term circuit and algorithm interchangeably.

1 Quantum Adiabatic Computing (Section 2.3.1).

7.1.1 Circuit

The QAOA circuit take in input n *qubits*, these are the boolean veriables that we want to optimize to minimize the energy represented by $\mathbf{H}$. At the beginning of the circuit all the *qubits* are set to zero ($|0\rangle$).

The QAOA algorithm stats preparing the state $|s\rangle$ with maximum superposition. It is possible to obtain this state starting from the state with n *qubits* set to zero applying the Hadamard gate (H) to each of them [49]; we denote this operator with $H^{\otimes n}$. State $|s\rangle$ is prepared by the blue gate in Figure 7.1.

QAOA gates

The QAOA circuit then consists of p identical levels. Each level is composed of two operators (two quantum gates):

- *phase separation operator* (gates orange in Figure 7.1): $U_P(\gamma) = e^{-i\gamma\mathbf{H}}$ where $\mathbf{H}$ is the Hamiltonian of the system, and $\gamma \in [0, 2\pi]$ is a parameter;
- *mixing operator* (gates green in Figure 7.1): $U_M(\beta) = e^{-i\beta B}$ where B is the operator that applies the Pauli Gates X to every *qubit*, and $\beta \in [0, \pi]$ is a parameter.

The operator $U_P(\gamma)$ simulates $\mathbf{H}$, while $U_M(\beta)$, which does not commute with $U_P(\gamma)$[2], allows modifying the probability distribution of the output [49].

Using Pauli Gates X is not the only way to construct the mixing operator; in [50, 51] other mixing operators are explored.

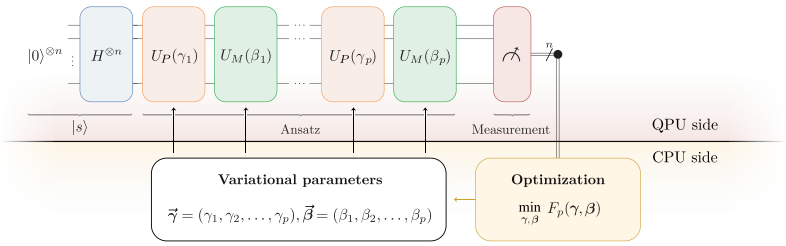


Figure 7.1: QAOA scheme [52]

Figure 7.1 shows the general scheme of the QAOA algorithm considering p levels.

Given $\vec{\gamma} = (\gamma_1, \gamma_2, \ldots, \gamma_p)$ and $\vec{\beta} = (\beta_1, \beta_2, \ldots, \beta_p)$, we denote with $\left|\psi(\vec{\gamma}, \vec{\beta})\right\rangle$ the *ansatz* (an estimate of the solution) of QAOA. The ansatz is defined as:

$$\left|\psi(\vec{\gamma}, \vec{\beta})\right\rangle = U_M(\beta_p)U_P(\gamma_p)\cdots U_M(\beta_1)U_P(\gamma_1)\,|s\rangle\,. \tag{7.1}$$

The circuit ends with the measurement; the expected value of this measurement is:

Energy of the system

$$F(\vec{\gamma}, \vec{\beta}) = \left\langle \psi(\vec{\gamma}, \vec{\beta}) \right| \mathbf{H} \left| \psi(\vec{\gamma}, \vec{\beta}) \right\rangle. \tag{7.2}$$

$F(\vec{\gamma}, \vec{\beta})$ represents the expected value of $\mathbf{H}$ in the state $\left|\psi(\vec{\gamma}, \vec{\beta})\right\rangle$ [49].

Once implemented, the expected value is estimated by executing the quantum circuit many times. Each run of the circuit retrieves an assignment for

2 $U_P(\gamma) \times U_M(\beta) \neq U_M(\beta) \times U_P(\gamma)$.

the n input *qubits*, this assignment represents a possible solution to the problem with an associated energy. The average energy represents our estimation $F(\vec{\gamma}, \vec{\beta})$.

The QAOA's performance depends on the number of levels; given optimal parameters of QAOA, the algorithm asymptotically converges to optimal solutions as the number of levels grows[48].

7.1.2 Parameter optimization

QAOA requires tuning the parameters to minimize the expected objective value for a given number of levels. Formally the best parameter ($\vec{\gamma^*}$ and $\vec{\beta^*}$) are defined as:

$$(\vec{\gamma^*}, \vec{\beta^*}) = \arg\min_{\vec{\gamma},\vec{\beta}} F(\vec{\gamma}, \vec{\beta}) \tag{7.3}$$

Limits of optimization

This parameter tuning problem is nonconvex and **NP**-hard [49]. Even with one level, the gradient of $F(\vec{\gamma}, \vec{\beta})$ vanishes rapidly; this phenomenon is known as the barren plateau [53, 54]. The barren plateau expands exponentially as the number of qubits grows, making the tuning process even more challenging [54].

The parameter optimization process is executed on a classical CPU rather than on a quantum computer. This makes QAOA a hybrid algorithm [49].

CONCLUSION: Despite the difficulty in optimizing the parameters, QAOA algorithm presents numerous interesting aspects and research in this field is very active. In our work it gives us the possibility to change the reference architecture, and attempt to solve a QUBO-formulated problem —obtained with the QA-Prolog pipeline— using quantum gate based computers. In the next sections we will see how to practically implement the QAOA algorithm.

7.2 FROM QAC TO QAOA

In the previous section we focused on the theoretical aspects of the QAOA algorithm. In this section instead we show how to use IBM's Qiskit Framework to solve a minimization problem such as those obtained at the end of the QA-Prolog pipeline.

7.2.1 Objective

At the end of the QA-Prolog pipeline we obtain a problem in QUBO form: in this representation the variables x_i to be optimized belong to the set $\{0, 1\}$. When we move to the quantum gate based architecture, the measured value of the *qubits* belongs to the set $\{-1, 1\}$. We must therefore move from a representation in QUBO form to a problem in Ising form. Qiskit also requires that the matrix $\mathbf{H}$ is expressed as Pauli operators.

With these steps we have made the output of QA-Prolog compatible with the input of the QAOA algorithm. To complete the implementation we must then optimize the parameters defined in Equations 7.1 and 7.2 in order to

minimize the expected value of **H**. Finally, after finding the optimal parameters we will sample the result of the quantum circuit one last time to obtain the result of our problem.

To summarize, to implement the QAOA algorithm we must:

- move from QUBO form to Ising form;
- express the observable **H** using Pauli operators;
- optimize the parameters of Equation 7.2;
- sample the optimized quantum circuit.

7.2.2 From QUBO to Ising

In Section 2.3.3 we saw that a translation from a problem in QUBO form to Ising form always exists. It is possible to implement this translation manually following [9]; in this case we will use the implementation provided by the Python library `qubovert`.

```
qubo = qubovert.utils.matrix_to_qubo(qubo_mat)
ising = qubovert.utils.qubo_to_quso(qubo)
ising_dict = dict(ising)
ising_dict.pop(())
ising_mat = qubovert.utils.qubo_to_matrix(ising_dict)
```

Listing 7.1: From QUBO form to Ising

Listing 7.1 shows how it is possible to move from one formulation to the other using the `qubovert` library:

1. we build a QUBO problem using as input the output of the QA-Prolog pipeline (`qubo_mat`);
2. we transform the QUBO problem into Ising form;
3. we extract the dictionary that assigns the numerical coefficients Q_{ij} in Formula 2.11;
4. we discard the constant term;
5. we construct the matrix with the weights of the problem in Ising form.

Discarding energy offset

The most interesting aspect of this procedure is the removal of the constant term; this value is used to align the ground state of the problem in QUBO form and Ising form (Equation 2.12) to the same energy. Since our objective is to find the ground state, regardless of its value, we can ignore this offset and consider only the coefficients of the variables.

7.2.3 Hamiltonian as Pauli operator

Consider the upper triangular Hamiltonian **H** that describes our problem in Ising form. Each element belonging to the diagonal $h_{i,i}$ is translated into a Pauli Gate Z that measures the z component of the *qubit* i's spin and applies

the identity transformation to all the other *qubits*[3]. The elements $h_{i,j}$[4] are transformed into gates that measure the spin along the Z axis of the *qubits* i and j, and apply the identity operator to the other *qubits*.

Following the IBM tutorial [55] we transform the matrix that describes the problem in Ising form into Pauli operators. Listing 7.2 shows how it is possible to automate this rewriting. At line 4 we generate the observable for the elements on the diagonal and at line 6 the observable for the other elements of the matrix.

```
1 def build_paulis(matrix):
2     pauli_list = []
3     for i in range(len(matrix)):
4         pauli_list.append(("Z", [i], matrix[i][i]))
5         for j in range(i+1, len(matrix)):
6             pauli_list.append(("ZZ", [i, j], matrix[i][j]))
7     return pauli_list
```

Listing 7.2: Genearating Pauli operator

We can finally construct a single observable starting from the list of Pauli operators that we have just built. To do this we use a library function provided by Qiskit. Listing 7.3 shows how to obtain the cost function, where `pauli_list` are the Pauli operators that we generated with Listing 7.2, and `n_qubits` is the number of *qubits* that describe our problem.

```
cost_hamiltonian = SparsePauliOp.from_sparse_list(pauli_list, n_qubits)
```

Listing 7.3: Defining cost function

7.2.4 Final circuit

As seen in Section 7.1.1 the QAOA algorithm does not directly use the cost function, but two parametric operator (gates $U_P(\gamma)$ and $U_M(\beta)$) repeated a certain number of times.

Qiskit provides the programmer with a constructor to generate the quantum circuit that implements the QAOA algorithm; this constructor allows specifying:

- cost function: the one obtained in Listing 7.3;
- `reps`: the number p of levels;
- initial state: in the form of a quantum circuit. If no initial state is specified, the Hadamard gate is applied to each *qubit*;
- mixing operator: this parameter can be specified as a base operator or as a quantum circuit. If no parameter is specified the Pauli X gate is applied.

3 Recalling the observation made in Paragraph 2.2.2 where we pointed out that Pauli Gate Z is both unitary (a valid quantum gate), both Hermitian (therefore represent an observable)

4 With $j > i$ because $\mathbf{H}$ is upper triangular.

The circuit ends by measuring the value of the *qubits*. Listing 7.4 shows the construction of the circuit.

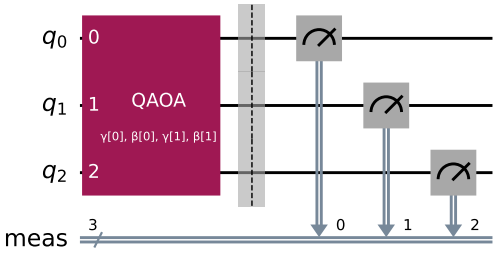
```
qc = QAOAAnsatz(cost_operator=cost_funcrion, reps=n_reps)
qc.measure_all()
```

Listing 7.4: Circuit construction

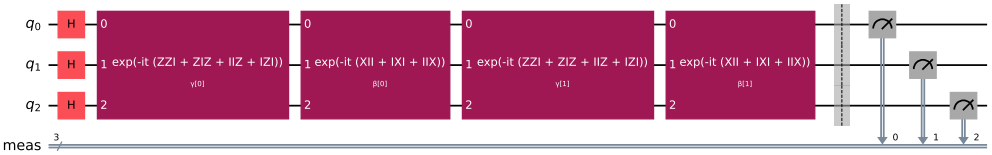
EXAMPLE: Considering a problem described by three *qubits* and the QAOA with two levels. Figure 7.2 shows: in a the matrix that describes the problem; in b the circuit in compact form; in c the circuit expanded to observe how the coefficients of the matrix have been translated into observables.

$$\mathbf{H} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

(a) Hamiltonian matrix



(b) Compact form



(c) Expanded form

Figure 7.2: QAOA circuit

Reversed qubits string

By carefully examining the operators that constitute the gate $U_P(\gamma)$ (first and third block in Figure 7.2c) we can observe that the order of the qubits is inverted. For example, the operator IIZ represents the element of $\mathbf{H}$ in position $(0,0)$: I is the identity gate applied to the third and second *qubits*, and Z is the Pauli Gate Z applied to the first *qubit*. The operator ZZI represents the element in position $(1,2)$. This observation must be kept in mind when reading the results of the circuit.

7.2.5 Parameter optimization

We define a procedure to obtain the cost $F(\vec{\gamma}, \vec{\beta})$ (Equation 7.2) as the expected value of the result of executing the quantum circuit defined earlier. This cost must be minimized by optimizing the parameters $\vec{\gamma}$ and $\vec{\beta}$. Following [55], we use the Python library `scipy` to minimize $F(\vec{\gamma}, \vec{\beta})$.

Listing 7.5 shows how to obtain $F(\vec{\gamma}, \vec{\beta})$ as an expected value:

- at line 6 we create a *primitive unified block* (`pub`): a triple containing the circuit to execute (`ansatz`), the observable to measure (`isa_hamiltonian`) and the circuit parameters (`params`);

```
1  def cost_func_estimator(params, ansatz, hamiltonian, estimator):
2      # transform the observable defined on virtual qubits to
3      # an observable defined on all physical qubits
4
5      isa_hamiltonian = hamiltonian.apply_layout(ansatz.layout)
6      pub = (ansatz, isa_hamiltonian, params)
7      job = estimator.run([pub])
8      results = job.result()[0]
9      cost = results.data.evs
10     objective_func_vals.append(cost)
11
12     return cost
```

Listing 7.5: $F(\vec{\gamma}, \vec{\beta})$ estimation

- at line 7 we estimate $F(\vec{\gamma}, \vec{\beta})$ by executing multiple runs of the circuit[5];
- from line 9 to 12 we retrieve and return the result.

```
1  with Session(backend=backend) as session:
2      estimator = Estimator(mode=session)
3      estimator.options.default_shots = 1000
4
5      result = minimize(
6          cost_func_estimator,
7          init_params,
8          args=(candidate_circuit, cost_hamiltonian, estimator),
9          method="COBYLA",
10         tol=1e-2,
11     )
12 optimized_circuit = candidate_circuit.assign_parameters(result.x)
```

Listing 7.6: $\vec{\gamma}$ and $\vec{\beta}$ optimization

Listing 7.6 shows the minimization process: we create an estimator that executes 1000 runs of the circuit (lines 2 and 3) and use the library function `scipy.optimize.minimize`[6] to minimize $F(\vec{\gamma}, \vec{\beta})$ by optimizing the parameters $\vec{\gamma}$ and $\vec{\beta}$. Finally we create the optimized circuit by assigning to the parametric circuit the value of the optimized parameters (line 12). In this listing, `candidate_circuit` is the transpiled[7] version of the circuit `qc` obtained in Listing 7.4.

7.2.6 Retrieving the results

Now that we have optimized the value of the parameters we can, one last time, perform sampling of the quantum circuit. This time we are not interested in estimating the minimum value of $F(\vec{\beta}, \vec{\gamma})$, but in determining the value of the *qubits* that generate this value.

5 The shot's number is an attribute of the `estimator` object that we define in Listing 7.6.
6 Its documentation is available at https://docs.scipy.org/doc/scipy/reference/generated/scipy.optimize.minimize.html.
7 Section 4.2.2

```
1 sampler = Sampler(mode=backend)
2 sampler.options.default_shots = 10000
3 pub = (optimized_circuit,)
4 job = sampler.run([pub], shots=int(1e4))
5 counts_bin = job.result()[0].data.meas.get_counts()
```

Listing 7.7: Circuit sampling

In Listing 7.7 we create a `Sampler` (line 1) that runs the optimized circuit 10000 times (line 4). In this case we use a `Sampler`, and not an `Estimator` as in Listing 7.6, because we are interested in the value of the *qubits* rather than the expected value of an observable. At line 5 we retrieve the results.

7.3 EXPERIMENTS

We apply the QAOA algorithm to solve a problem in QUBO form obtained with the QA-Prolog pipeline.

7.3.1 Problem

Consider the satisfiability problem $y = x_1 \wedge \neg(x_2 \vee x_3)$ (Equation 5.1). This problem is well suited for the application of QAOA because we can read the solution directly from the value of the *qubits*[8].

As shown in Example 5.2.3, starting from Prolog code to solve the satisfiability problem introduces a significant amount of overhead. To minimize the number of *qubits* required to represent the problem we focus only on the last part of the pipeline, starting from the Verilog code of Example 5.2.2.

QUBO and Ising matrices

The QUBO problem that we obtain is represented by an 11×11 matrix, this means that we need eleven *qubits* to encode our problem.

$$\mathbf{H} = \begin{pmatrix} 4 & 2 & -4 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -8 \\ 0 & 4 & -4 & 0 & -8 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 4 & 0 & 0 & -8 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 6 & 2 & -4 & -8 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 6 & -4 & 0 & -8 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 6 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 4 \end{pmatrix} \qquad \mathbf{H} = \begin{pmatrix} 0.5 & 0.5 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -2 \\ 0 & 0.5 & -1 & 0 & -2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -0.5 & 0.5 & -1 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -0.5 & -1 & 0 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

(a) QUBO Hamiltonian **(b)** Ising Hamiltonian

Figure 7.3: Problem's Hamiltonian

Figure 7.3 shows the matrix representation of the problem both in QUBO and the corresponding Ising form. The two representations have ground states with different values, but they correspond to the same configuration of the *qubits*.

8 We will discuss the possibility of using QA-Prolog to make the output of QAOA readable in Chapter 8.

7.3.2 Execution

To solve the satisfiability problem we construct a QAOA with $p = 10$ levels. By increasing the number of levels, the algorithm converges increasingly well to the optimal solution.

However, increasing the number of levels also leads to longer execution time and, if executed on quantum hardware (and not through simulators), the deeper the circuit, the more one encounters the phenomenon of decoherence [6, p. 278][9]. Moreover, the number of parameters to optimize is equal to twice the number of levels, therefore increasing the number of levels also makes the parameter optimization more complex[10].

Parameter optimization

Figure 7.4 shows the parameter optimization process: as better parameters are found, the estimated cost decreases. In this case, the parameters were optimized after approximately 130 iterations. Each iteration involves 1000 runs of the quantum circuit. During these 1000 shots, the circuit remains unchanged, and the parameters $\vec{\gamma}$ and $\vec{\beta}$ are fixed. After each iteration (i.e., after 1000 shots), the expected value of the system's energy is estimated, and the parameters $\vec{\gamma}$ and $\vec{\beta}$ are updated.

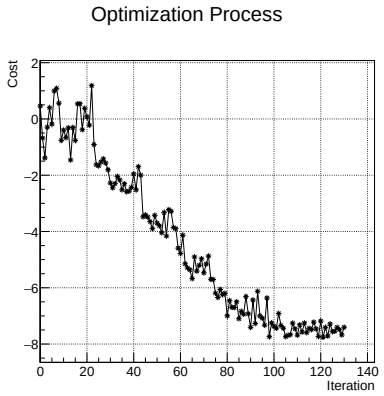


Figure 7.4: Cost minimization

After the optimization phase, we obtain an approximation of the optimal parameters $\vec{\gamma}^*$ and $\vec{\beta}^*$ (Equation 7.3). We can then sample the optimized circuit to obtain the solution to our problem. Each shot, both in the optimization phase and in the final sampling phase uses the same initial *qubit* configuration. The algorithm starts with all *qubits* in state $|0\rangle$, and then prepares the state $|s\rangle$ by applying the Hadamard gate H to each *qubit*.

Figure 7.5a shows the sampling results (10000 runs of the quantum circuit), highlighting in purple the four most frequent outcomes.

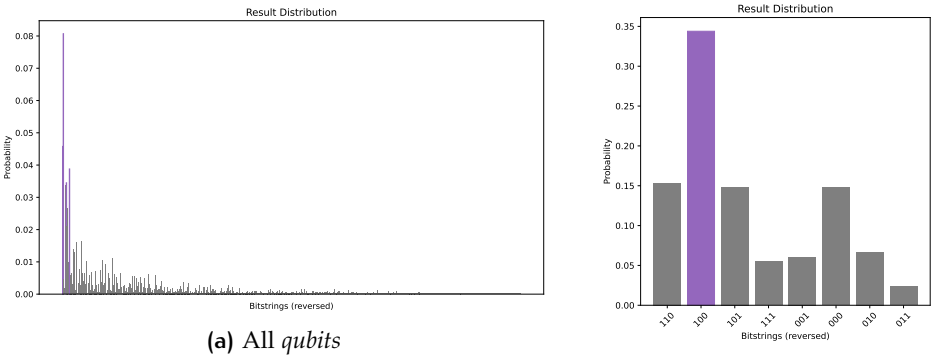
Retrieving the results

The histogram in Figure 7.5a is difficult to interpret; moreover it shows the result of all eleven *qubits* that form the polynomial to be minimized. We are interested only in the value of the variables x_1, x_2 and x_3, the others are ancillary. Figure 7.5b shows the sampling result where we consider only the variables of interest. These were extracted by checking which *qubits* had been mapped by QMASM to the variables x_1, x_2 and x_3, and considering that the obtained bitstring must be read in reverse.

The obtained result is the expected one; in fact we obtained the string 1 0 0 as the most probable sample. These values must be associated respectively with the variables x_1, x_3 and x_2, and we can easily verify that this assignment satisfies Equation 5.1.

9 Even 10 levels are already too many for the today noisy quantum computer, however, it is an appropriate number for our simulations.

10 As said in Section 7.1.2.



(a) All *qubits*

(b) Interesting *qubits*

Figure 7.5: Circuit sampling

7.3.3 Results

Both the sampling process and the parameter optimization process introduce stochastic components into the execution of QAOA. We therefore want to verify, by executing the parameter optimization and the final sampling multiple times,how often we obtain the expected result[11].

Since the execution time of each experiment needs to be taken into consideration, the number of experiments performed is limited and does not allow a reliable statistical analysis. However, it is sufficient to show how the QAOA algorithm is not always able to find the correct answer to the minimization problem.

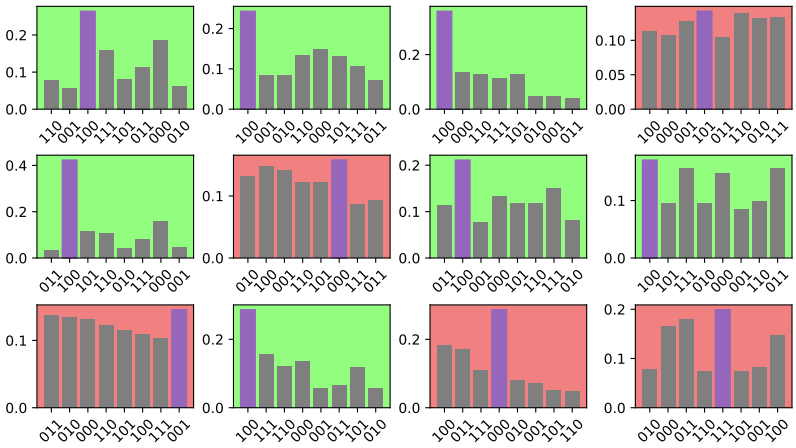


Figure 7.6: Multiple experiments' result

Figure 7.6 shows the results (of the relevant *qubits*) of 12 repetitions of the experiment. In the image, all the cases in which the correct assignment of the variables x_1, x_2 and x_3 was identified are highlighted with a green. The cases, in which the most probable result is incorrect, are highlighted in red.

The number of experiments is very low and advanced statistical analyses would not be reliable. Nevertheless, we can make some qualitative obser-

11 The code that automates the execution of the experiments is available at https://github.com/DavideCamino/Thesis/blob/main/Code/3_QAUA/circ_sat/circ_sat_qaoa_multiple_run.ipynb.

vations. Typically, in the cases in which the correct solution is found, the probability of obtain the correct bit-string is noticeably higher than that of the bits' configuration. When, instead, the solution is incorrect, the different results are distributed in a rather uniform way in over half of the cases (with some evident outlier).

7.4 CONCLUSION

In this chapter we presented a new algorithm for solving a minimization problem in QUBO form. From a theoretical point of view, we saw that QAOA guarantees a monotonic improvement in performance as the number of levels increases (provided that optimal parameters have been found), while QAC does not guarantee this monotonicity when increasing the annealing time. We then presented the critical aspects of QAOA, highlighting how the problem of finding optimal parameters is, in general, NP-Hard.

The discussion then moved to the practical implementation, and we saw how to use the Qiskit Framework to solve on IBM quantum gate based machines a problem originally in QUBO form. We presented Python scripts to automate the translation, we saw how to retrieve the results and how to interpret them. Finally, we discussed the quality of the solutions provided by QAOA by repeating the parameter tuning and the sampling of the optimized circuit, discussing the obtained results.

In the previous chapter we showed how to add a rewriting step upstream of the QA-Prolog pipeline. This chapter can also be considered as a further rewriting step of a problem into different but equivalent formats; this time downstream of the pipeline we took the output provided by QMASM and rewrote it into a format compatible with QAOA.

8 CONCLUSION

In this chapter we briefly review the results achieved in this work, and propose some directions for future work and investigations.

8.1 RESULTS ACHIEVED

8.1.1 Updating QA-Prolog

The QA-Prolog software stack had remained without maintenance while the frameworks on which it relied were updated. As a result, the pipeline initially did not work and did not correctly interface with the D-Wave Ocean framework.

Part of the work of this thesis consisted in restoring compatibility between the software written *ad hoc* for the QA-Prolog pipeline and the other software used to complete the entire translation process.

At present, the updated version of QMASM is available[1] and allows the use of the entire QA-Prolog pipeline, that is, starting from a Prolog program, obtaining an equivalent problem in QUBO form, and retrieving the results in a readable format after solving the QUBO problem.

8.1.2 Ontology in Prolog

The work of this thesis continued with the translation of an ontology expressed in OWL (Section 3.1.3) into an equivalent knowledge base expressed in Prolog. We started by following the literature and implementing the main concepts offered by OWL in Prolog. We focused on the concepts of class and subclass, relations between individuals, and the identification of inconsistencies in the KB. We also mentioned that in the literature there are works showing how to translate more advanced concepts such as the open world assumption and enumerable classes.

QA-Prolog does not implement all the functionalities of Prolog; the main limitation is that it does not allow the definition of recursive rules. The contribution of this thesis was, therefore, to propose solutions to overcome the restrictions imposed by QA-Prolog, managing to obtain a KB equivalent to the original ontology and compatible with the QA-Prolog pipeline.

The proposed solution to address the problem of recursive calls was to perform recursion unfolding. We then focused on the drawbacks of this technique. We observed that developing many levels of recursion through unfolding leads to a significant increase in the size of the problem, and therefore in the number of *qubits* required to represent it. We estimated and

1 a https://github.com/DavideCamino/qmasm.

discussed the quadratic increase in the size of the problem as a function of the number of unfolded recursion levels and observed that this can be problematic considering the current state of quantum machines, both gate-based and quantum annealers.

8.1.3 QAOA Algorithm

The last chapter of the thesis presents an alternative algorithm to annealing for solving an optimization problem in QUBO form such as the one obtained at the end of the QA-Prolog pipeline.

We presented the QAOA algorithm, briefly discussing its characteristics from a theoretical point of view. We then focused on automating the rewriting from the QUBO weight matrix (Section 2.3.3) to a format compatible with IBM's Qiskit framework.

After obtaining the translation, we performed some tests starting from the output of QA-Prolog and attempting to find the solution through QAOA. We carried out various tests and commented on the obtained results, observing that QAOA is not always able to return the correct solution.

8.1.4 A rewriting process

This work can be summarized as a long chain of rewritings of problems into equivalent problems. The goal is to move from one format to another, from one language to another, while preserving certain properties of interest, the most important one, of course, is that the solution of the problem must never change.

This rewriting process makes it possible to have different versions of the same problem, to observe it from different perspectives, and to choose the most convenient format to manipulate the problem depending on the objective. For example, we would like to describe the problem in the most convenient way possible, using a high-level language; on the other hand, we would like to solve it as quickly as possible, and if this means using a quantum machine, we would like to be able to move from one representation to another easily. All intermediate steps are also important because they offer different views of the problem and give us the possibility to extract properties that are not obvious at other levels of abstraction and, consequently, to optimize the problem or identify the best solving strategy.

8.2 FUTURE WORK

We present some directions for future research that can benefit from the tools developed in this work and from the results achieved.

8.2.1 Testing on quantum hardware

In this thesis we limited ourselves to testing the proposed algorithms and solutions on the simulators provided by D-Wave and IBM. The reasons for

this choice are mainly related to licensing issues. D-Wave does not allow the use of its platforms without a subscription; IBM instead offers a certain amount of computational time on its quantum machines, but without a paid plan the priority assigned to tasks is minimal, and due to time constraints it was not possible to perform satisfactory tests on quantum computers.

A possible future direction would be to verify the experimental results achieved in this thesis also on quantum hardware. On one hand, we can expect a speedup in performance due to the exploitation of quantum properties rather than their simulation on classical hardware. On the other hand, an improvement in the precision of the obtained solutions is not sure, due to noise issues that are still significant in current machines.

8.2.2 Increasing QMASM compatibility

QMASM (Section 5.2.1) is the last stage of the QA-Prolog pipeline and was developed to interface with the D-Wave Ocean framework. Thanks to QMASM it is therefore possible to submit a problem in QUBO form directly to D-Wave quantum annealers or to solve it using one of the simulators provided within the Ocean framework. These simulators are designed to test code on small instances in order to be more confident in correct execution when moving to quantum hardware. In our experiments we encountered the limitations of these simulators in Section 5.2.3.

Making QMASM compatible with other frameworks and libraries for solving QUBO problems would allow for greater flexibility in choosing the solving tool, potentially enabling both classical and quantum approaches. This would open the possibility of using simulated annealing, simulated quantum annealing, or the QAOA algorithm, in addition to quantum annealing. It would also remove the dependency on the D-Wave framework.

Interfacing QMASM with other tools is an approach fundamentally different from the one adopted in Chapter 7. There, we started from the matrix representation of the QUBO problem and translated it into a format compatible with Qiskit, thus ending all interactions with QMASM. If QMASM itself were able to interface with other solvers, we could also benefit from traversing the pipeline in reverse, that is, once the results are obtained, translating them back into a high-level language to facilitate interpretation.

In Chapter 7 we chose to implement the QAOA algorithm in Qiskit to solve the satisfiability problem, since we could directly read the solution from the final configuration of the *qubits*. Encoding a KB and a query would have been possible, but interpreting the results in binary format would have been extremely complex. If QMASM interfaced with Qiskit in the same way it does with Ocean, it would be possible to obtain results in a human-readable format.

Part IV

APPENDIX

A CODE

```
// Verilog version of Prolog program friends.pl
// Conversion by QA-Prolog, written by Scott Pakin <pakin@lanl.gov>
//
// This program is intended to be passed to edif2qmasm, then to qmasm, and
// finally run on a quantum annealer.
//
// Note: This program uses 3 bit(s) for atoms and 1 bit(s) for (unsigned)
// integers.

// Define all of the symbols used in this program.
`define alice   3'd0
`define bob     3'd1
`define charlie 3'd2
`define enemies 3'd3
`define friends 3'd4
`define hates   3'd5

// Define hates(atom, atom).
module \hates/2 (A, B, Valid);
  input [2:0] A;
  input [2:0] B;
  output Valid;
  wire [1:0] $v1;
  assign $v1[0] = A == `alice;
  assign $v1[1] = B == `bob;
  wire [1:0] $v2;
  assign $v2[0] = A == `bob;
  assign $v2[1] = B == `charlie;
  assign Valid = &$v1 | &$v2;
endmodule

// Define enemies(atom, atom).
module \enemies/2 (A, B, Valid);
  input [2:0] A;
  input [2:0] B;
  output Valid;
  wire $v1;
  \hates/2 \hates_pujkx/2 (A, B, $v1);
  wire $v2;
  \hates/2 \hates_DoUbH/2 (B, A, $v2);
  assign Valid = &$v1 | &$v2;
endmodule

// Define friends(atom, atom).
module \friends/2 (A, B, Valid);
```

```
  input [2:0] A;
  input [2:0] B;
  output Valid;
  (* keep *) wire [2:0] C;
  wire [2:0] $v1;
  \enemies/2 \enemies_GcbiL/2 (A, C, $v1[0]);
  \enemies/2 \enemies_GNLnA/2 (C, B, $v1[1]);
  assign $v1[2] = A != B;
  assign Valid = &$v1;
endmodule

// Define Query(atom, atom).
module Query (P1, P2, Valid);
  input [2:0] P1;
  input [2:0] P2;
  output Valid;
  wire $v1;
  \friends/2 \friends_KMGjP/2 (P1, P2, $v1);
  assign Valid = &$v1;
endmodule
```

Listing A.1: QA-Prolog compilation result

```
ancestor1(I1, I2) :-
    hasproperty(I2, sonOf, I1).
ancestor1(I1, I2) :-
    hasproperty(I2, sonOf, I3).
    hasproperty(I1, marry, I3).
ancestor2(I1, I2) :-
  hasproperty(I1, grandParentOf, I2).
ancestor2(I1, I2) :-
    ancestor1(I1, I3),
    hasproperty(I2, sonOf, I3).
ancestor2(I1, I2) :-
    ancestor1(I3, I2),
    hasproperty(I3, sonOf, I4).
    hasproperty(I1, marry, I4).
ancestor3(I1, I2) :-
  ancestor1(I3, I2)
  hasproperty(I1, grandParentOf, I2).
ancestor3(I1, I2) :-
  ancestor2(I3, I2)
  hasproperty(I3, sonOf, I1).
ancestor3(I1, I2) :-
  ancestor2(I3, I2)
  hasproperty(I3, sonOf, I4).
  hasproperty(I1, marry, I4).
  ...
```

Listing A.2: Ancestor definition extended to family relation

LIST OF FIGURES

LIST OF LISTINGS

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